



Transforming Mobility: From Next-Visit Prediction to a Mobility Foundation Model

<http://imsc.usc.edu/>

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Spatial Sciences*

University of Southern California (USC)

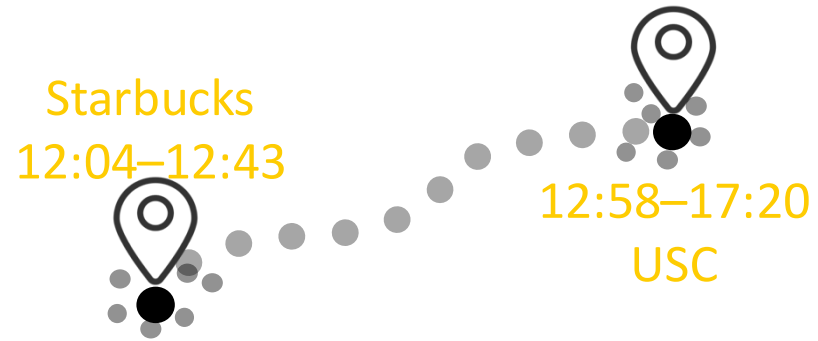
shahabi@usc.edu

OUTLINE

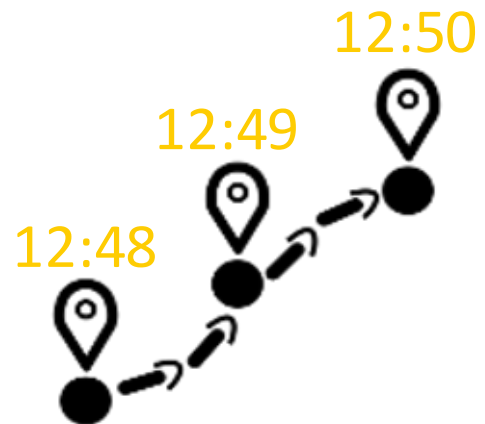


- **Motivation**
 - Mobility data is important to many applications
- **Background**
 - Prior Work on Synthetic Generation
 - Visit representation: location, arrival, and departure times
 - Next-visit prediction as a core task
- **TrajGPT: Transformer for Next-Visit Prediction**
- **Downstream Applications**
 - Filling missing visits (gap imputation)
 - POI attribution
 - Mobility outlier detection
- **Towards a Mobility-Foundation Model (Open Problems)**

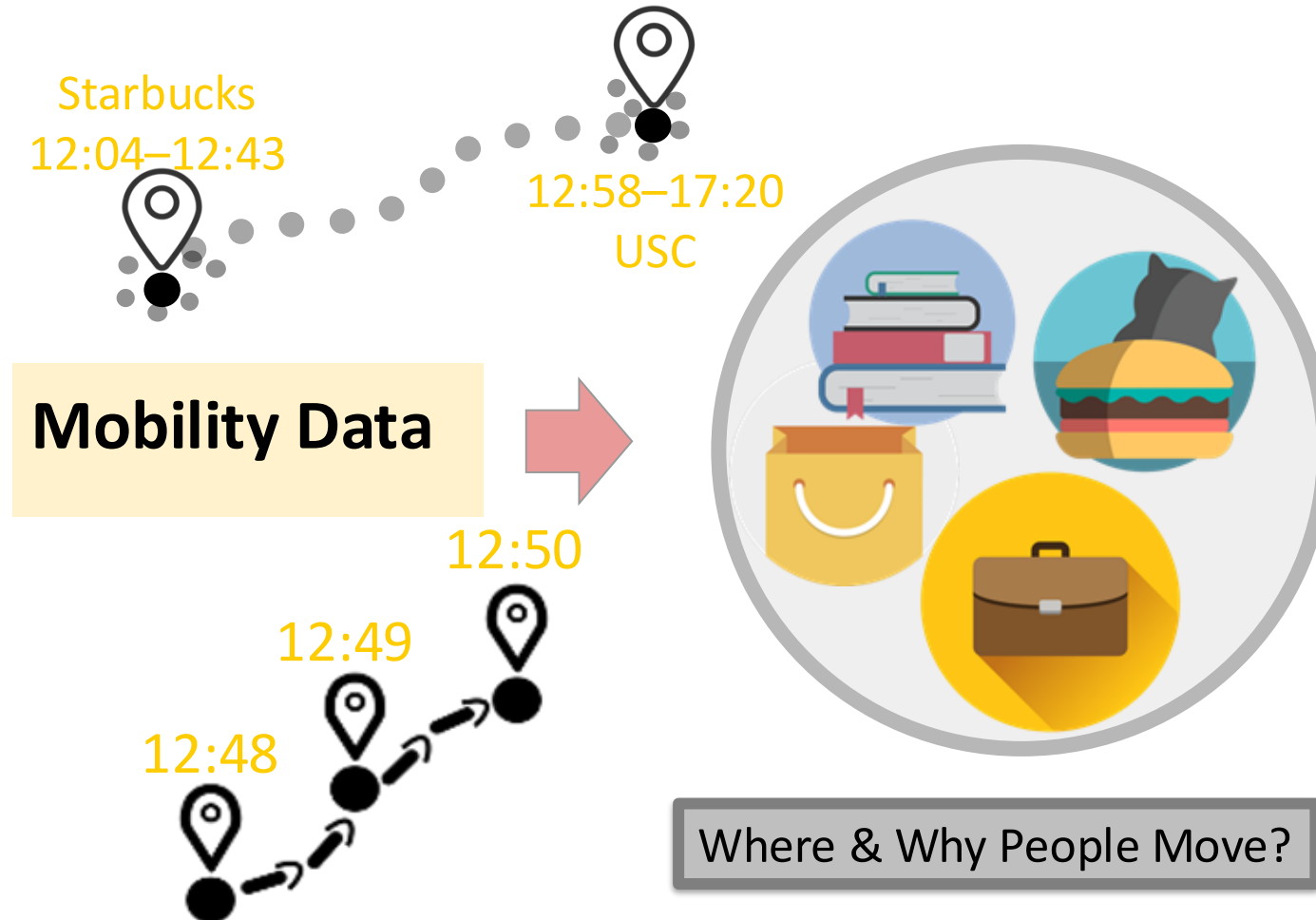
Motivation



Mobility Data



Motivation



Motivation



Example Applications



Example Applications



- Synthetic Trajectory Generation
 - Traffic Forecasting and Routing
 - Pollution Monitoring/Forecasting



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 - Recommendation (e.g., where to park)
 - Intervention (e.g., healthy vs. junk food)
 - Agentic AI (e.g., suggest fastest Paths)
 - Target advertising (e.g., coffeeshop discount)



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- Anomaly Detection
 - Suspicious Activity
 - Intervention (e.g., alcohol consumption midday)



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- Anomaly Detection

- Suspicious Activity
- Intervention (e.g., alcohol consumption midday)



- Social Relationship Inference

- Mental Health
- Recommendation and Marketing



OUTLINE

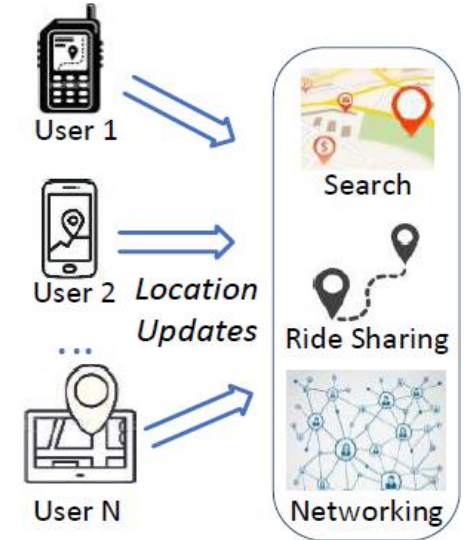


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Scarcity of “Public” Mobility Data



Name	# Objects	Location	Duration	Year	# Points
Go!Track [7]	163 trips	various	2 years	2014–2016	18,107
CabSpotting [1]	536 taxis	San Francisco, CA	24 days	2008	11,219,955
Chengdu Taxi Data [9]	-	Chengdu, China	2 months	2018	3,032,212
Xian Taxi Data [9]	-	Xian, China	2 months	2018	18,267,443
Porto Taxi Data [10]	442 taxis	Porto, Portugal	5 months	2013	12,852,750
LaDe [11]	21K couriers	5 cities	6 months	2018	10,677,000
T-Drive [6]	10,357 taxis	Beijing, China	1 week	2008	15,000,000
Rome Taxi Data [5]	316 taxis	Rome, Italy	1 month	2014	21,817,851
DeepGTT [2]	12,475 taxis	Heilongjiang, China	5 days	2015	58,834,550
GeoLife [3]	182 users	Beijing, China	3 years	2007–2012	24,876,978
Mobile Data Challenge [4]	185 users	Lausanne/Geneva, Switzerland	1 year	2009–2011	26,152,673
Foursquare-TKY [13]	2,321 users	Tokyo, Japan	10 months	2012–2013	537,703
Foursquare-NYC [13]	1,083 users	New York, NY	10 months	2012–2013	227,428
Gowalla [12]	196,591 users	worldwide	20 months	2009–2010	6,442,890
Veraset CA [8]	29M devices	California, USA	89 days	2019	13.8B

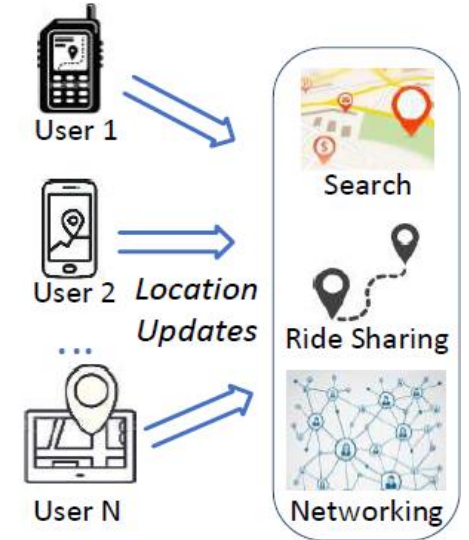


- Free datasets – not large enough
- Veraset – paid service

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- Privacy
- Confidentiality
- Cost
- Missing observations



Synthetic Trajectory Generation



Synthetic Trajectory Generation



Computational

Synthetic Trajectory Generation



Computational

- Moving-object generators (e.g., Brinkhoff 2002) → Travel surveys; hand-crafted rules

Synthetic Trajectory Generation



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- Agent-based microsimulations → Lots of parameter setting

Synthetic Trajectory Generation



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Limitation: Require human labor and domain knowledge to convert real-world data into the parameters/rules of the closed (artificial) environment.

Synthetic Trajectory Generation



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Data-Driven



- Learn directly from the data distribution

Synthetic Trajectory Generation



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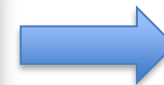
Data-Driven



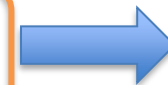
- Learn directly from the data distribution



Real Data



Learning
Algorithm



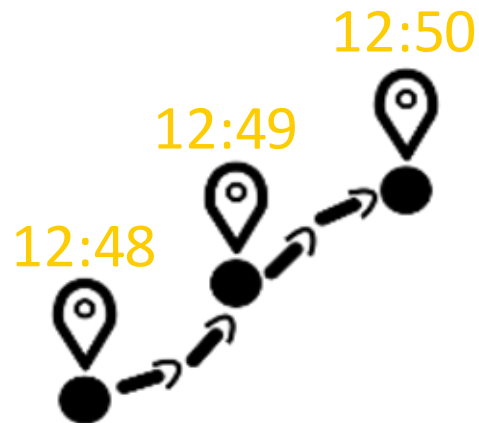
Generated Data



Forms of Trajectory Data

Sequences of **GPS logs**

- Location: (latitude, longitude)
- Regular Timestamp

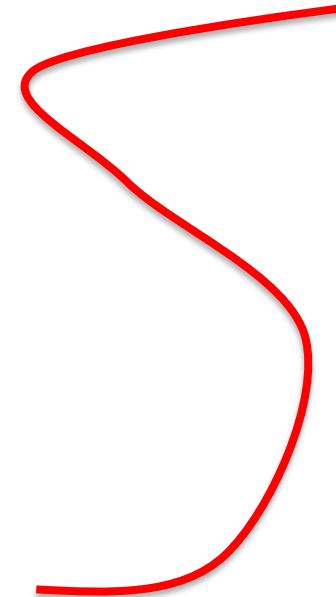
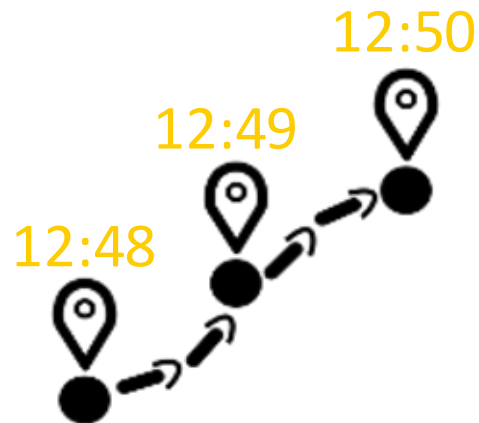




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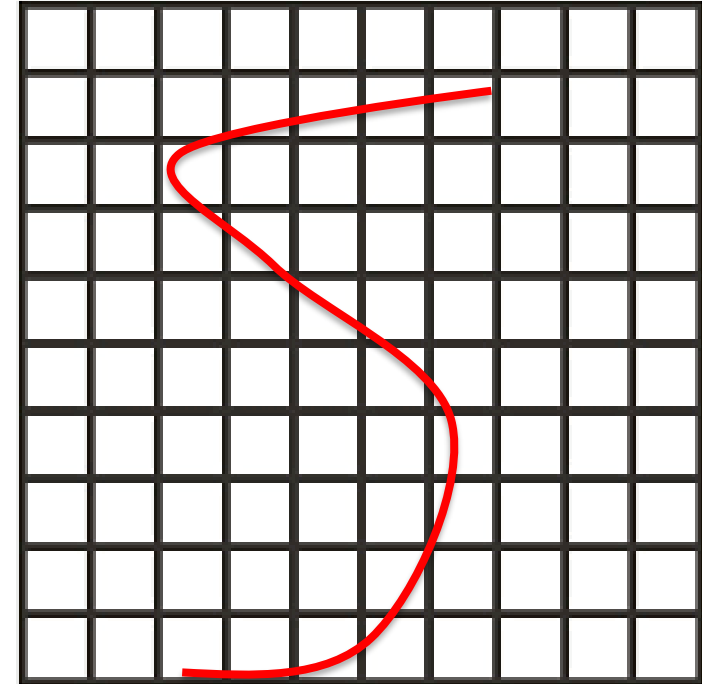
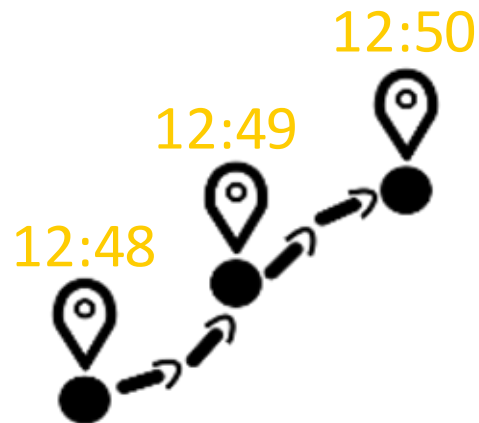




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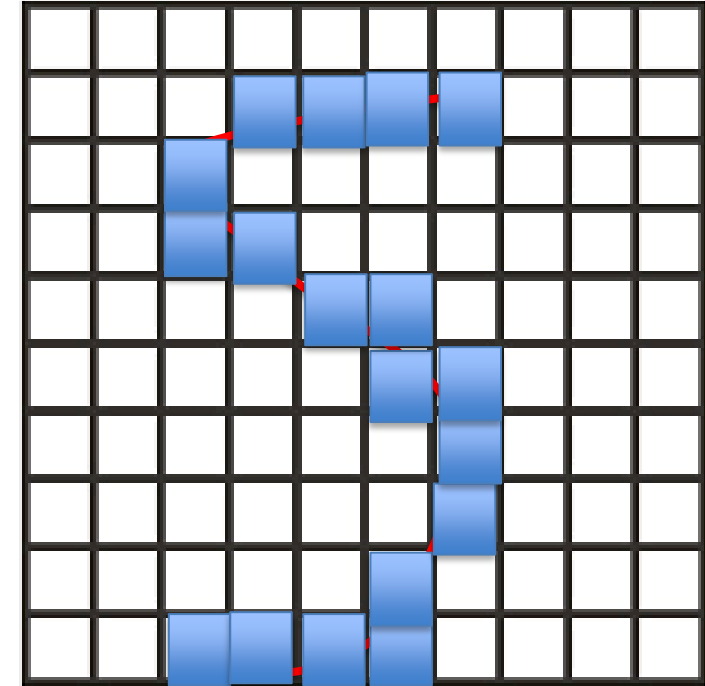
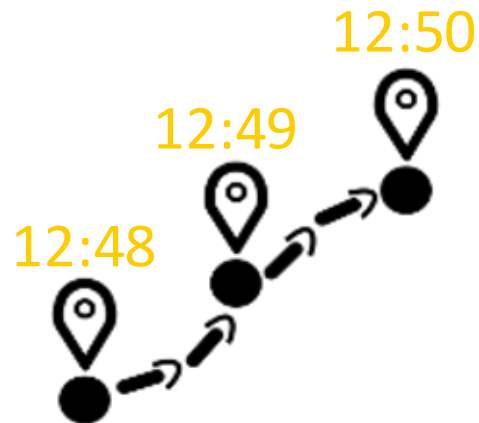




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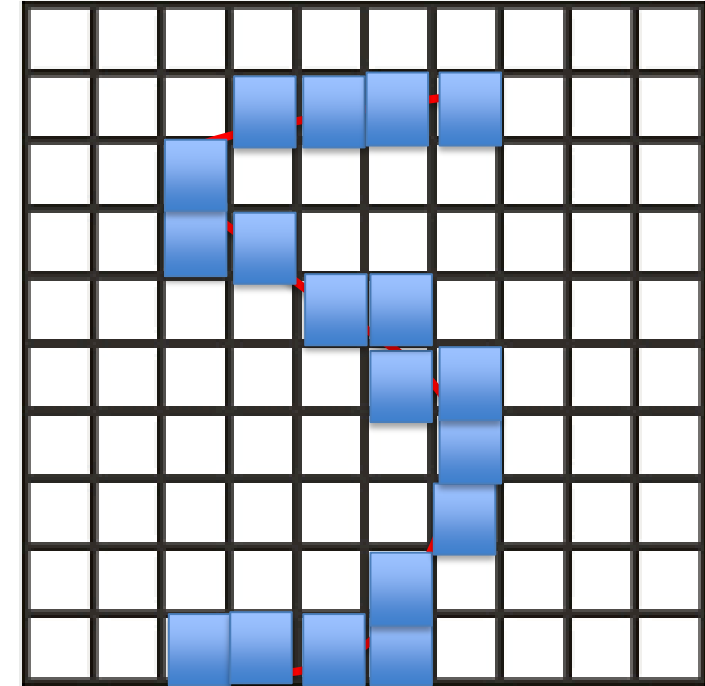
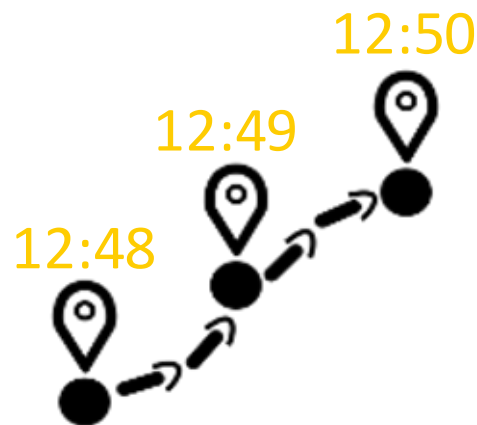
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Forms of Trajectory Data

Sequences of GPS logs

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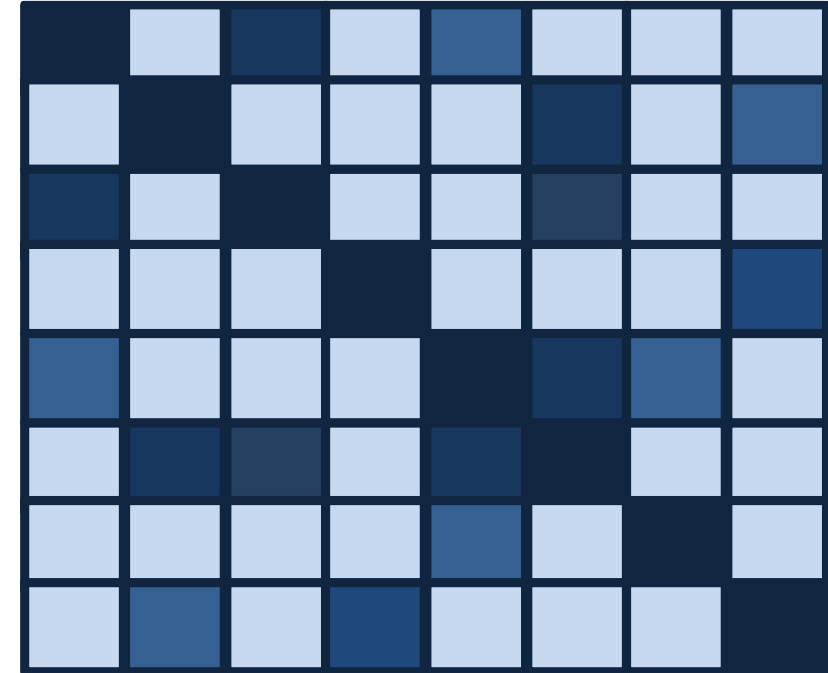
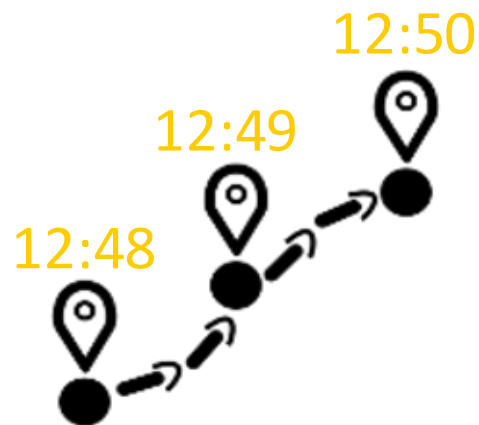
- A computer vision problem: GAN, VAE, Diffusion
- Large “worlds” lead to sparser trajectory images, making learning difficult



Forms of Trajectory Data

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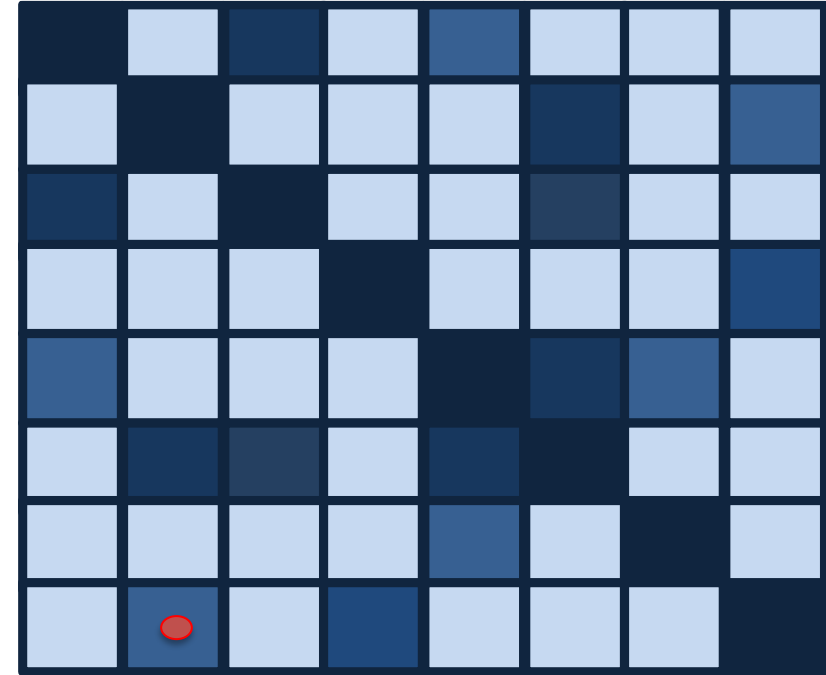
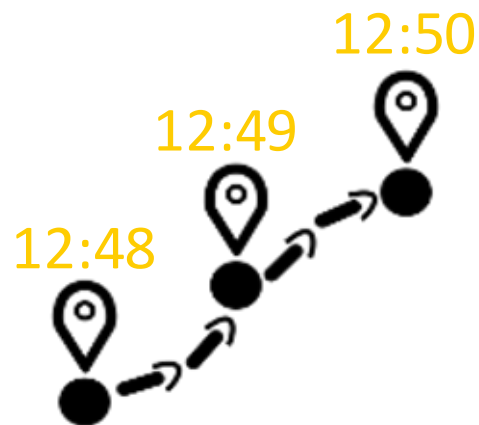
- A sequence generation problem: SeqGAN



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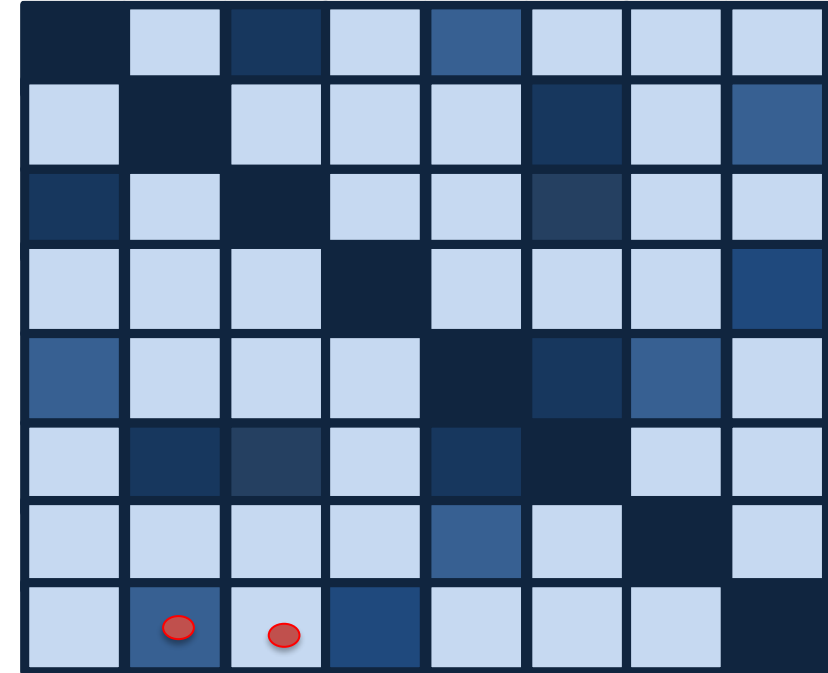
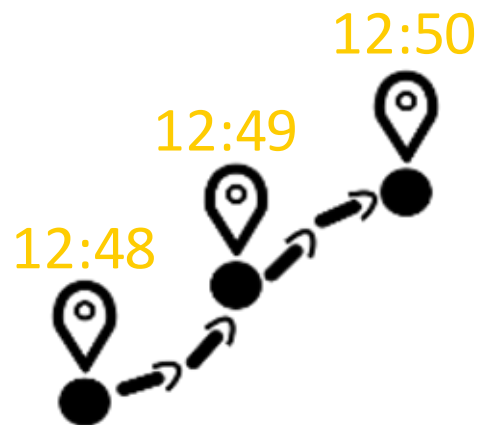
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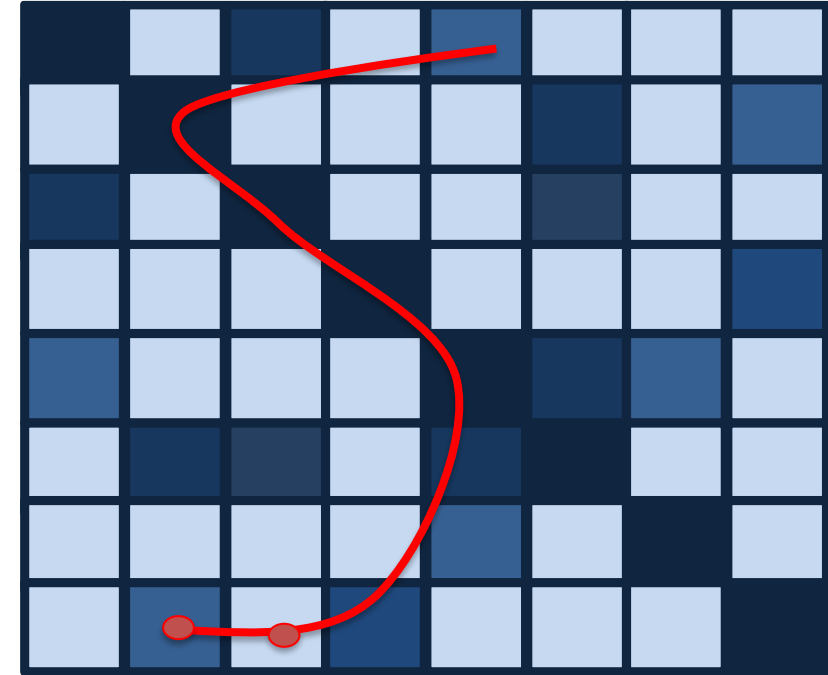
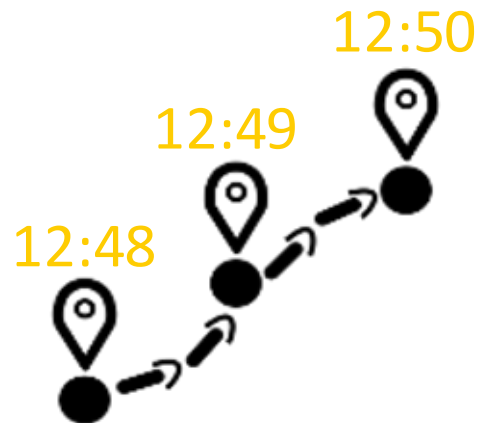
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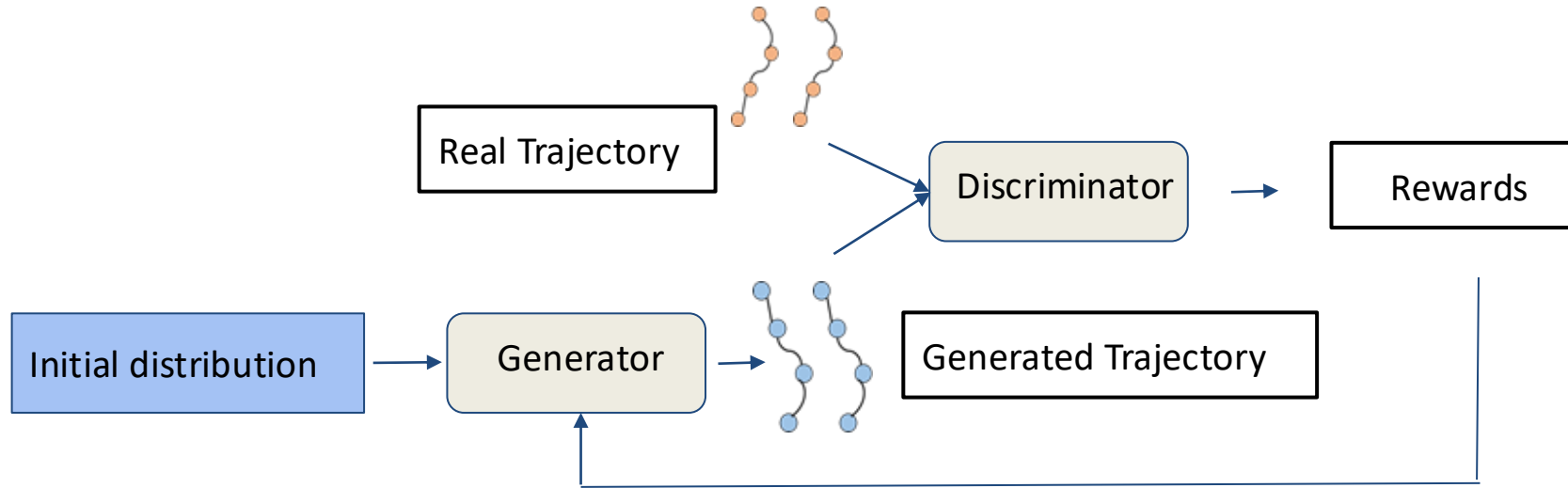
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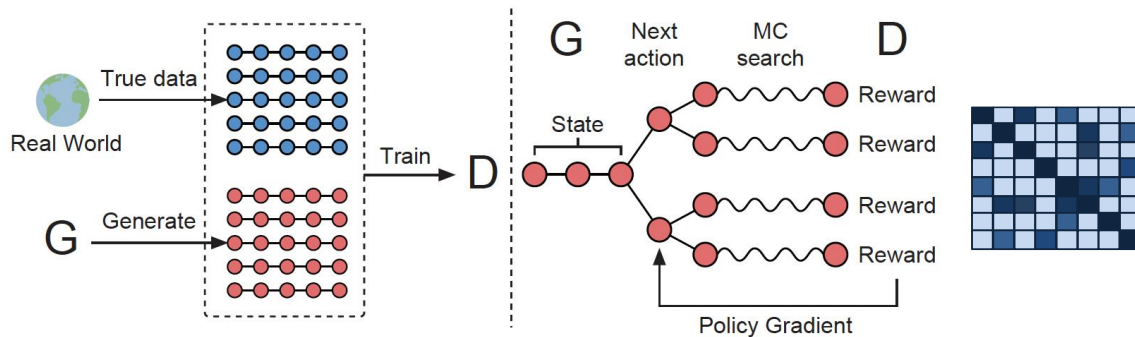
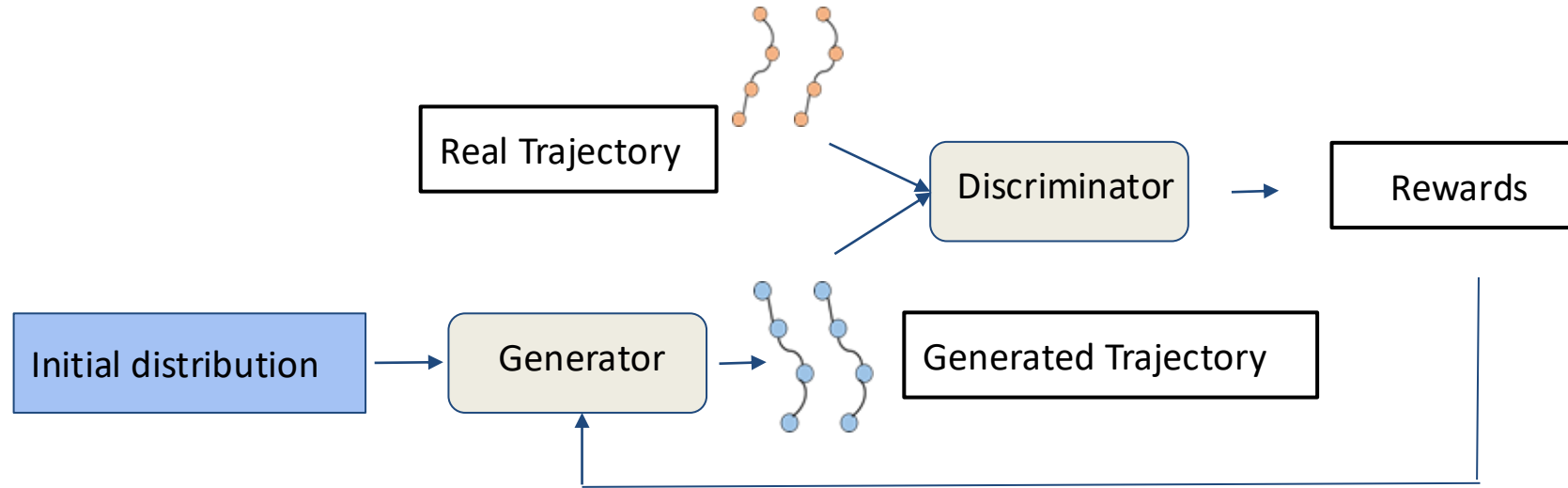
Seq Generative Adversarial Network



$$\min_{G_\theta} \max_{D_\phi} E_{Y \sim p_d(Y)} [\log(D_\phi(Y))] + E_{\hat{Y} \sim G_\theta(\hat{Y})} [\log(1 - D_\phi(\hat{Y}))]$$

Compete with each other in a zero-sum game framework.

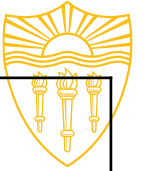
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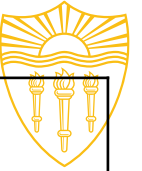
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Related Work



	Name	Methods	Dataset
Non Deep Learning Model	TimeGeo [PNAS16]	Statistics modeling	CDR(Boston)
DNN Methods	TrajGAN [IJCAI18]	GAN	Nokia Lausanne location
	MoveSim [KDD 20]	GAN	GeoLife (Beijing)
	TrajGAIL [ICDM20]	GAIL	Taxi data (Shenzhen)
	DeltaGAN [ICLR 21]	GAN	GeoLife (Beijing)
	NEXTGAIL [SIGSPATIAL21]	GAIL	Taxi data (Shenzhen)
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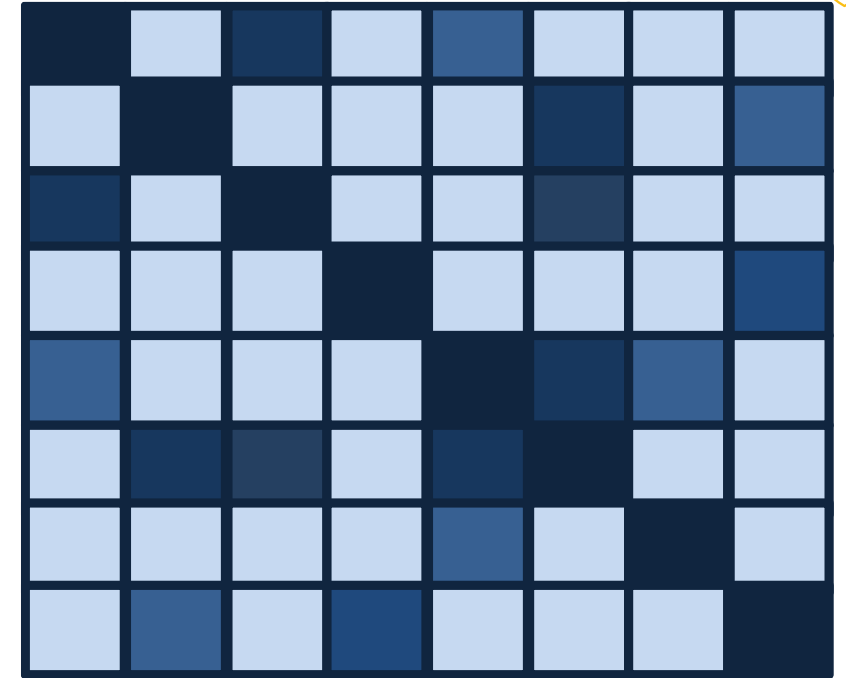
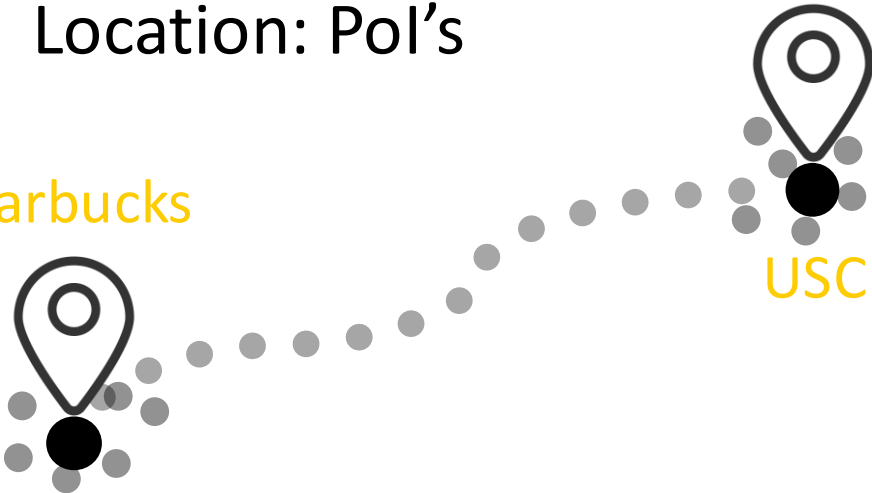
Forms of Trajectory Data



Sequences of Locations

- Location: Pol's

Starbucks



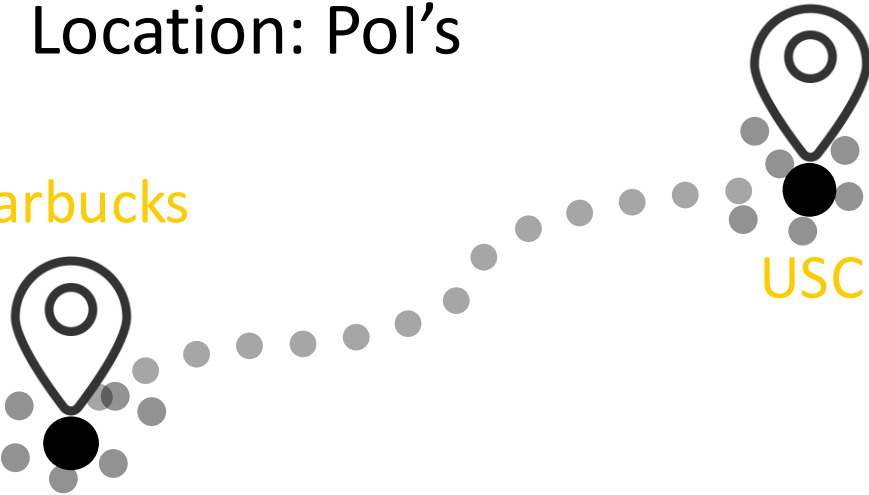
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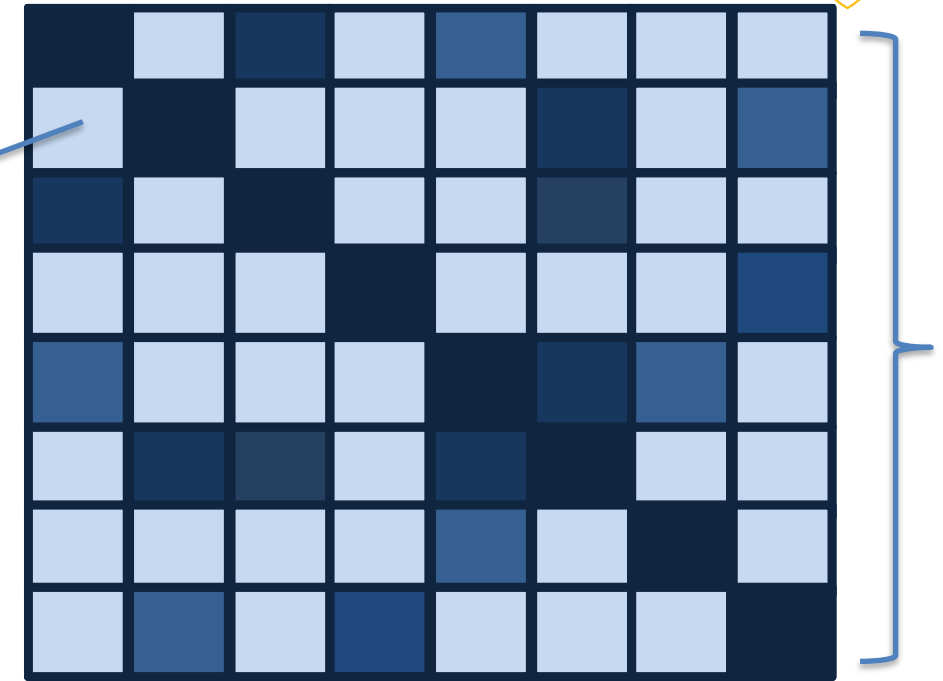
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A language problem

Words



Vocabulary

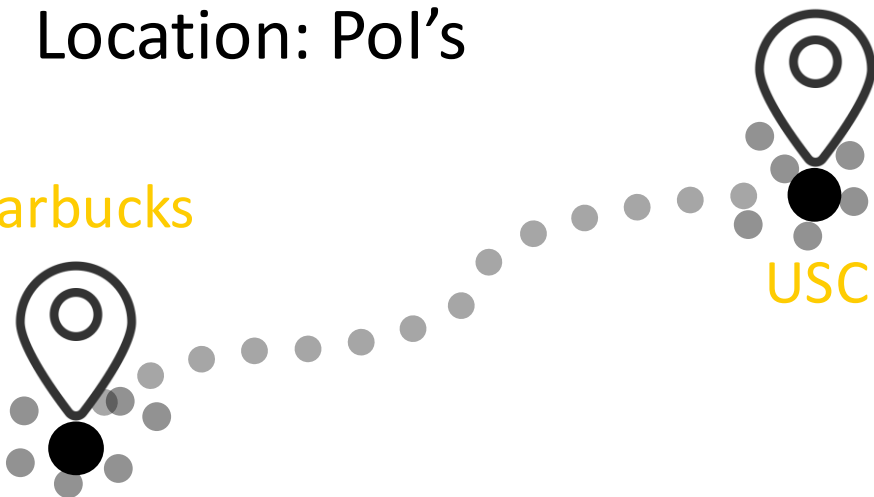
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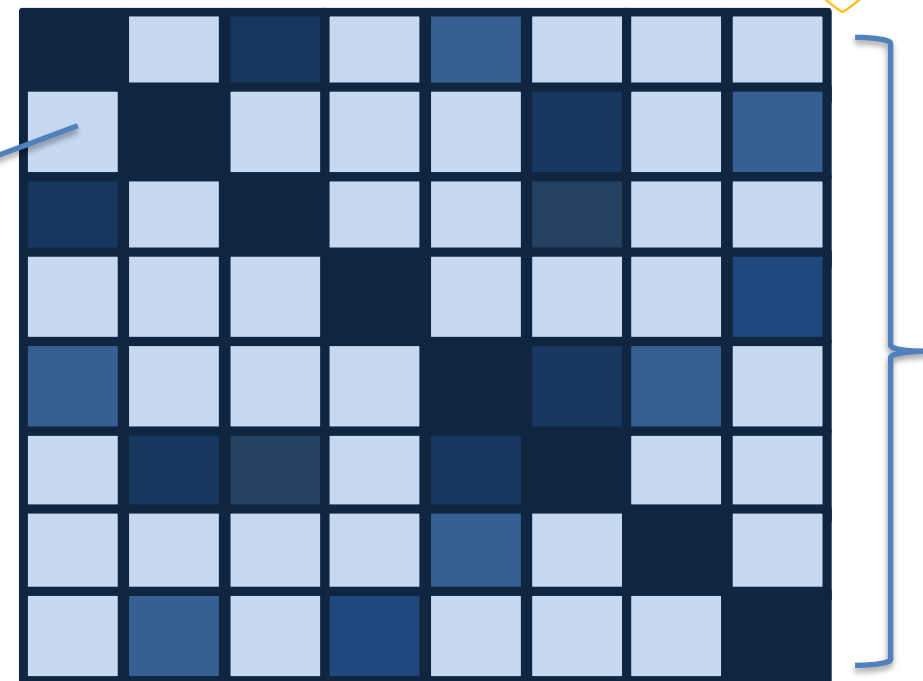
Starbucks



A language problem

- Transformers -- Auto Regressive **Next Location Prediction**

Words



Vocabulary

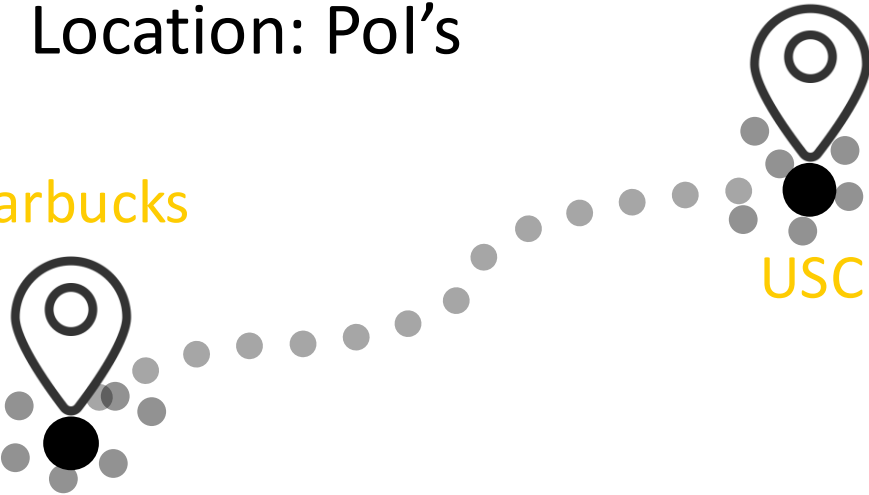
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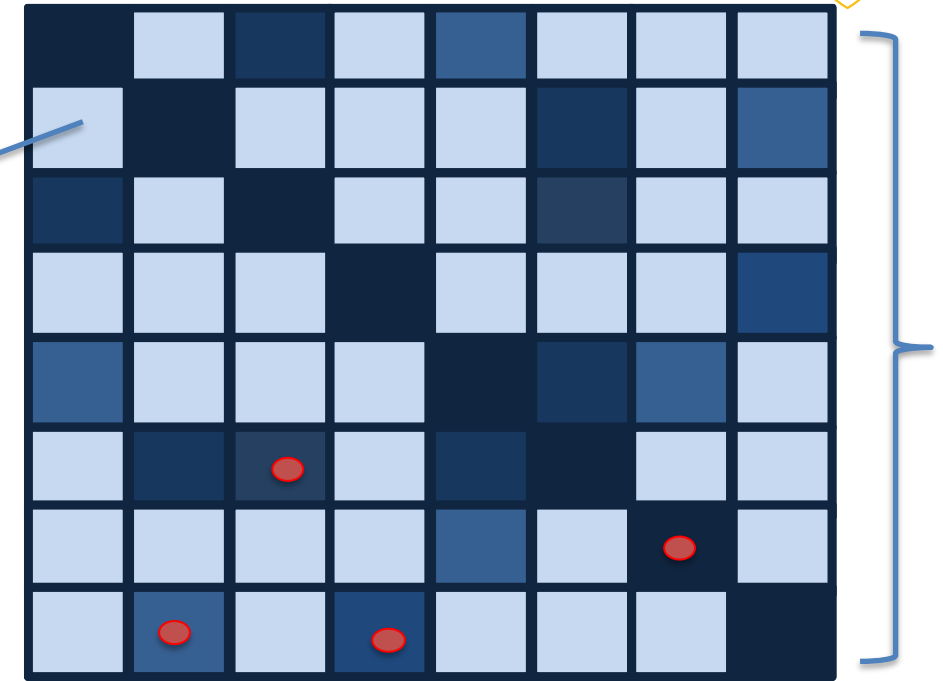
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Sentence ← [1 , 3 , 14, 21]

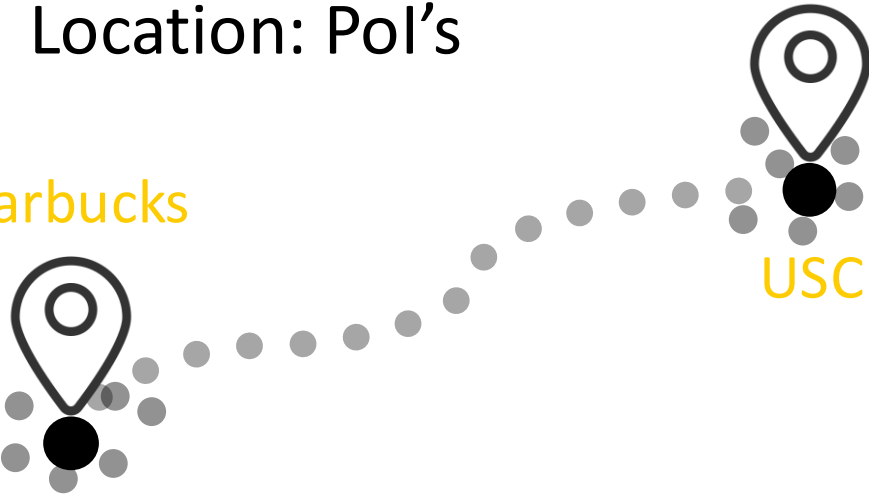
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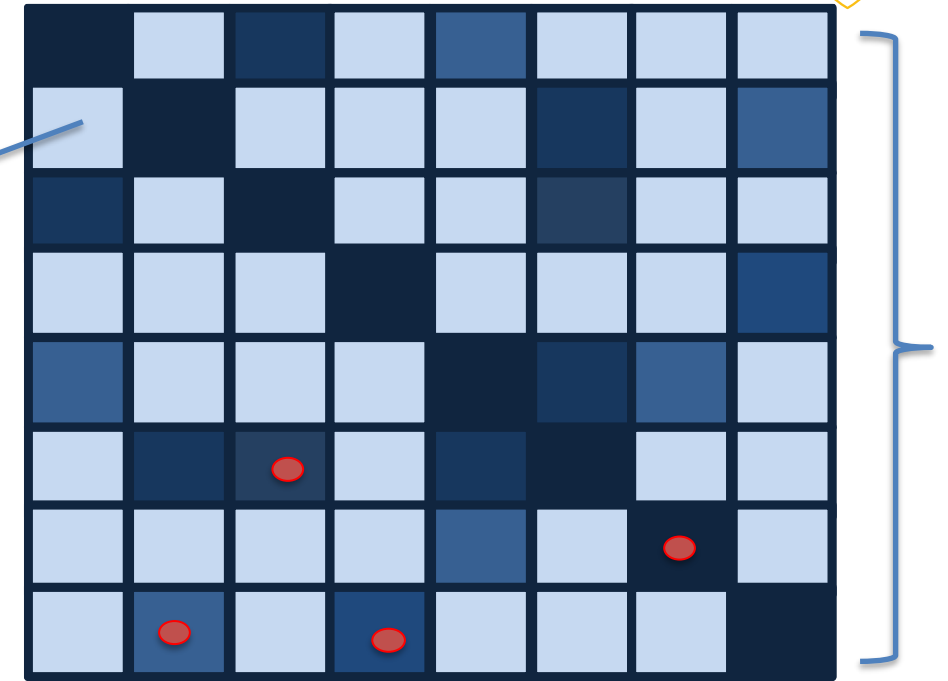
Sequences of Locations

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Vocabulary

Sentence [1 , 3 , 14, 21]

A language problem

- Transformers -- Auto Regressive **Next Location Prediction**
 - A deep learning architecture, relying on the attention mechanism
 - Widely used LLM (e.g. ChatGPT)

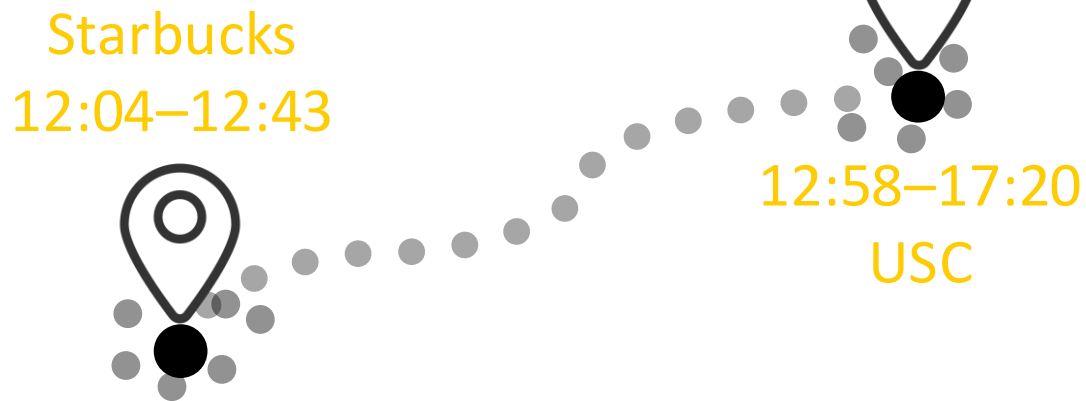
Data-Driven Approach	Related Work	Location Logs	Pol's 
Statistical	TimeGeo [PNAS 16]	✓	
	NEXTGAIL [SIGSPATIAL 21]	✓	
Adversarial/Variational	ActSTD [KDD 21]	✓	
	FVAE [SIGSPATIAL 22]	✓	
	Geo-CETRA [ICDM 24]	✓	
Transformer	DeepJMT [WSDM 20]		✓
	MobTCast [NeurIPS 21]		✓
	GETNext [SIGIR 22]		✓
	STAR-HiT [TOIS 23]		✓
Large Language Model	LLMmove [CAI 24]		✓
	LLM4POI [SIGIR 24]		✓



Forms of Trajectory Data

Sequences of visits

- Location: Pol's
- Arrival time
- Departure time

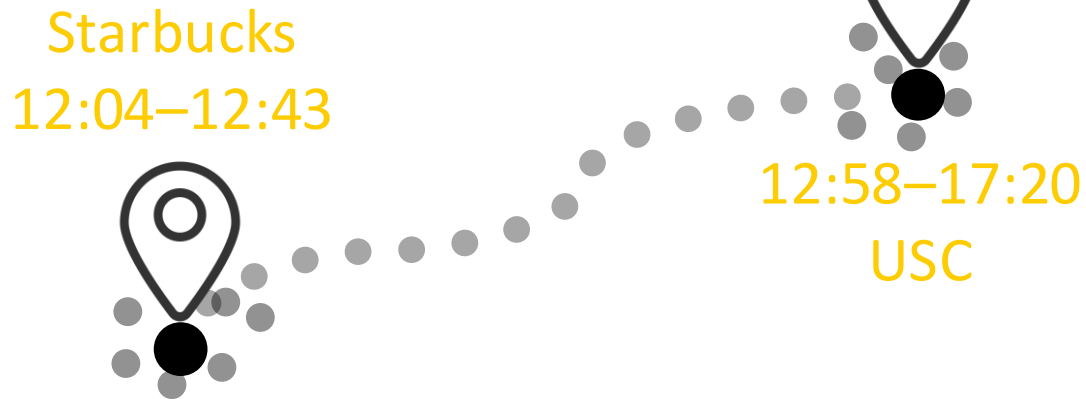




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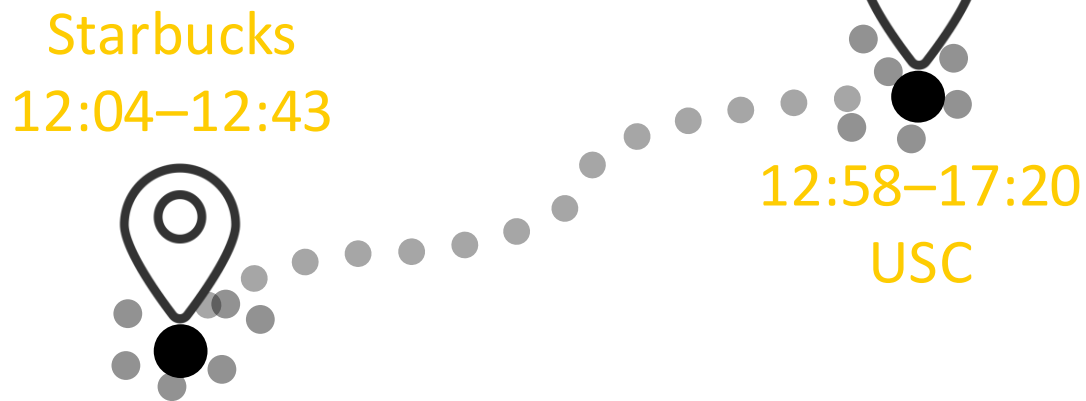
Prior Work Missing the “temporal” aspect:
Cannot generate visit sequences



Forms of Trajectory Data

Sequences of visits

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Can we adapt transformers for Auto Regressive **Next Location Visit Prediction?**

OUTLINE



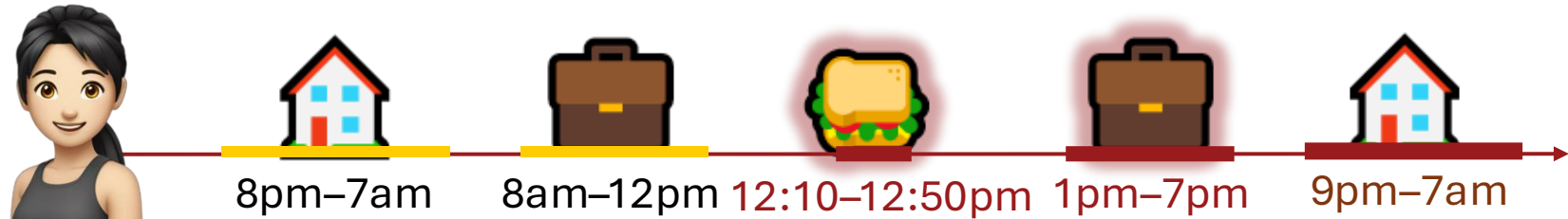
- **Motivation**
- **Background**
- **TrajGPT: Transformer for Next-Visit Prediction**
 - Why irregular, real-world visit data is challenging
 - Conditional loss with Gaussian Mixture Model (GMM) approximation
 - Predicting arrival/departure as probability distributions
- **Downstream Applications**
 - Filling missing visits (gap imputation)
 - POI attribution
 - Mobility outlier detection
- **Towards a Mobility-Foundation Model (Open Problems)**

TrajGPT: A Transformer-Based Approach to Synthetic Trajectory Generation

Input



Output

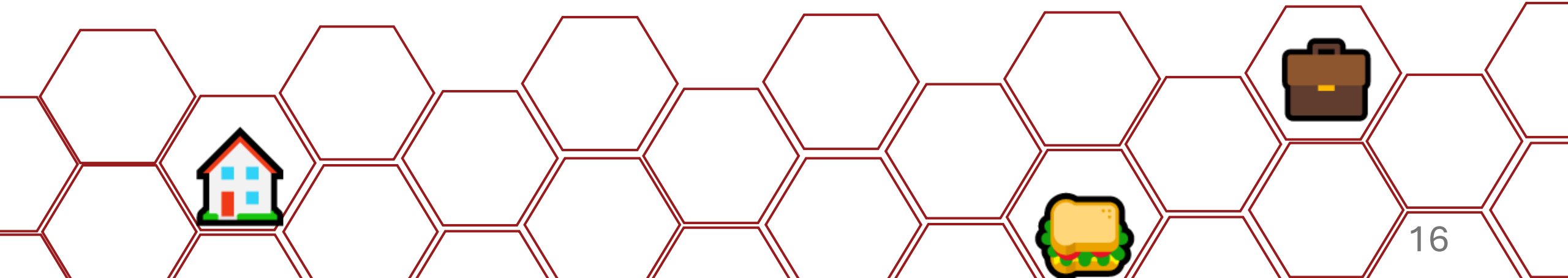
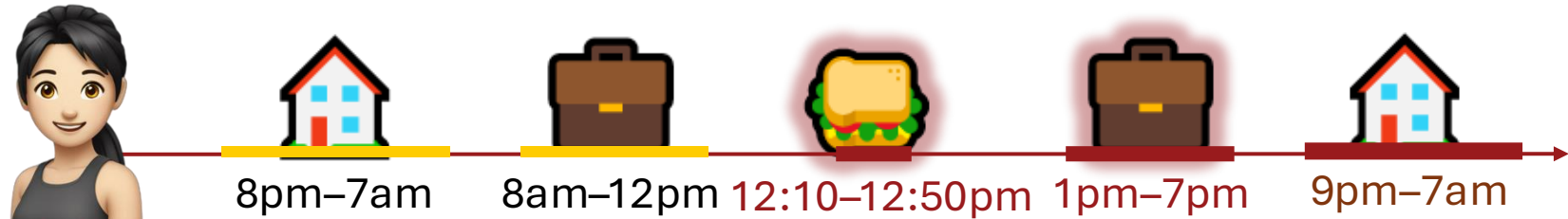


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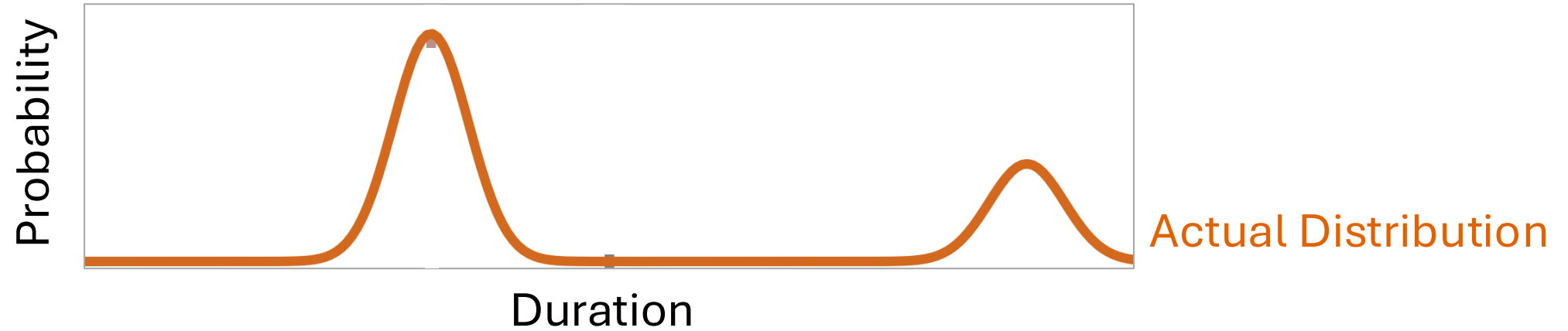
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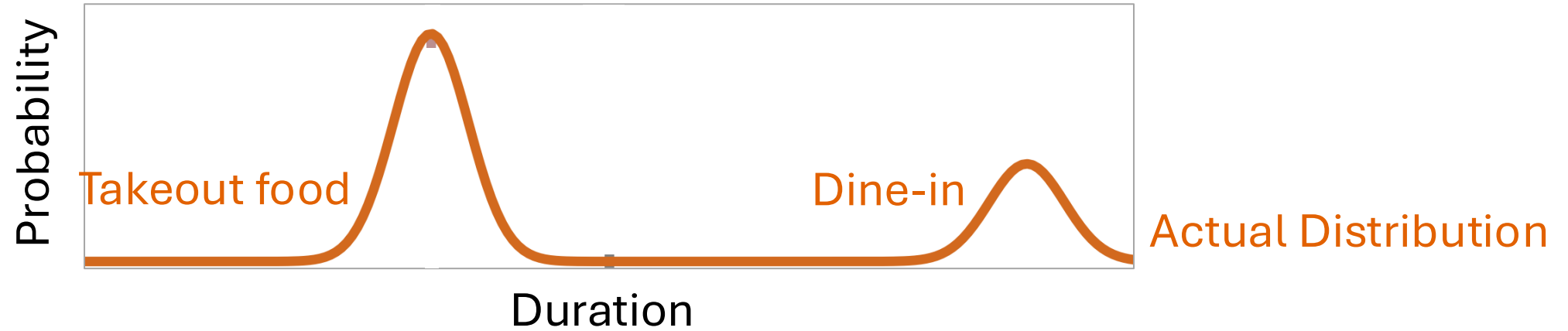
Output



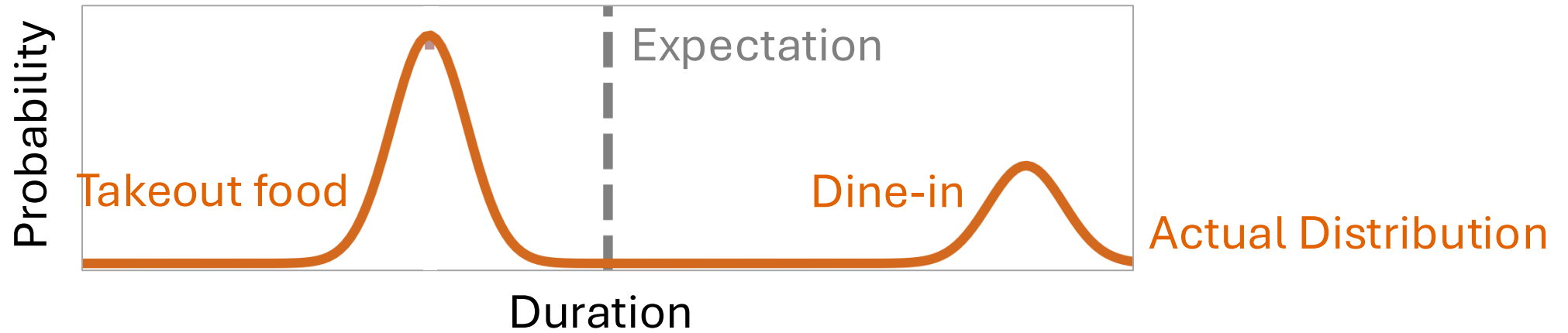
Temporal Challenge



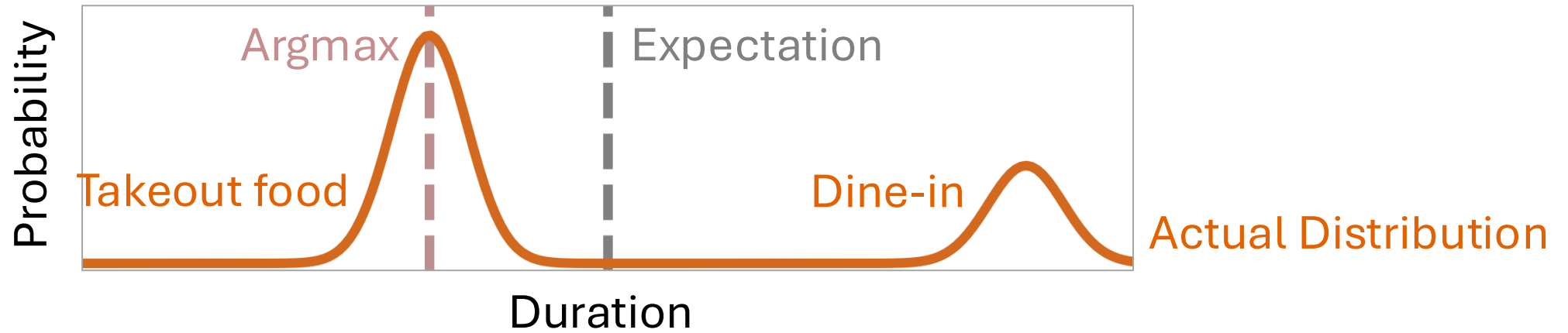
Temporal Challenge



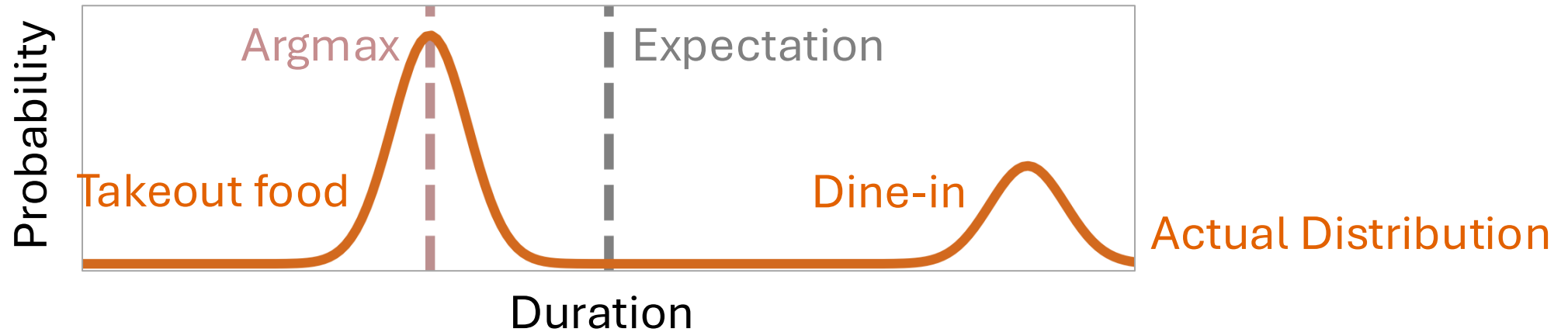
Temporal Challenge



Temporal Challenge



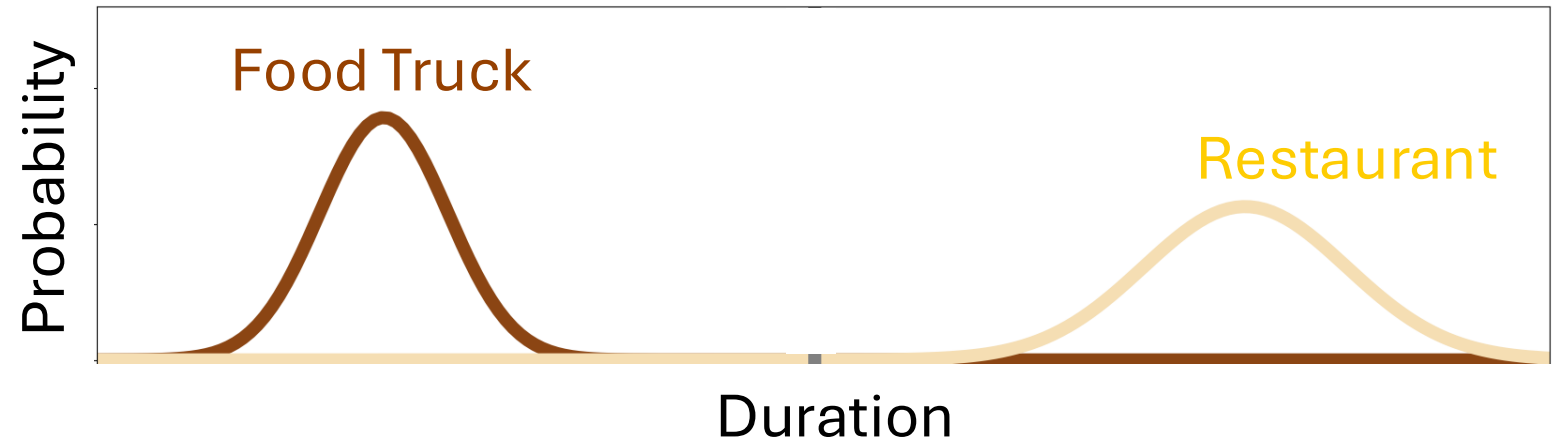
Temporal Challenge



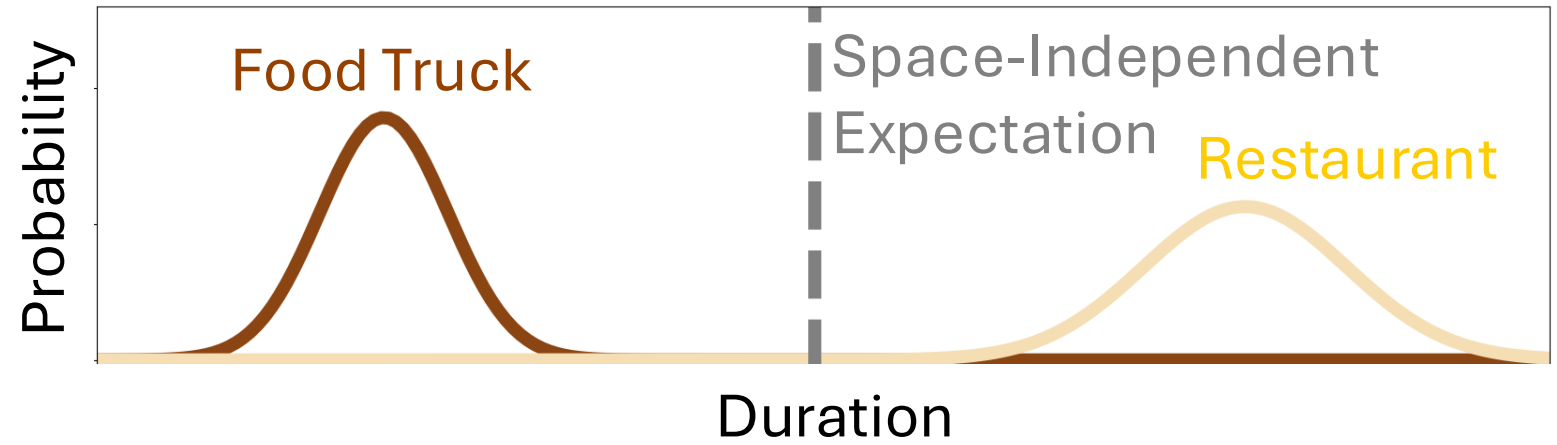
Solution: Temporal Gaussian Mixture Model

Approximate distributions instead of predicting times

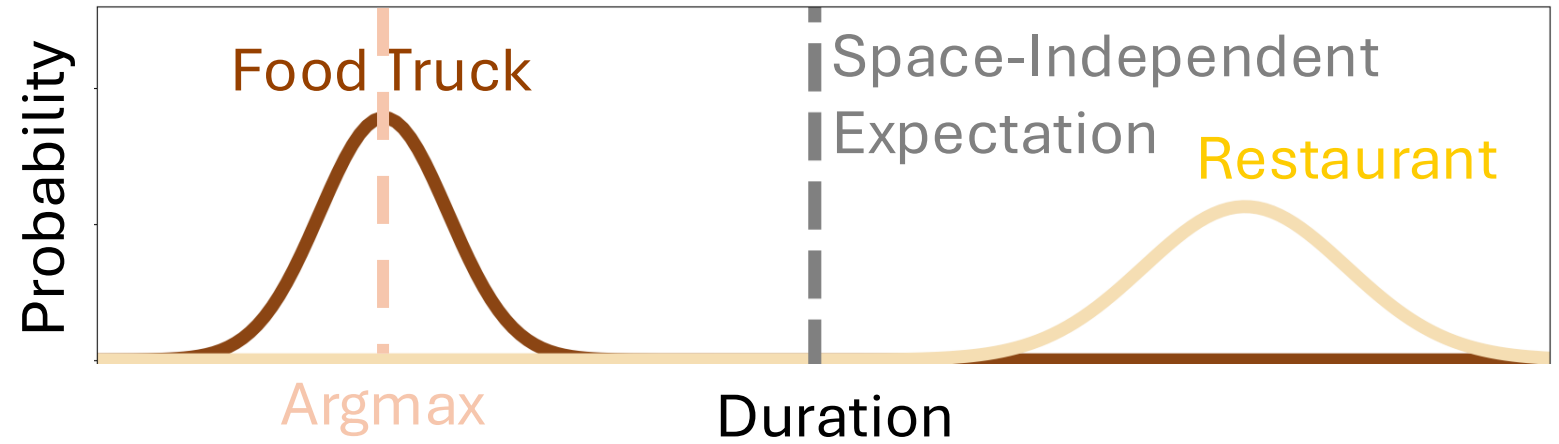
Spatiotemporal Challenge



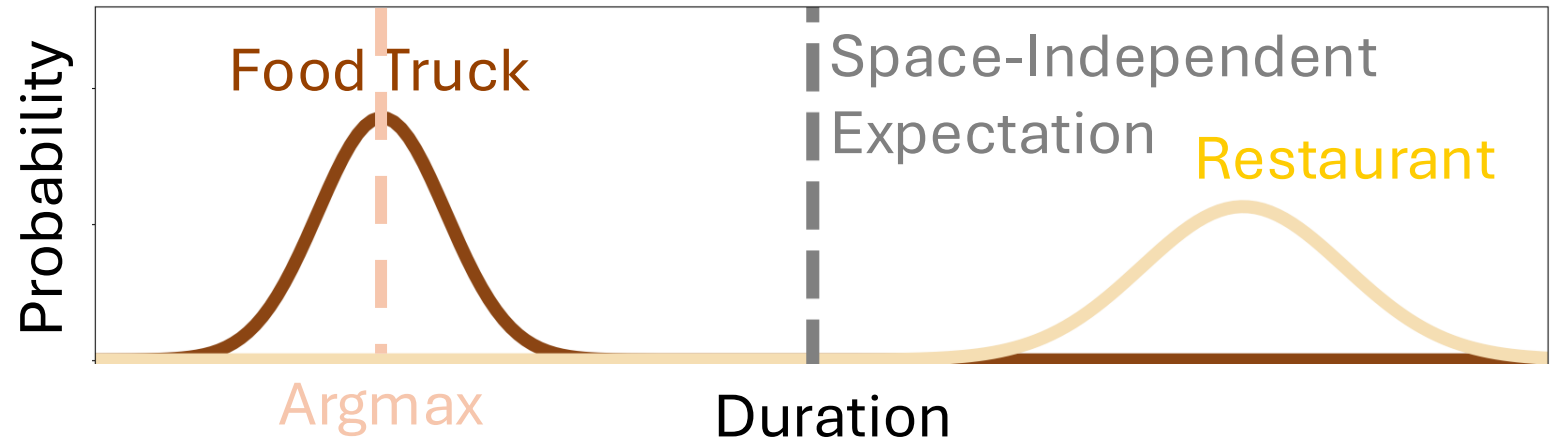
Spatiotemporal Challenge



Spatiotemporal Challenge



Spatiotemporal Challenge



Solution:

Spatiotemporal

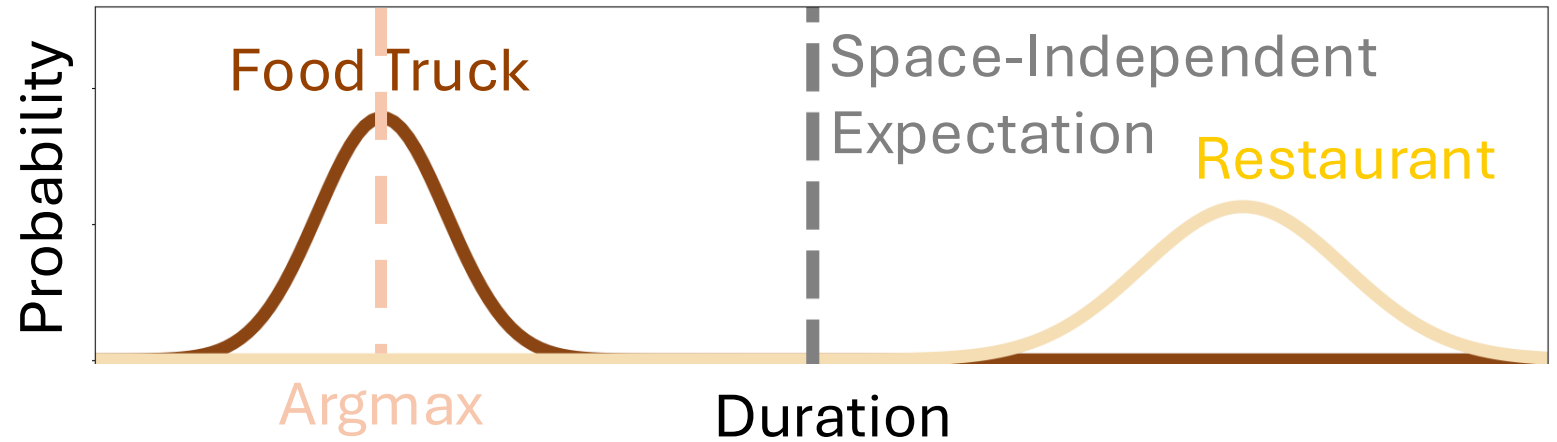
Joint Modeling

Input sequence

$$P(x \mid X') = P(r, t^a, t^d \mid X')$$

Next visit

Spatiotemporal Challenge



Solution:
Spatiotemporal
Joint Modeling

Input sequence

$$P(x \mid X') = P(r, t^a, t^d \mid X')$$

Next visit

Region

Arrival

Departure

Next Visit Prediction: Training Objective

Goal: Loss Function

$$\mathcal{L} = -\log \hat{P}(x \mid X')$$

Negative Log Likelihood loss

Next Visit Prediction: Training Objective

Goal: Loss Function

$$\mathcal{L} = -\log \hat{P}(x | X')$$

Likelihood of truth

Negative Log Likelihood loss

Next visit

Input sequence

Next Visit Prediction: Training Objective

Goal: Loss Function

$$\mathcal{L} = -\log \hat{P}(x | X')$$

Likelihood of truth

Negative Log Likelihood loss

Next visit

Input sequence

Chain Rule

$$\hat{P}(x | X') = \hat{P}(r, t^a, t^d | X')$$

$$= \hat{P}(r | X') \hat{P}(t^a | X', r) \hat{P}(t^d | X', r, t^a)$$

Next Visit Prediction: Training Objective

Goal: Loss Function

$$\mathcal{L} = -\log \hat{P}(x | X')$$

Likelihood of truth

Negative Log Likelihood loss

Next visit

Input sequence

Chain Rule

$$\hat{P}(x | X') = \hat{P}(r, t^a, t^d | X')$$

Joint distribution

$$= \hat{P}(r | X') \hat{P}(t^a | X', r) \hat{P}(t^d | X', r, t^a)$$

Next Visit Prediction: Training Objective

Goal: Loss Function

$$\mathcal{L} = -\log \hat{P}(x | X')$$

Likelihood of truth

Negative Log Likelihood loss

Next visit

Input sequence

Chain Rule

$$\hat{P}(x | X') = \hat{P}(r, t^a, t^d | X')$$

Joint distribution

$$= \hat{P}(r | X') \hat{P}(t^a | X', r) \hat{P}(t^d | X', r, t^a)$$

Region

.1	0	0
.4	.1	0
.2	0	.2

Next Visit Prediction: Training Objective

Goal: Loss Function

$$\mathcal{L} = -\log \hat{P}(x | X')$$

Likelihood of truth

Negative Log Likelihood loss

Next visit

Input sequence

Chain Rule

$$\hat{P}(x | X') = \hat{P}(r, t^a, t^d | X')$$

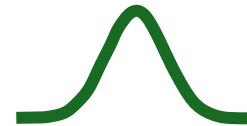
Joint distribution

$$= \hat{P}(r | X') \hat{P}(t^a | X', r) \hat{P}(t^d | X', r, t^a)$$

Region

.1	0	0
.4	.1	0
.2	0	.2

Arrival Time



Next Visit Prediction: Training Objective

Goal: Loss Function

$$\mathcal{L} = -\log \hat{P}(x | X')$$

Likelihood of truth

Negative Log Likelihood loss

Next visit

Input sequence

Chain Rule

$$\hat{P}(x | X') = \hat{P}(r, t^a, t^d | X')$$

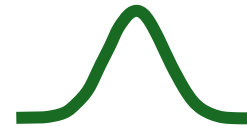
Joint distribution

$$= \hat{P}(r | X') \hat{P}(t^a | X', r) \hat{P}(t^d | X', r, t^a)$$

Region

.1	0	0
.4	.1	0
.2	0	.2

Arrival Time



Departure Time



Next Visit Prediction: Training Objective

Goal: Loss Function

$$\mathcal{L} = -\log \hat{P}(x | X')$$

Likelihood of truth

Negative Log Likelihood loss

Next visit

Input sequence

Chain Rule

$$\hat{P}(x | X') = \hat{P}(r, t^a, t^d | X')$$

Joint distribution

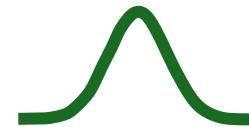
$$= \hat{P}(r | X') \hat{P}(t^a | X', r) \hat{P}(t^d | X', r, t^a)$$

Region

Arrival Time

Departure Time

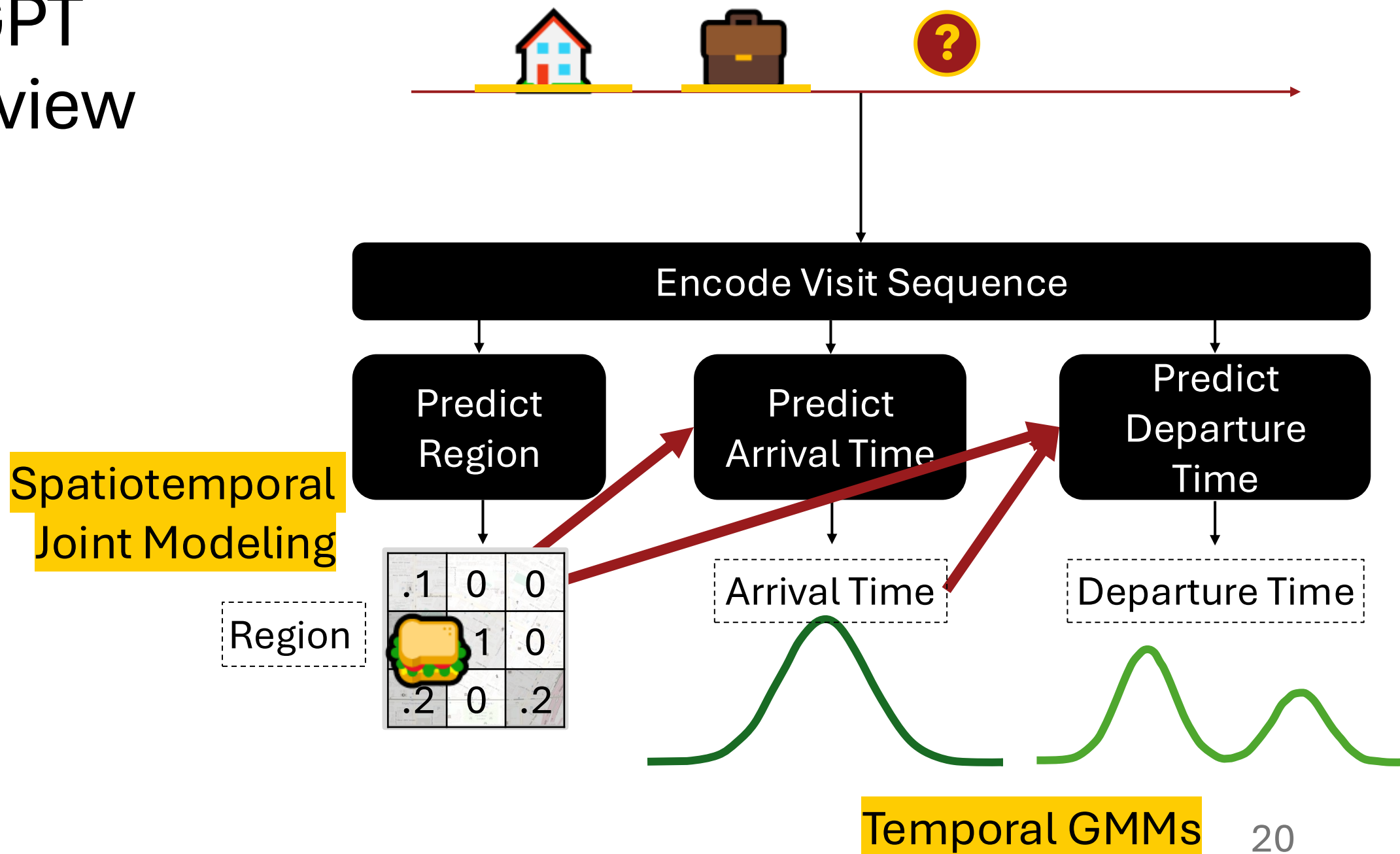
.1	0	0
.4	.1	0
.2	0	.2

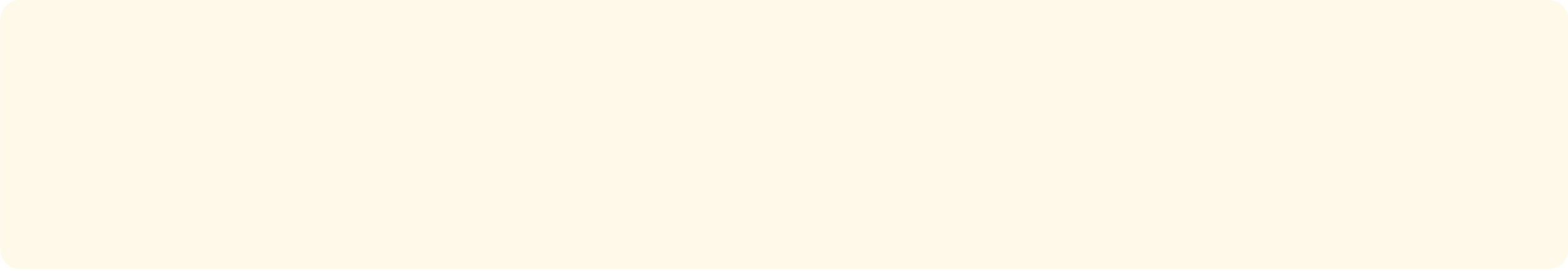


Loss Function

$$\mathcal{L} = - \sum \log \hat{P}(r | X') + \log \hat{P}(t^a | X', r) + \log \hat{P}(t^d | X', r, t^a)$$

TrajGPT Overview







Encode Visit Sequence

Space2Vec

Location Encoding

Time2Vec

Arrival Encoding

Departure Encoding

Region Embedder

Region Embedding



Encode Visit Sequence

Space2Vec

Location Encoding

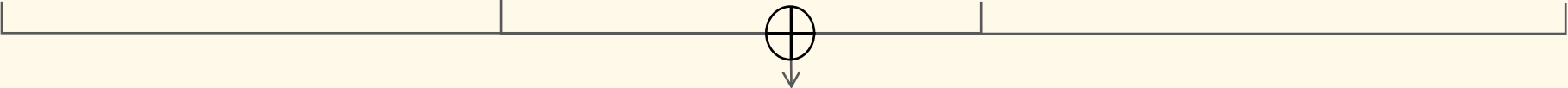
Time2Vec

Arrival Encoding

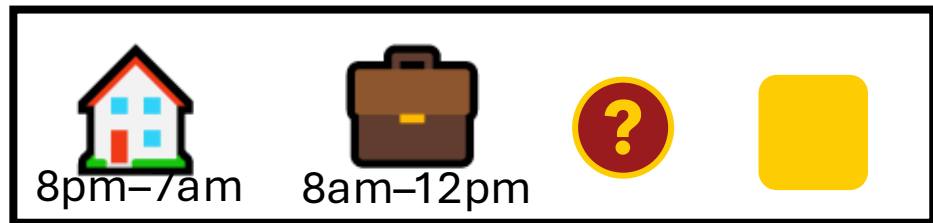
Departure Encoding

Region Embedder

Region Embedding



Positional & Transformer Encoders



Encode Visit Sequence

Space2Vec

Location Encoding

Time2Vec

Arrival Encoding

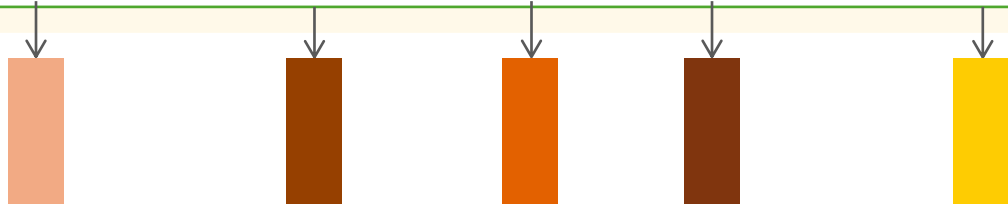
Departure Encoding

Region Embedder

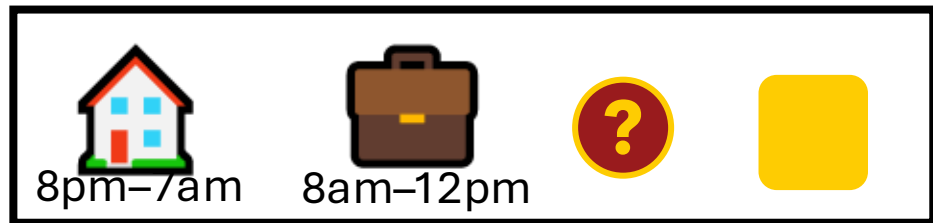
Region Embedding



Positional & Transformer Encoders



$$H := \text{TransformerEncoder}(\text{PositionalEncoder}(X'))$$



Encode Visit Sequence

Space2Vec

Location Encoding

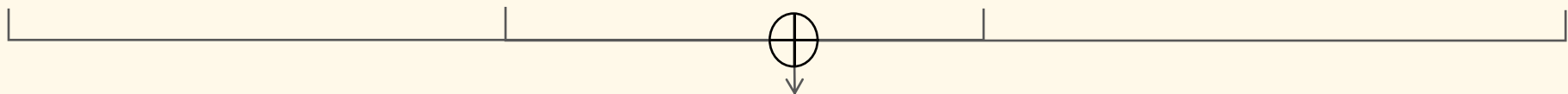
Time2Vec

Arrival Encoding

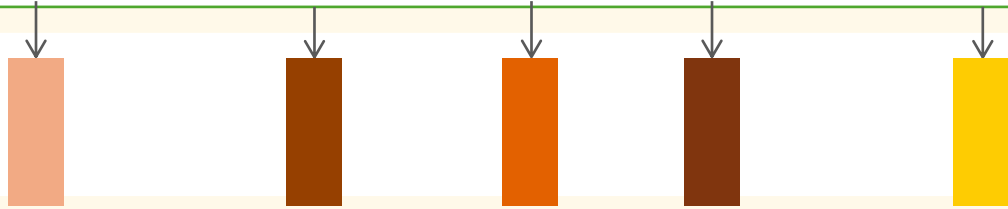
Departure Encoding

Region Embedder

Region Embedding



Positional & Transformer Encoders



$$H := \text{TransformerEncoder}(\text{PositionalEncoder}(X'))$$

Transformer Encoder

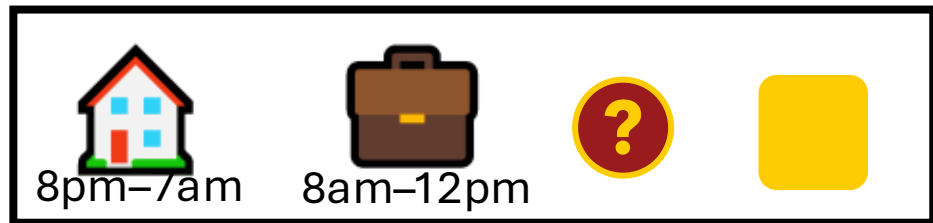
Linear

Softmax

.1	0	0
.4	.1	0
.2	0	.2

Predict Region

$$\hat{P}(r_i | X') := \text{Softmax}(\text{Linear}(\text{TransformerEncoder}(H)))$$



Encode Visit Sequence

Space2Vec

Location Encoding

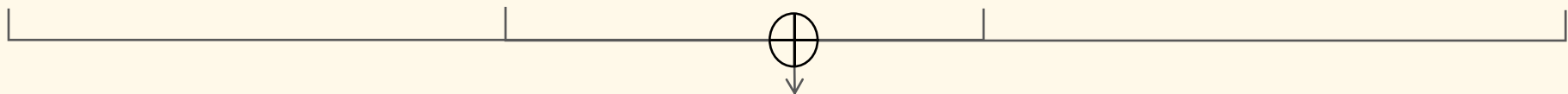
Time2Vec

Arrival Encoding

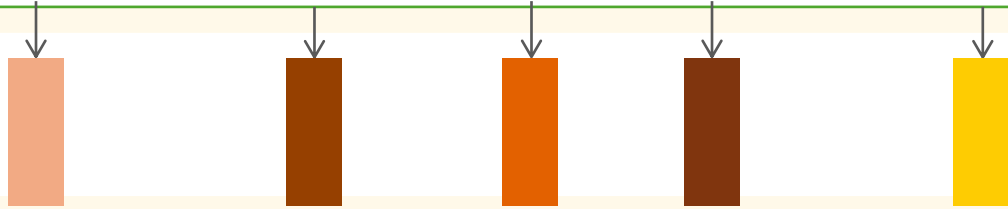
Departure Encoding

Region Embedder

Region Embedding



Positional & Transformer Encoders



$$H := \text{TransformerEncoder}(\text{PositionalEncoder}(X'))$$

Transformer Encoder

Linear

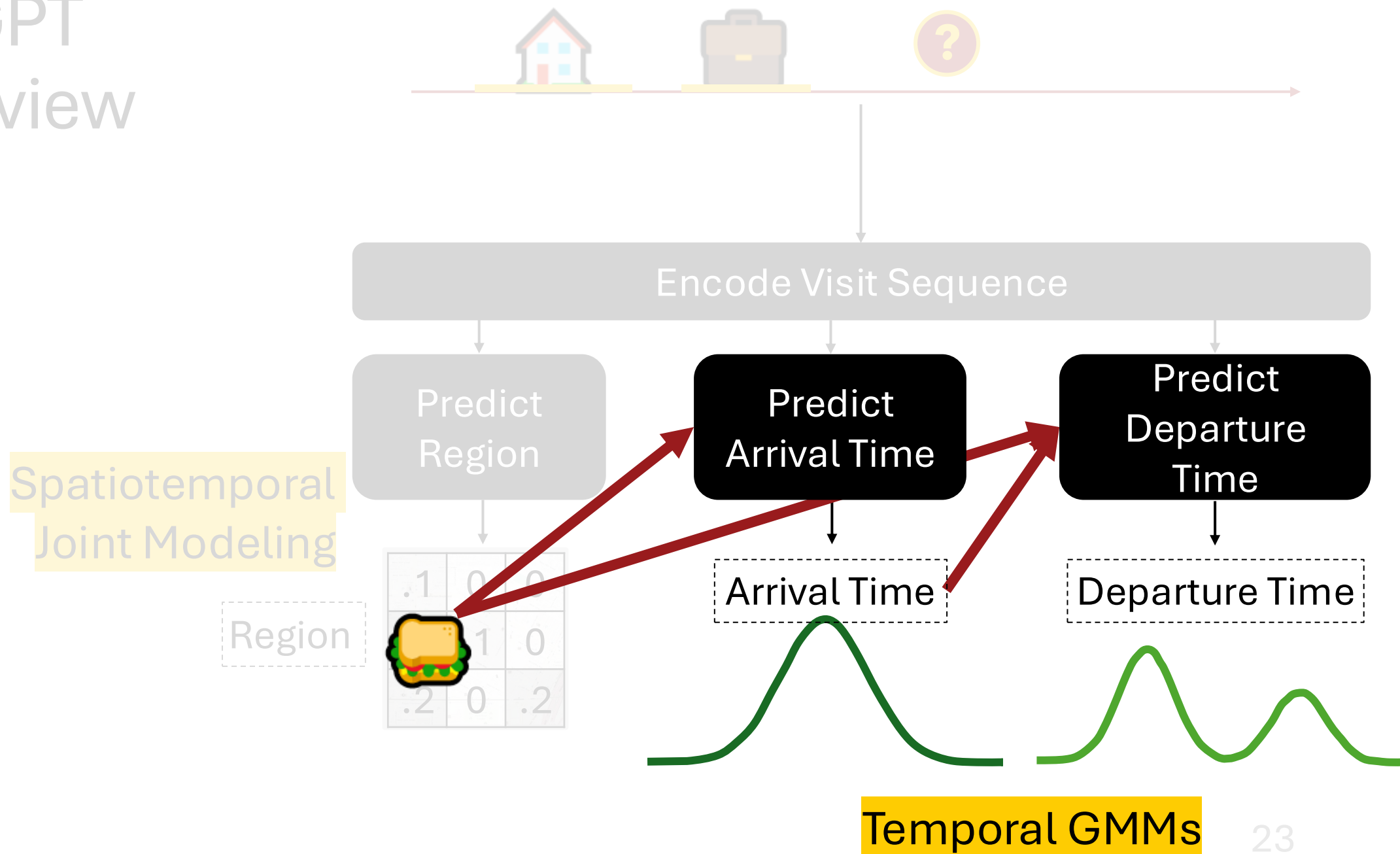
Softmax

.1	0	0
	1	0
.2	0	.2

Predict Region

$$\hat{P}(r_i | X') := \text{Softmax}(\text{Linear}(\text{TransformerEncoder}(H)))$$

TrajGPT Overview



Predict Arrival Time



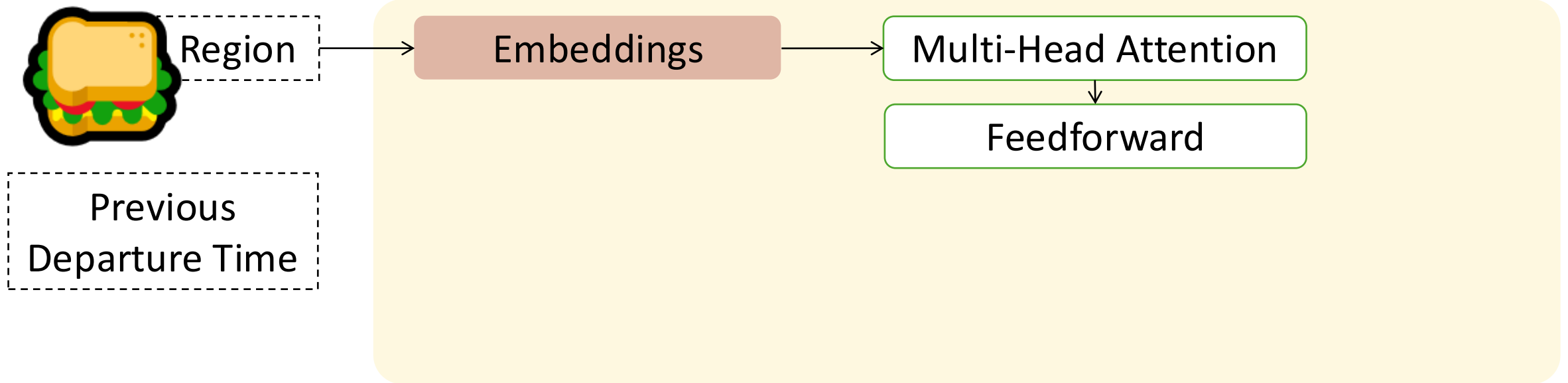
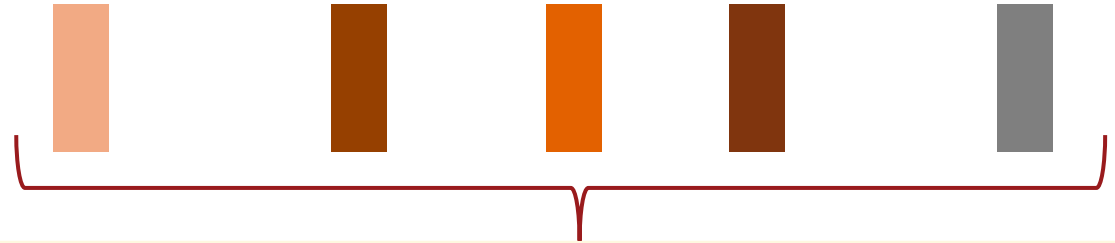
Region

Previous
Departure Time

Predict Arrival Time

$$\text{MHA}(e_i, H, H) := \text{Concat}(\text{head}_1, \dots, \text{head}_M)W^O$$

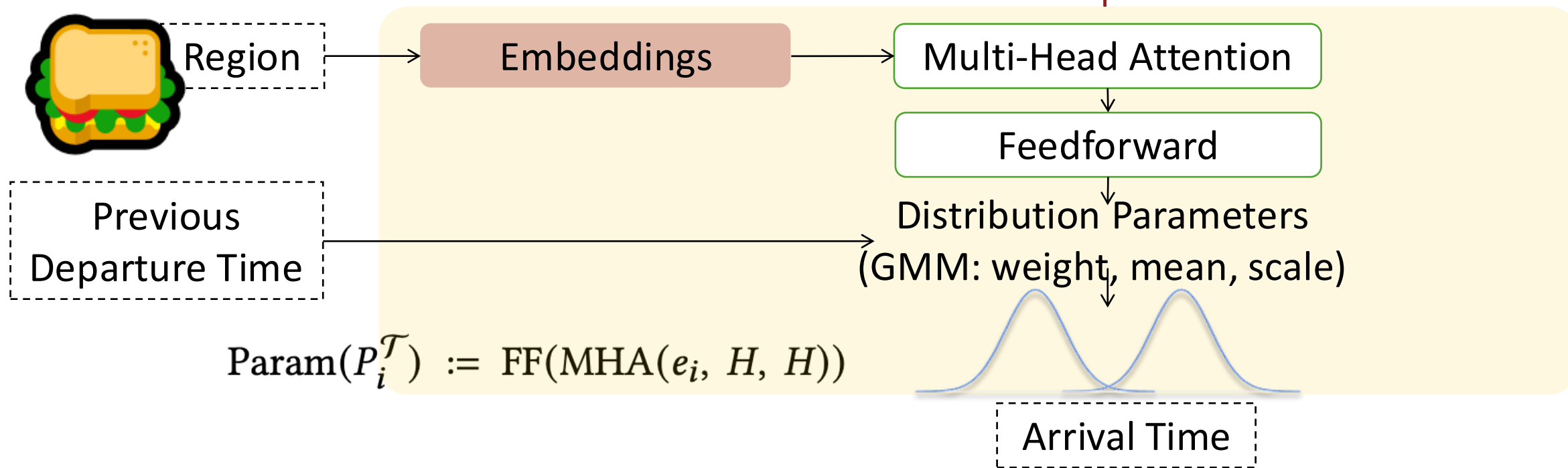
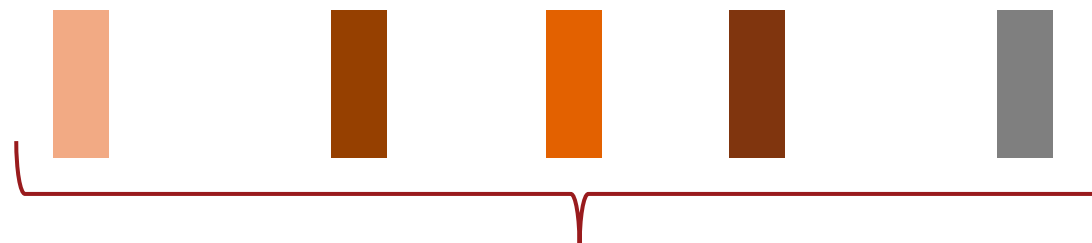
where $\text{head}_j := \text{Attention}(e_i W_j^Q, H W_j^K, H W_j^V)$



Predict Arrival Time

$$\text{MHA}(e_i, H, H) := \text{Concat}(\text{head}_1, \dots, \text{head}_M)W^O$$

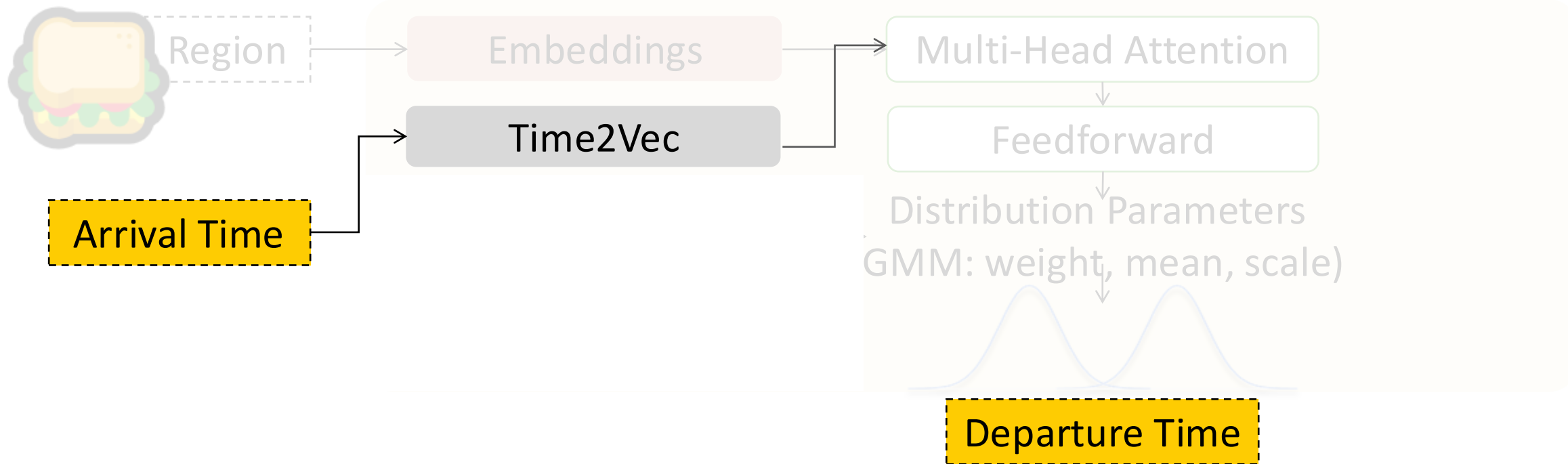
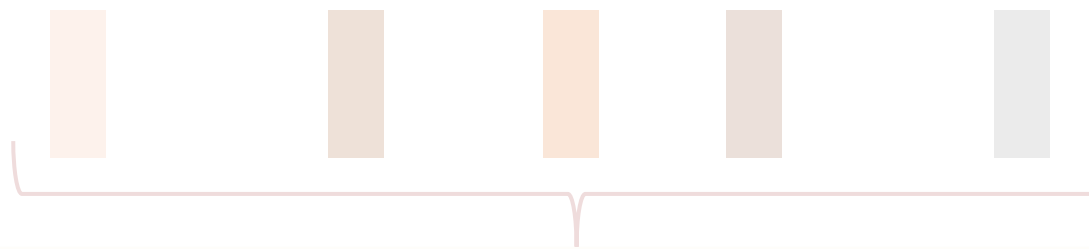
where $\text{head}_j := \text{Attention}(e_i W_j^Q, H W_j^K, H W_j^V)$



Predict ~~Arrival~~ Time Departure

$$\text{MHA}(e_i, H, H) := \text{Concat}(\text{head}_1, \dots, \text{head}_M) W^O$$

where $\text{head}_j := \text{Attention}(e_i W_j^Q, H W_j^K, H W_j^V)$





TrajGPT

Region | **Arrival Time** | **Departure Time**

.1	0	0
.4	.1	0
.2	0	.2





TrajGPT

Region Arrival Time Departure Time

Top-k
accuracy
("Acc@k")

.1	0	0
.4	.1	0
.2	0	.2





TrajGPT



Top-k
accuracy
("Acc@k")

.1	0	0
.4	.1	0
.2	0	.2



Top-t
accuracy
("P_{±t}")

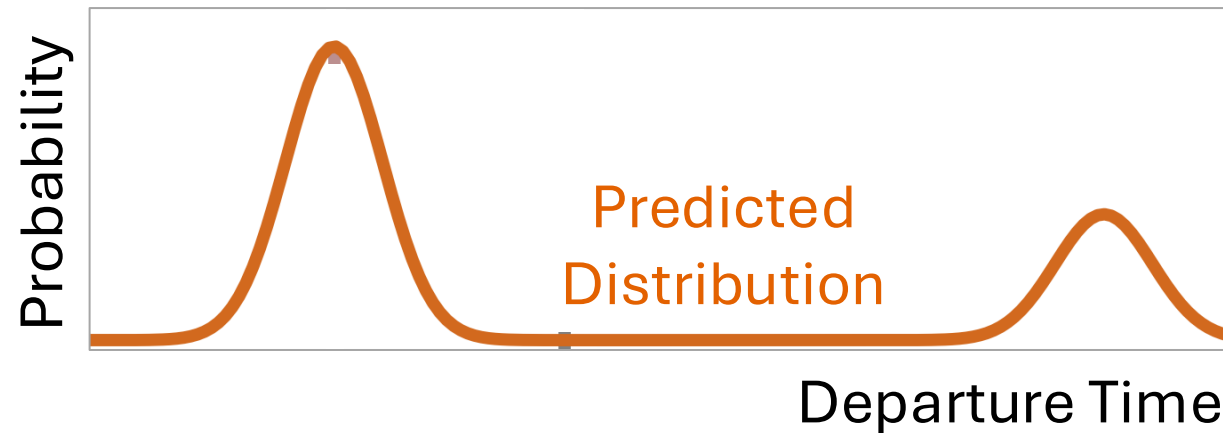
New

Top-t Accuracy (“ $P_{\pm t}$ ”) New

To compare points with distributions (unit of t : minute)

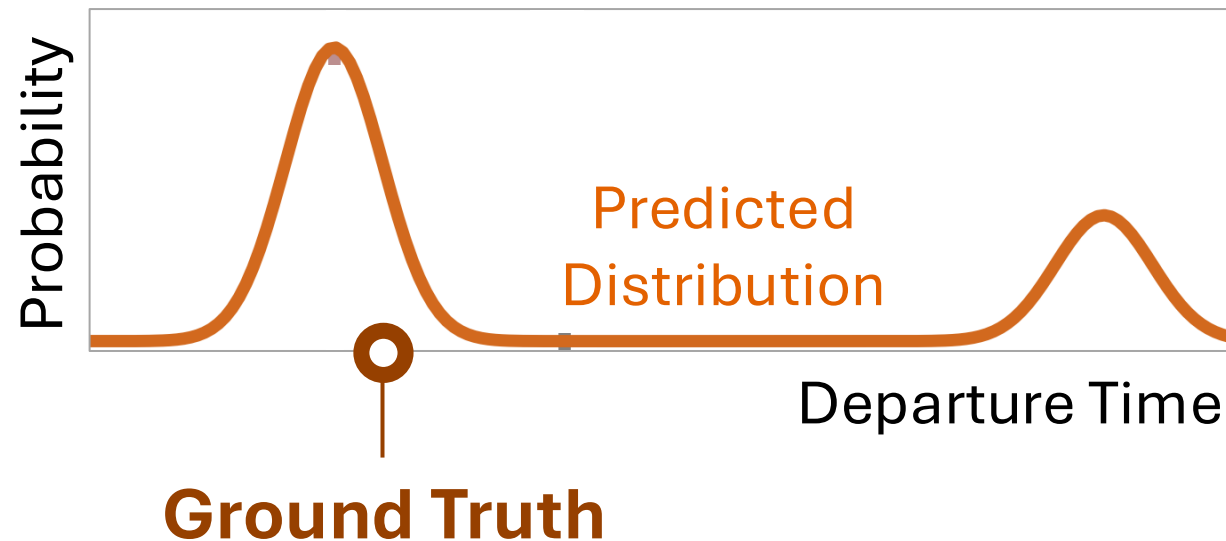
Top-t Accuracy (“ $P_{\pm t}$ ”) New

To compare points with distributions (unit of t : minute)



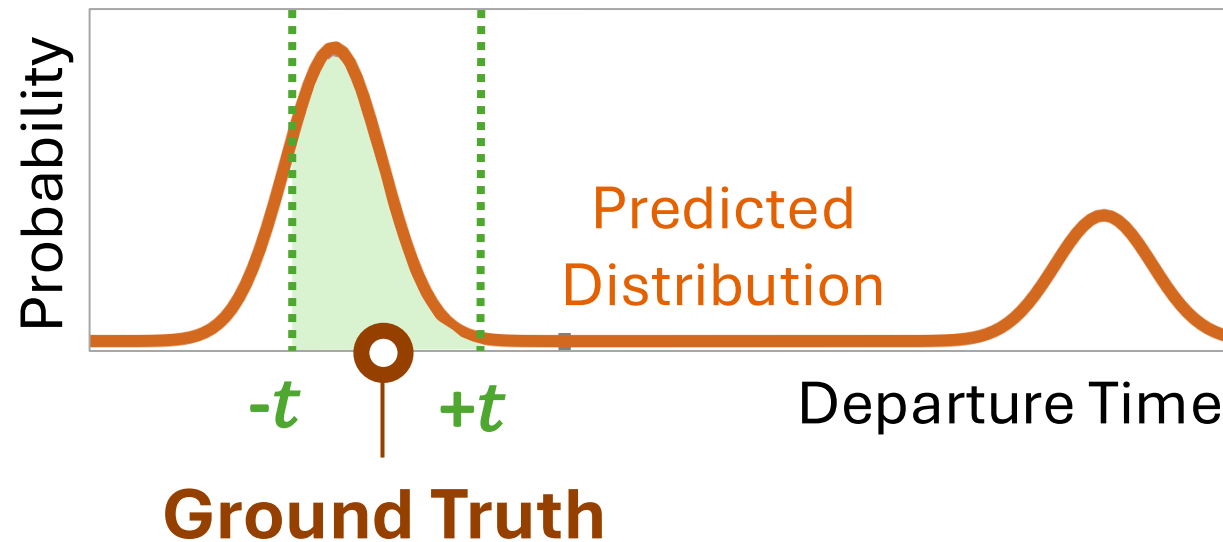
Top-t Accuracy (“ $P_{\pm t}$ ”) New

To compare points with distributions (unit of t : minute)



Top-t Accuracy (“ $P_{\pm t}$ ”) New

To compare points with distributions (unit of t : minute)



Experimental Setup

- Two human mobility datasets

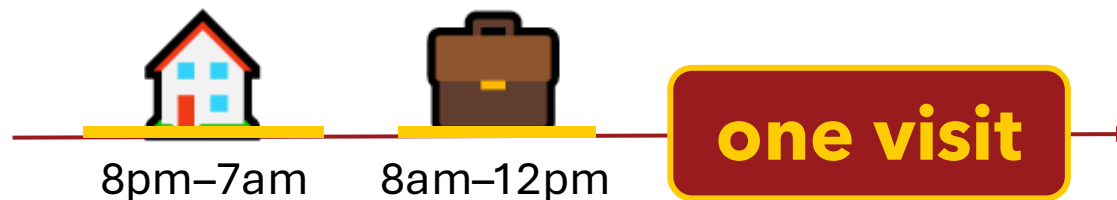
Dataset	#agent	#region	#visit	#trajectory*	Avg. #visit/agent	Avg. #visit/region
GeoLife	102	1,369	20,278	11,724	198.80	14.81
MobilitySim	2,000	3,481	191,963	6,178	95.98	55.15

- Region := Uber H3 Index cells

Next Visit Prediction

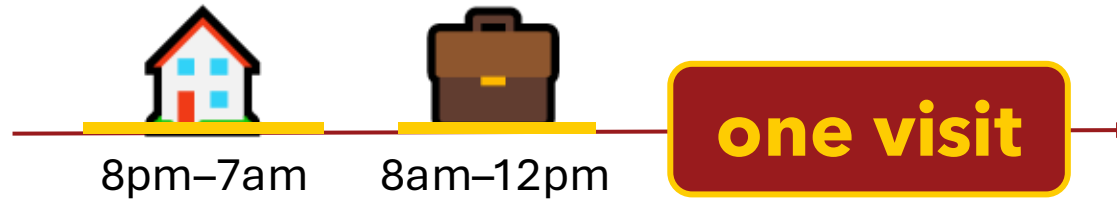


Next Visit Prediction



Dataset	Method	Region				Arrival Time			Departure Time		
		Acc@1	Acc@5	Acc@10	Acc@20	P _{±5}	P _{±10}	P _{±20}	P _{±5}	P _{±10}	P _{±20}
GeoLife	STAR-HiT	17.92	40.78	48.98	56.40	-	-	-	-	-	-
	GETNext	38.74	66.38	72.78	77.13	2.39	4.27	8.62	-	-	-
	TrajGPT	35.06	62.08	70.77	77.63	64.28	71.70	80.02	35.01	48.40	60.82
MobilitySim	STAR-HiT	42.79	62.87	70.25	75.74	-	-	-	-	-	-
	GETNext	51.46	80.91	91.59	94.50	1.29	2.91	5.50	-	-	-
	TrajGPT	54.53	80.26	92.88	94.66	89.33	94.01	98.07	52.57	62.05	71.20

Next Visit Prediction



Predicts complete visits

Dataset	Method	Region				Arrival Time			Departure Time		
		Acc@1	Acc@5	Acc@10	Acc@20	P _{±5}	P _{±10}	P _{±20}	P _{±5}	P _{±10}	P _{±20}
GeoLife	STAR-HiT	17.92	40.78	48.98	56.40	-	-	-	-	-	-
	GETNext	38.74	66.38	72.78	77.13	2.39	4.27	8.62	-	-	-
	TrajGPT	35.06	62.08	70.77	77.63	64.28	71.70	80.02	35.01	48.40	60.82
MobilitySim	STAR-HiT	42.79	62.87	70.25	75.74	-	-	-	-	-	-
	GETNext	51.46	80.91	91.59	94.50	1.29	2.91	5.50	-	-	-
	TrajGPT	54.53	80.26	92.88	94.66	89.33	94.01	98.07	52.57	62.05	71.20

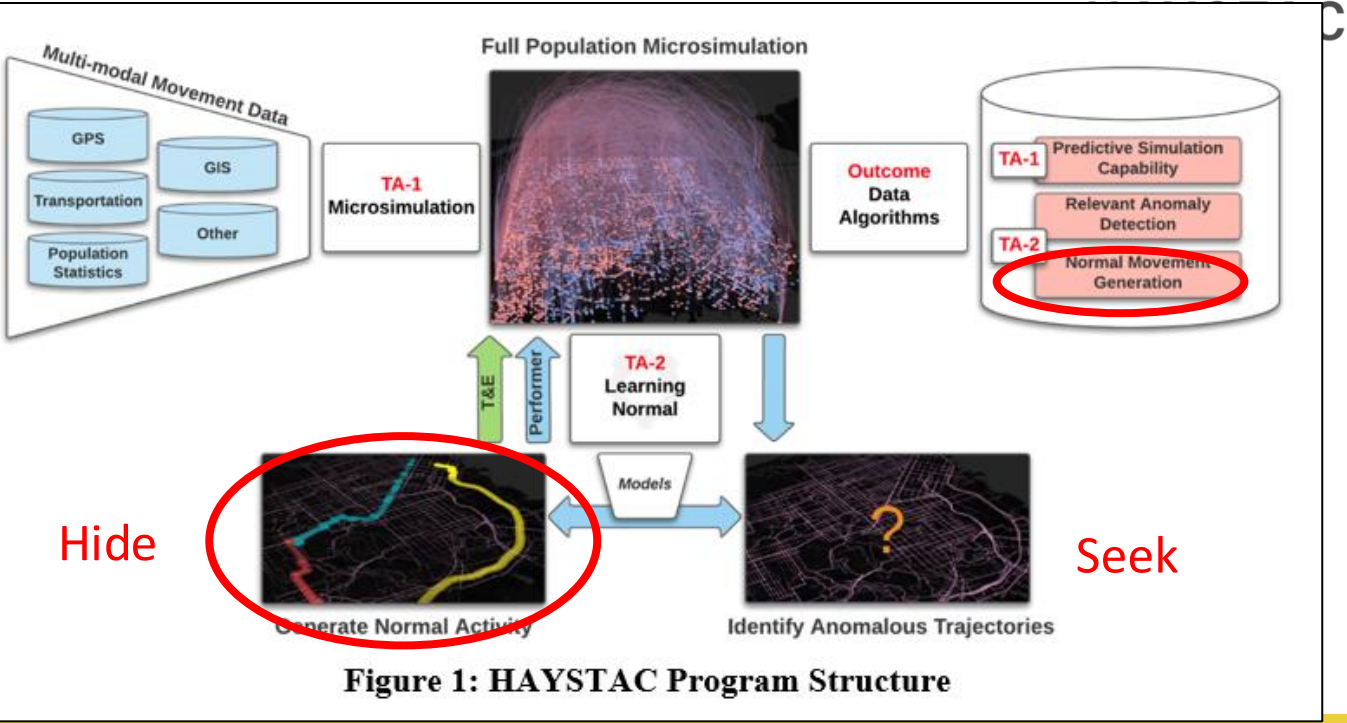
OUTLINE



- **Motivation**
- **Background**
- **TrajGPT: Transformer for Next-Visit Prediction**
- **Downstream Applications**
 - Filling missing visits (gap imputation)
 - POI attribution
 - Mobility outlier detection
- **Towards a Mobility-Foundation Model (Open Problems)**

HAYSTAC

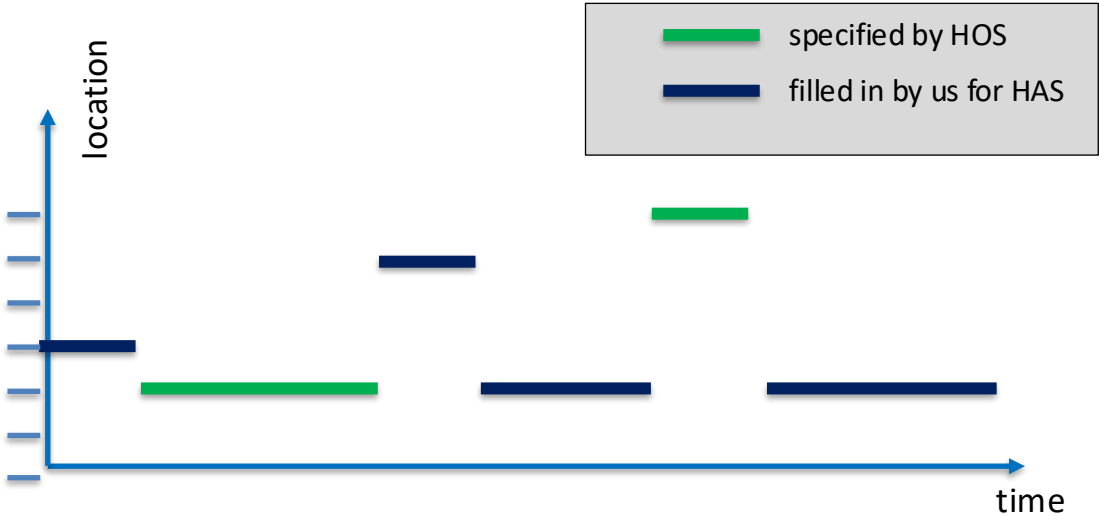
HIDDEN ACTIVITY SIGNAL AND TRAJECTORY
 ANOMALY CHARACTERIZATION



Motivation



- Create normal movement trajectories from partial specifications



Data-Driven Approach	Related Work	Location Logs	Visits
Statistical	TimeGeo [PNAS 16]	✓	
	NEXTGAIL [SIGSPATIAL 21]	✓	
Adversarial/Variational	ActSTD [KDD 21]	✓	
	FVAE [SIGSPATIAL 22]	✓	
Transformer	Geo-CETRA [ICDM 24]	✓	
	DeepJMT [WSDM 20]		✓
	MobTCast [NeurIPS 21]		✓
	GETNext [SIGIR 22]		✓
	STAR-HiT [TOIS 23]		✓
Large Language Model	LLMmove [CAI 24]		✓
	LLM4POI [SIGIR 24]		✓

Data-Driven Approach	Related Work	Location Logs	Visits
Statistical	TimeGeo [PNAS 16]	✓	
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	FVAE [SIGSPATIAL 22]	✓	
Transformer	Geo-CETRA [ICDM 24]	✓	Limitation 1: Cannot generate visit sequences
	DeepJMT [WSDM 20]		
	MobTCast [NeurIPS 21]		
	GETNext [SIGIR 22]		
	STAR-HiT [TOIS 23]		
Large Language Model	LLMmove [CAI 24]		✓
	LLM4POI [SIGIR 24]		✓

Data-Driven Approach	Related Work	Location Logs	Visits
Statistical	TimeGeo [PNAS 16]	✓	
	NEXTGAIL [SIGSPATIAL 21]	✓	
Adversarial/Variational	ActSTD [KDD 21]	✓	
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Transformer	Geo-CETRA [ICDM 24]	✓	Limitation 1: Cannot generate visit sequences
	DeepJMT [WSDM 20]		
	MobTCast [NeurIPS 21]		
	GETNext [SIGIR 22]		✓
	STAR-HiT [TOIS 23]		✓
Large Language Model	LLMmove [CAI 24]		✓
	LLM4POI [SIGIR 24]		✓

Controlled Generation

Masked Modeling (e.g. BERT)



Causal Modeling (e.g. GPT)



Controlled Generation

Masked Modeling (e.g. BERT)



Causal Modeling (e.g. GPT)

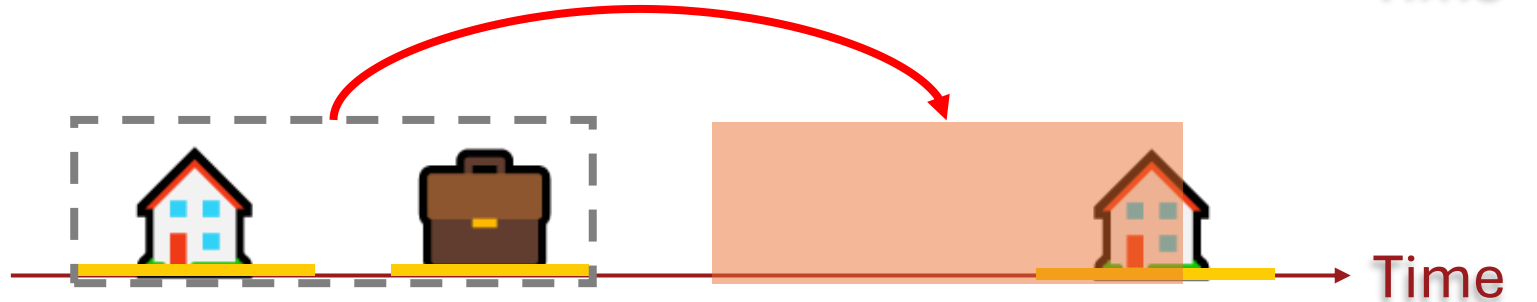


Controlled Generation

Masked Modeling (e.g. BERT)



Causal Modeling (e.g. GPT)

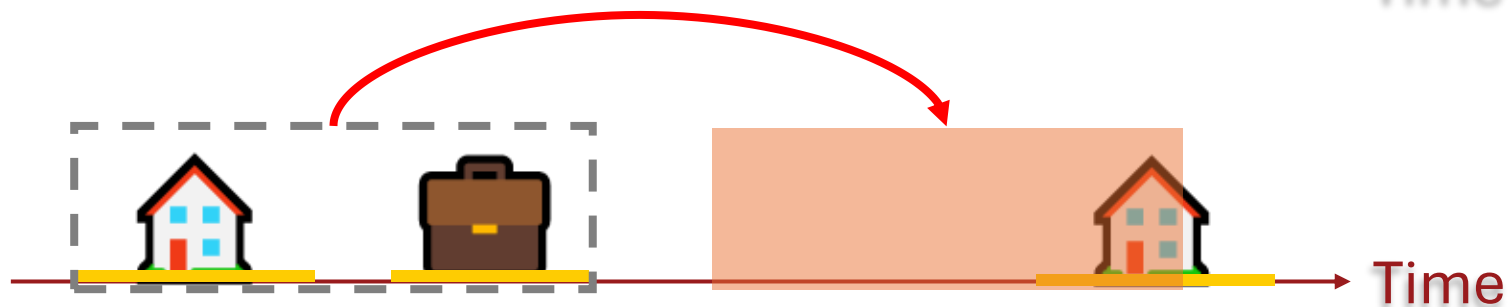


Controlled Generation

Masked Modeling (e.g. BERT)



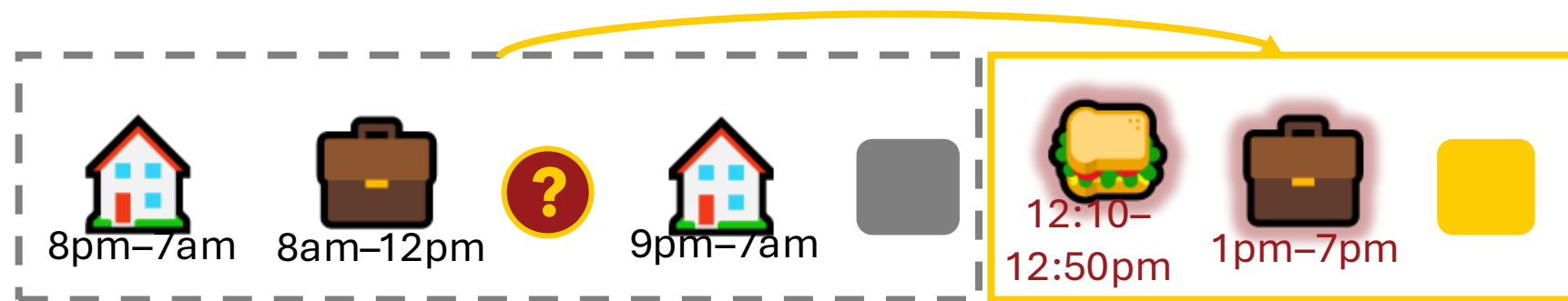
Causal Modeling (e.g. GPT)



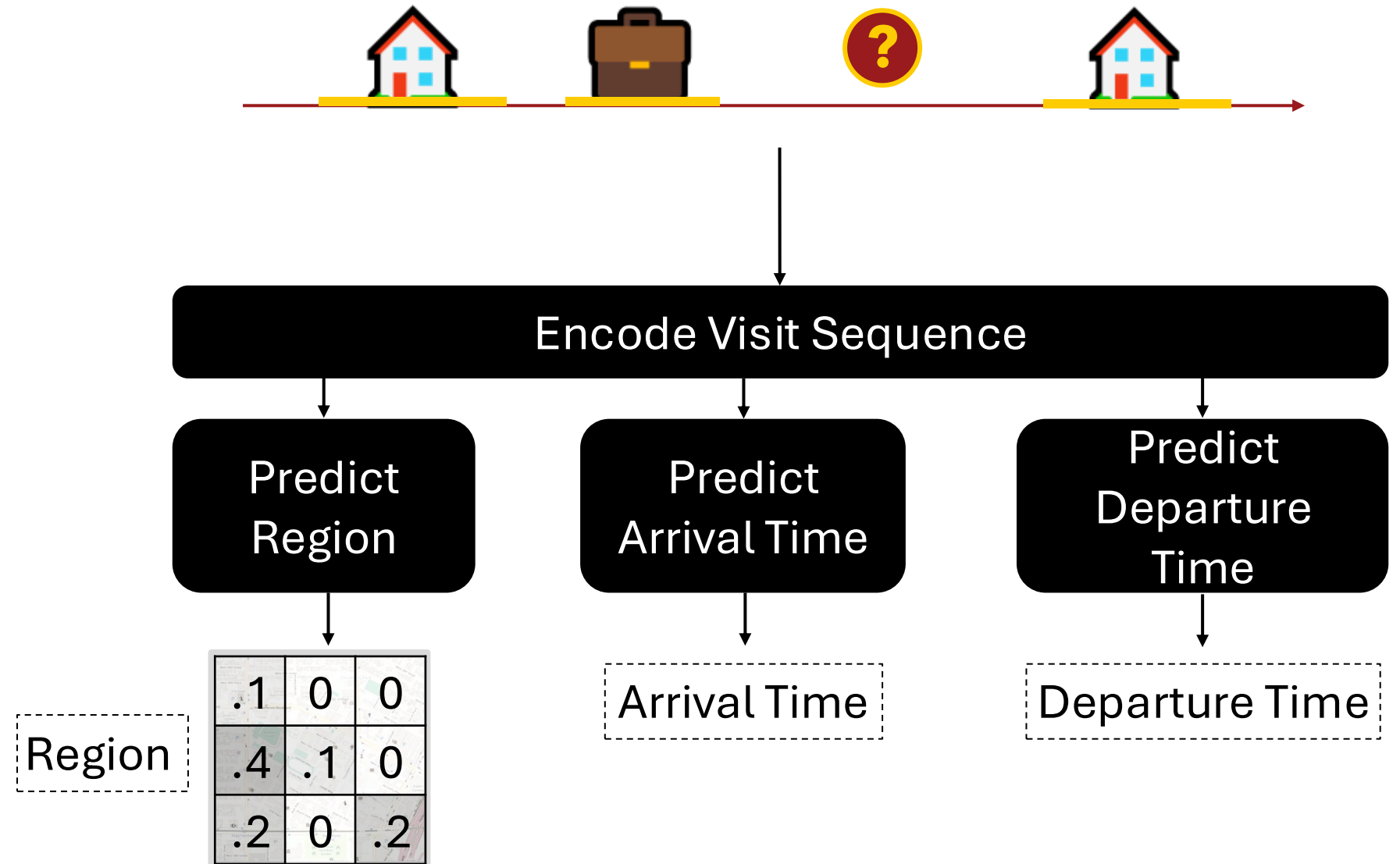
Approach:

Text Infilling

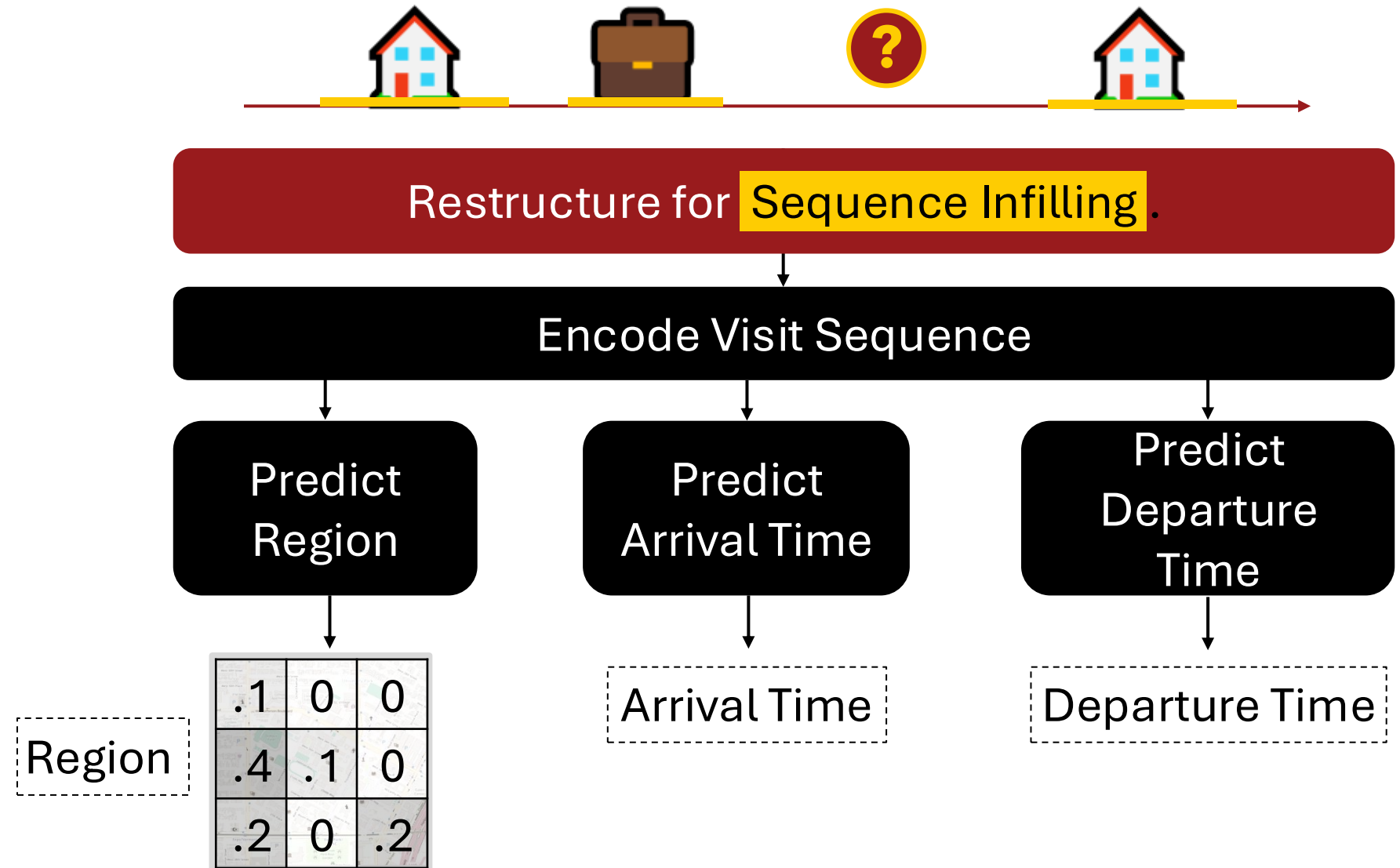
(Donahue et al., 2020)



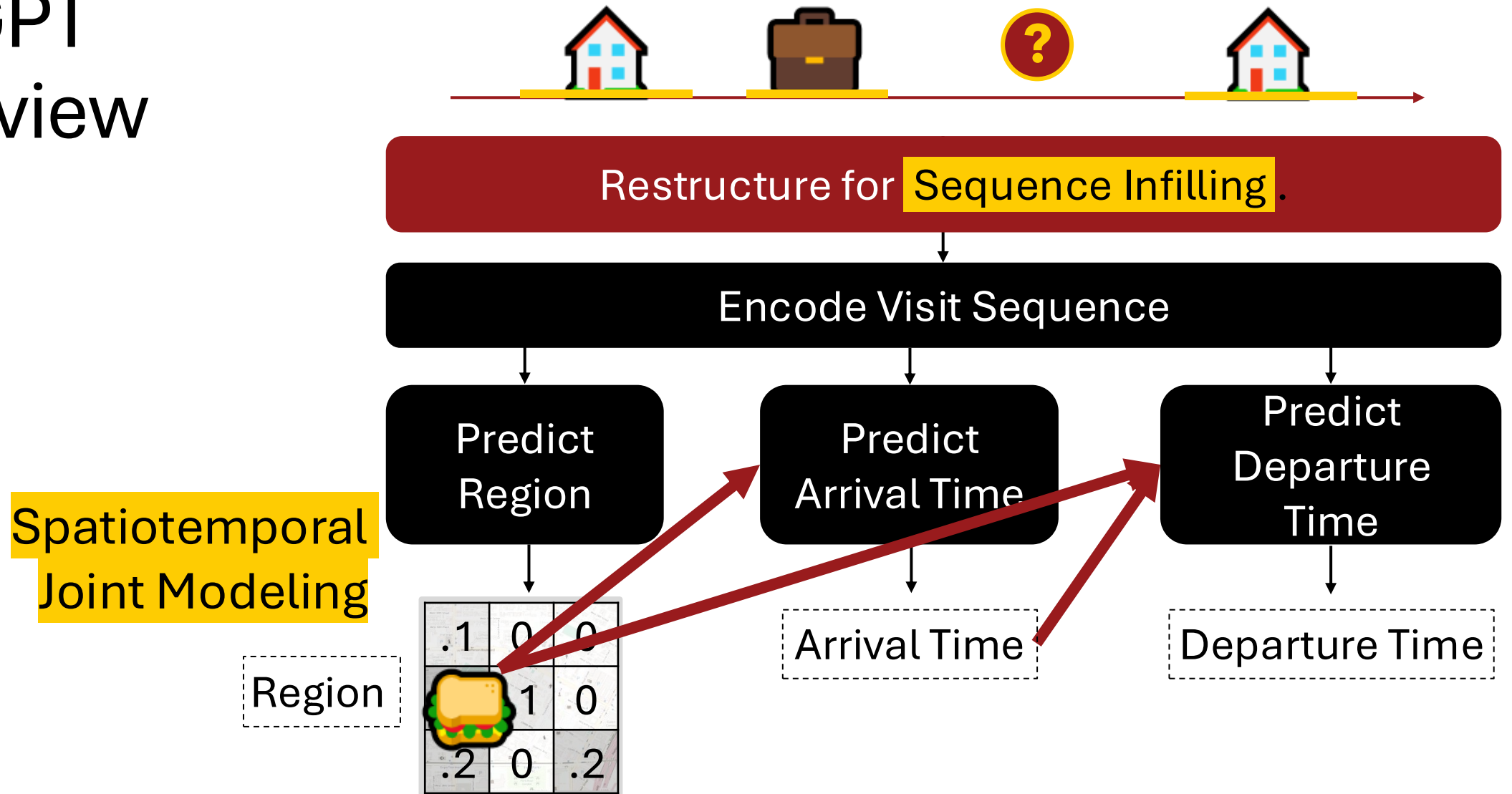
TrajGPT Overview



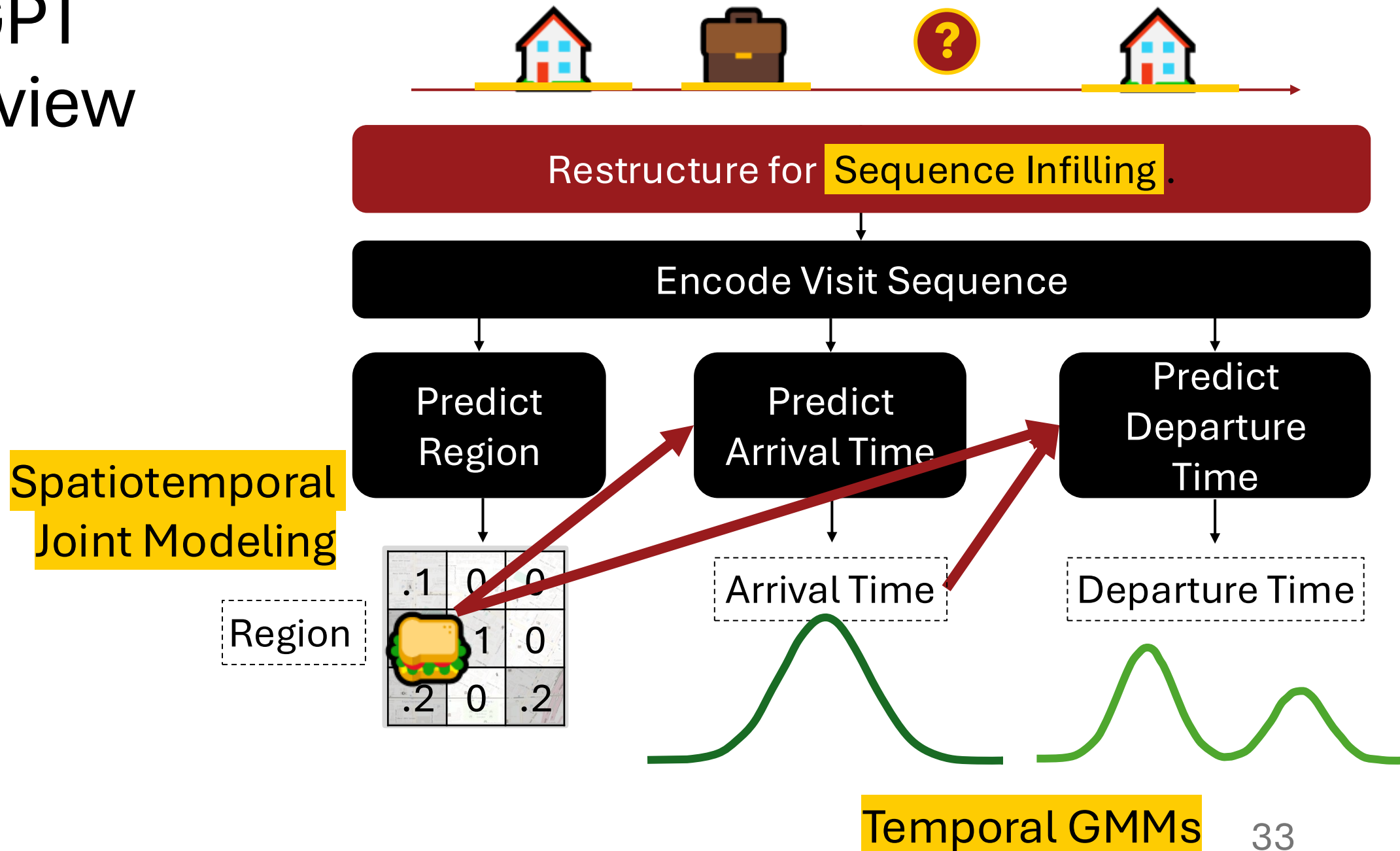
TrajGPT Overview



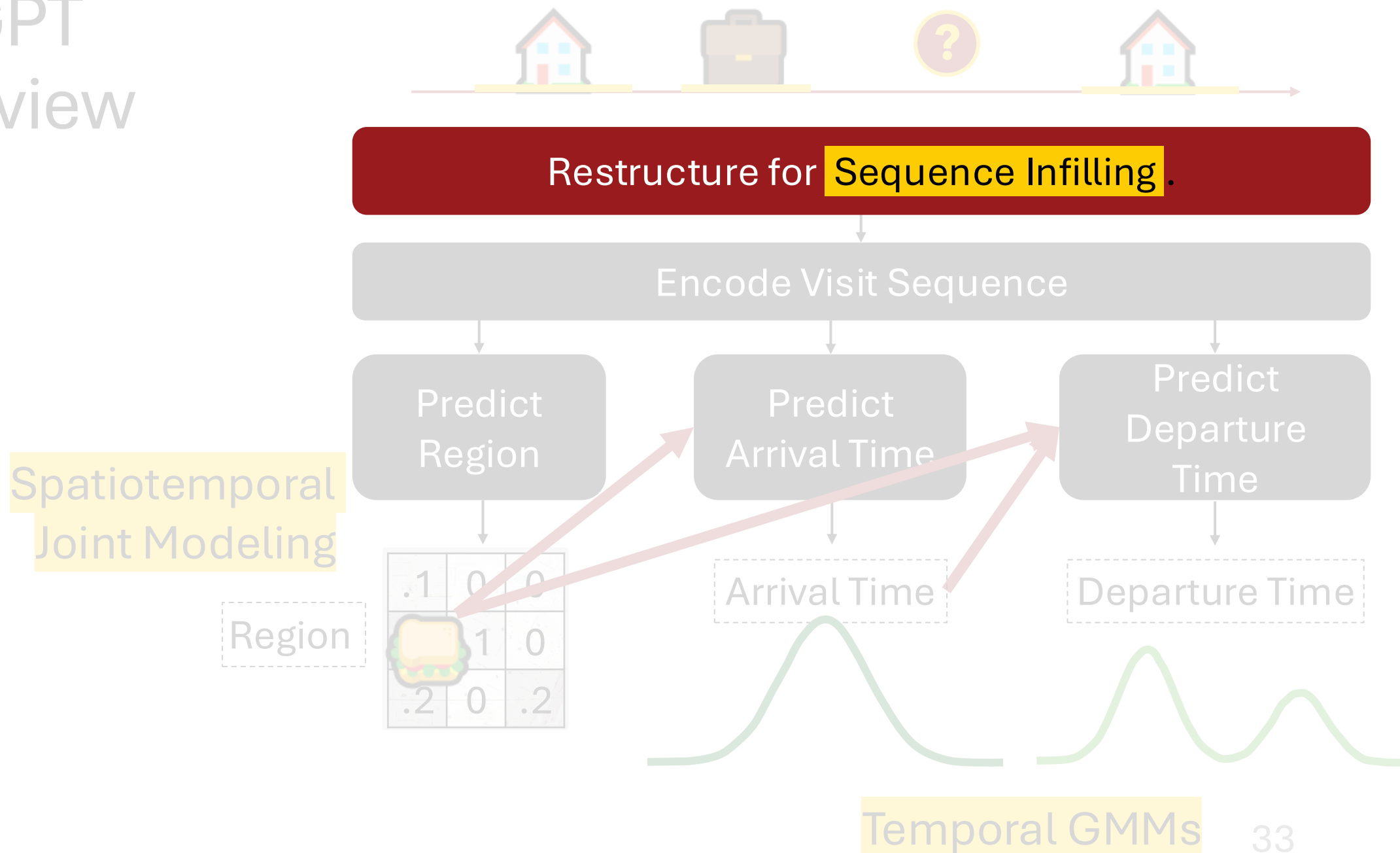
TrajGPT Overview



TrajGPT Overview



TrajGPT Overview



Infilling a *Visit* Sequence

Input



Output



Infilling a *Visit* Sequence

Input



Output



Restructured
Sequence



Infilling a *Visit* Sequence

Input



Output



Restructured
Sequence



Infilling a *Visit* Sequence

Input



Output



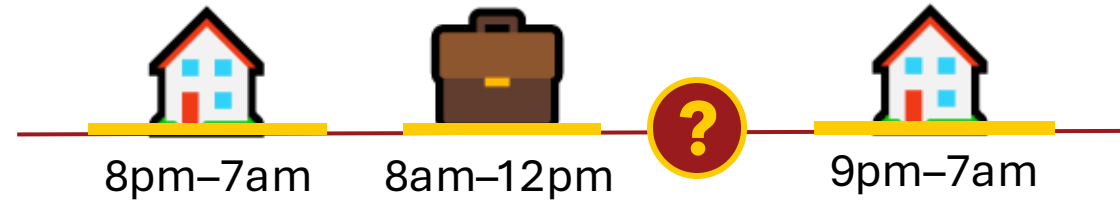
Restructured
Sequence



Controlled Generation [GeoLife Dataset]

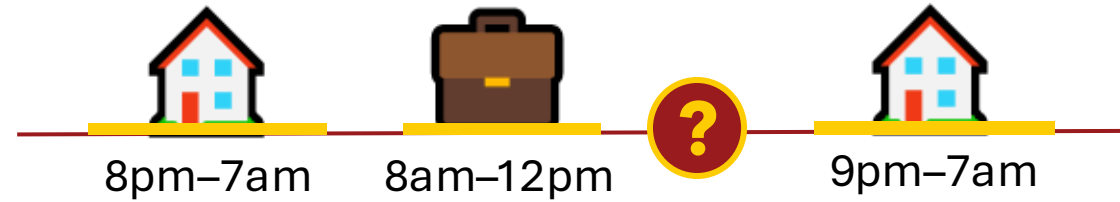


Controlled Generation [GeoLife Dataset]



Method	Region				Arrival Time			Departure Time		
	Acc@1	Acc@5	Acc@10	Acc@20	P _{±5}	P _{±10}	P _{±20}	P _{±5}	P _{±10}	P _{±20}
GETNext*	7.64	36.87	46.83	51.87	0.16	0.38	0.73	1.41	2.87	7.91
TrajGPT	19.87	40.57	44.78	52.21	65.72	75.53	85.31	36.08	50.81	64.23

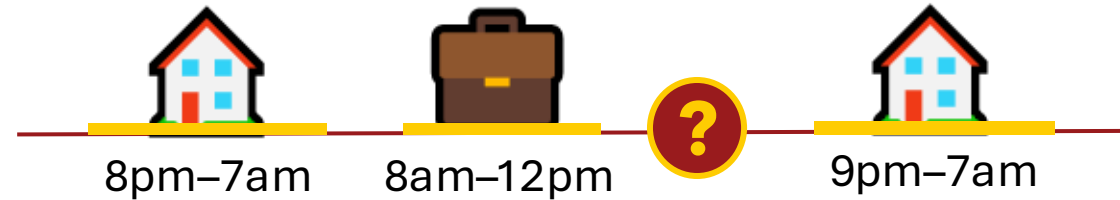
Controlled Generation [GeoLife Dataset]



- Transformer-based next POI recommender
- Modified for *controlled* generation

Method	Region				Arrival Time			Departure Time		
	Acc@1	Acc@5	Acc@10	Acc@20	P _{±5}	P _{±10}	P _{±20}	P _{±5}	P _{±10}	P _{±20}
GETNext*	7.64	36.87	46.83	51.87	0.16	0.38	0.73	1.41	2.87	7.91
TrajGPT	19.87	40.57	44.78	52.21	65.72	75.53	85.31	36.08	50.81	64.23

Controlled Generation [GeoLife Dataset]



- Transformer-based next POI recommender
- Modified for *controlled* generation

Method	Region				Arrival Time			Departure Time		
	Acc@1	Acc@5	Acc@10	Acc@20	P _{±5}	P _{±10}	P _{±20}	P _{±5}	P _{±10}	P _{±20}
GETNext*	7.64	36.87	46.83	51.87	0.16	0.38	0.73	1.41	2.87	7.91
TrajGPT	19.87	40.57	44.78	52.21	65.72	75.53	85.31	36.08	50.81	64.23

Much higher temporal accuracy

OUTLINE

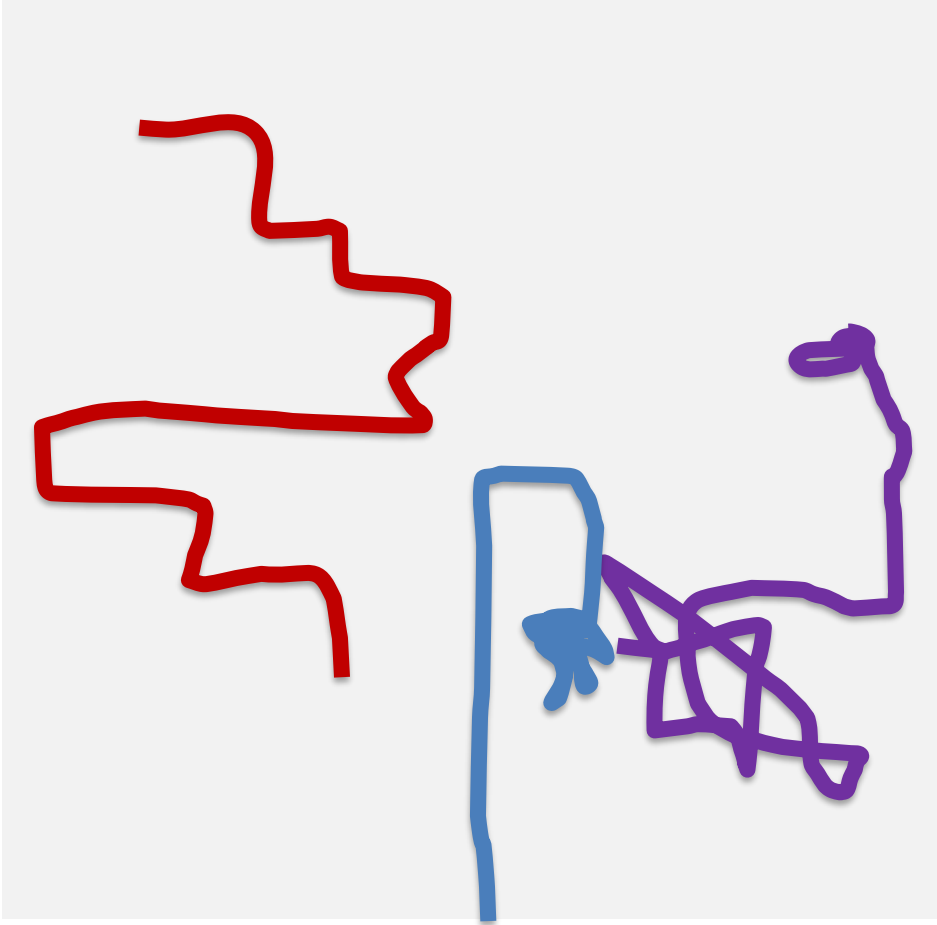


- **Motivation**
- **Background**
- **TrajGPT: Transformer for Next-Visit Prediction**
- **Downstream Applications**
 - Filling missing visits (gap imputation)
 - **POI attribution**
 - Mobility outlier detection
- **Towards a Mobility-Foundation Model**

Why is Point-of-Interest (POI) attribution important?



- Crucial for precise spatial understanding.



Why is Point-of-Interest (POI) attribution important?



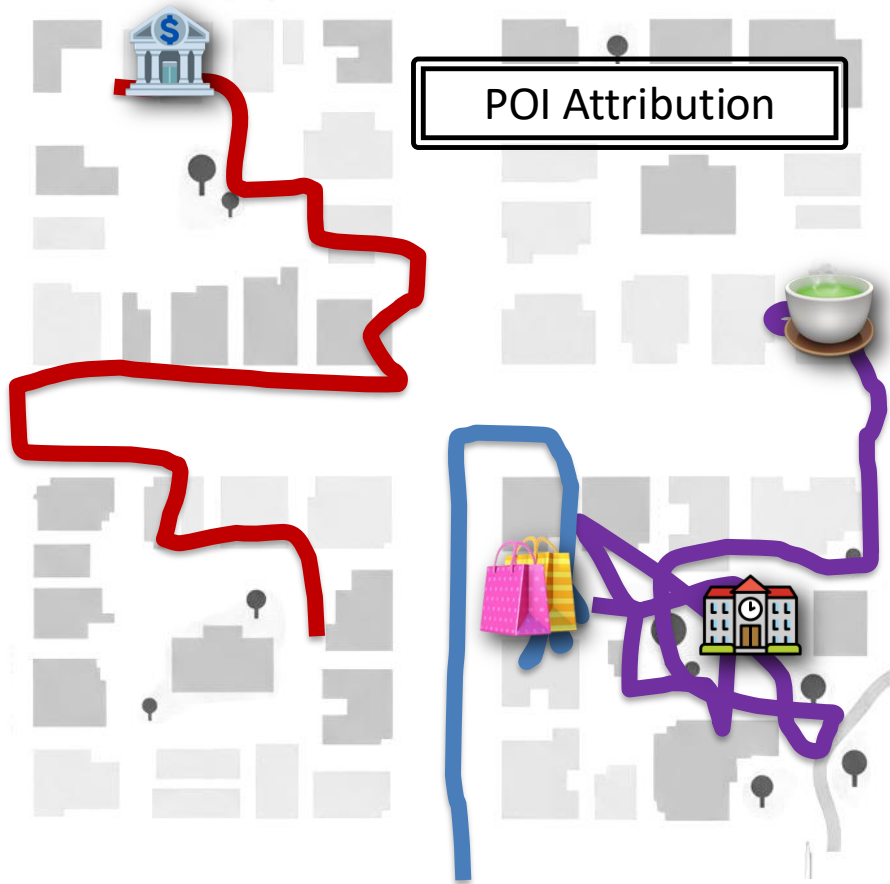
- Crucial for precise spatial understanding.



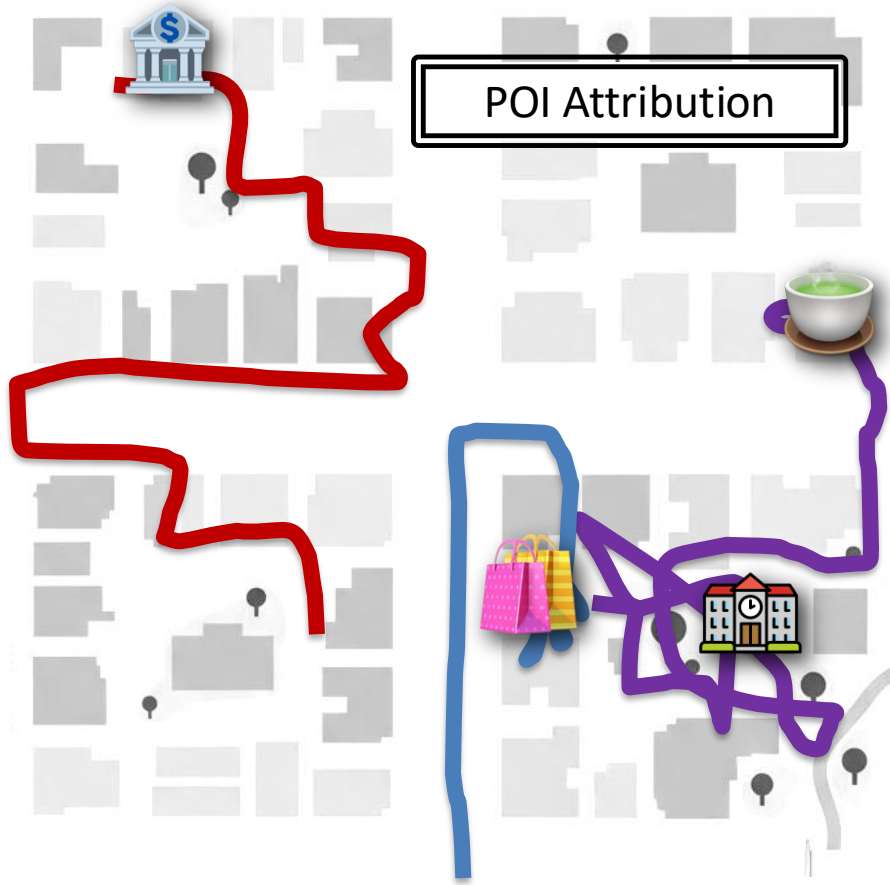
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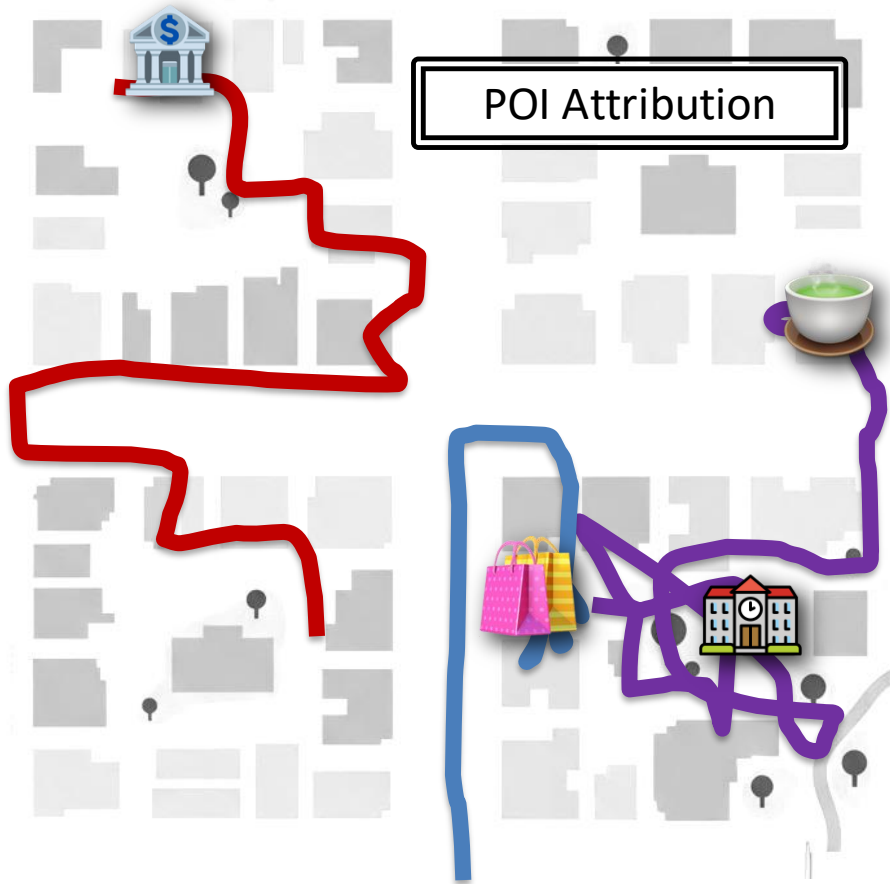


- Crucial for precise spatial understanding.
- High accuracy is required to prevent error propagation.

Downstream Tasks:

- Next location prediction
- Anomaly detection

Why is Point-of-Interest (POI) attribution important?



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Downstream Tasks:

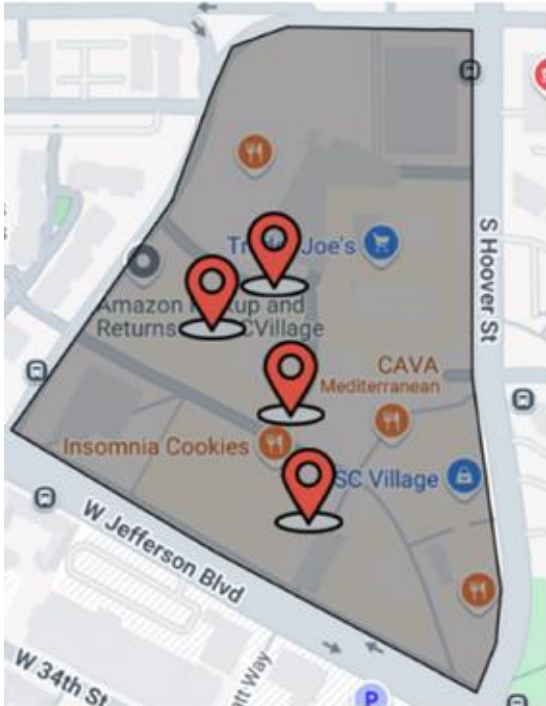
-  Next location prediction
-  Anomaly detection

Applications:

-  Urban planning
-  Public health interventions
-  Location-based marketing

Motivation: Why is POI attribution hard?

Low GPS Accuracy + High POI Density

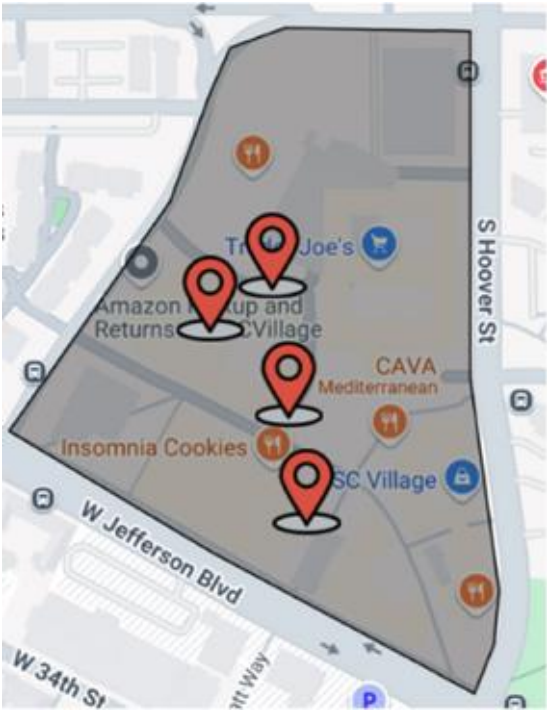


Method	How It Works	Avg. Accuracy Indoors / Ur- ban	Avg. Accuracy Out- doors / Rural	Coverage and Best Use Case
Assisted-GPS (A-GPS) (commonly used in smartphones)	Triangulates signals from multiple satellites via radio waves by measuring signal travel time. Uses cellular assistance to speed up location fixes.	~20 m [25]	Modern devices: 7–13 m [22, 28] Older devices (such as iPhone 3G 2008): 20–30 m [48]	Global. Best for outdoor use in open areas; signal degradation likely in urban canyons or indoors due to obstruction or multipath reflections.
Handheld GPS Devices (e.g., Garmin GPSMAP 60Cx)	Uses satellite triangulation with advanced antennas and typically supports dual-frequency GPS for improved precision.	3–5 m [28, 48]	1.6 m (horizontal), 3.1 m (vertical) [28, 48]	High accuracy. Global coverage with good performance in most terrains.
Wi-Fi Positioning	Estimates location by matching nearby Wi-Fi signals to a known database of access points.	10–20 m [5, 24, 42, 44, 48]	15–40 m in suburban or semi-urban environments [24, 48]	Best in urban or dense environments; limited performance in rural or uncalibrated areas.
Bluetooth	Estimates location by measuring direction and strength of received signal	2-5m	up to 700m for direction estimation w/ an antenna array [39]	Susceptible to interference and multipath effects, making it less reliable in complex indoor environments.
Cellular Network Positioning	Determines location using timing and signal strength from nearby cell towers.	~245 m (dense urban) [23, 48]	~626 m (rural) [23, 48]	Wide coverage. Low precision; best as a fallback method.
Hybrid (GPS, Wi-Fi, Cellular)	Combines GPS, Wi-Fi, and cellular data to enhance positioning robustness and adaptability.	~ 4.62m in ideal conditions (i.e., sufficient Wi-Fi and cell tower signals) [10]	Accuracy degrades with weaker Wi-Fi and cellular network signals, reverting to 7–13 m [22, 28] when relying solely on GPS	Most resilient across environments; balances accuracy and coverage.



Motivation: Why is POI attribution hard?

Low GPS Accuracy + High POI Density

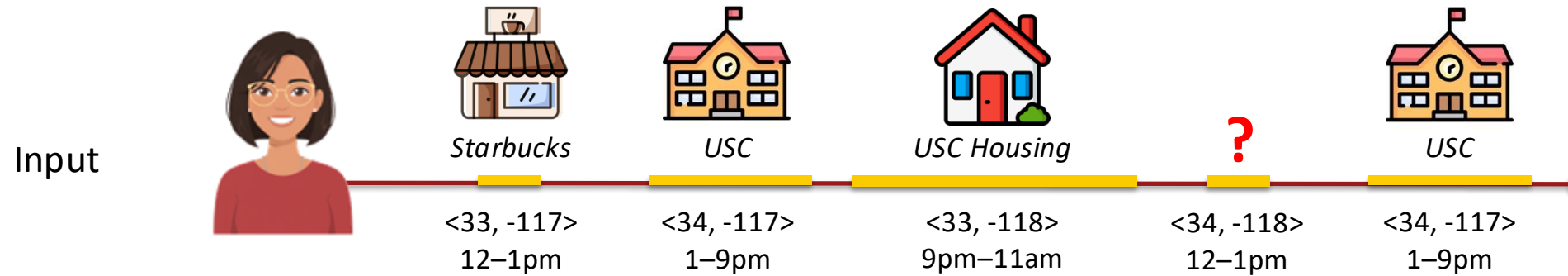


753 retail businesses / mile²
in Downtown Los Angeles

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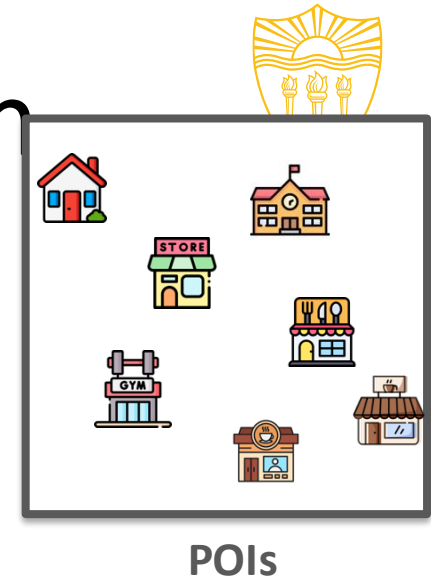
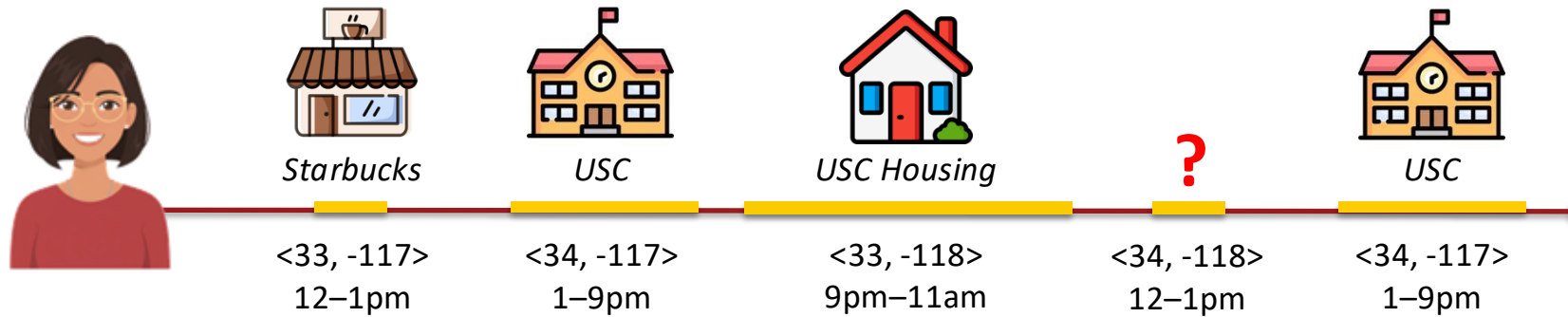


Problem Definition: POI Attribution



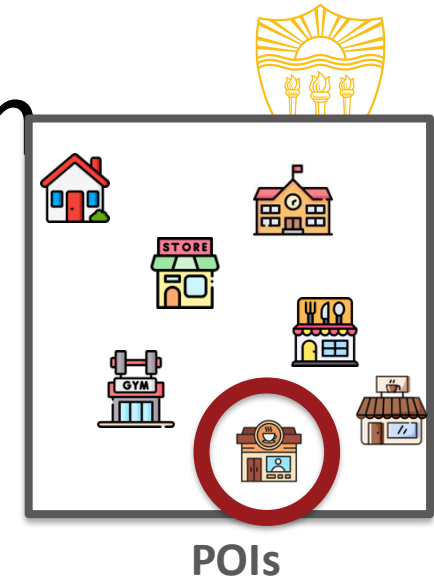
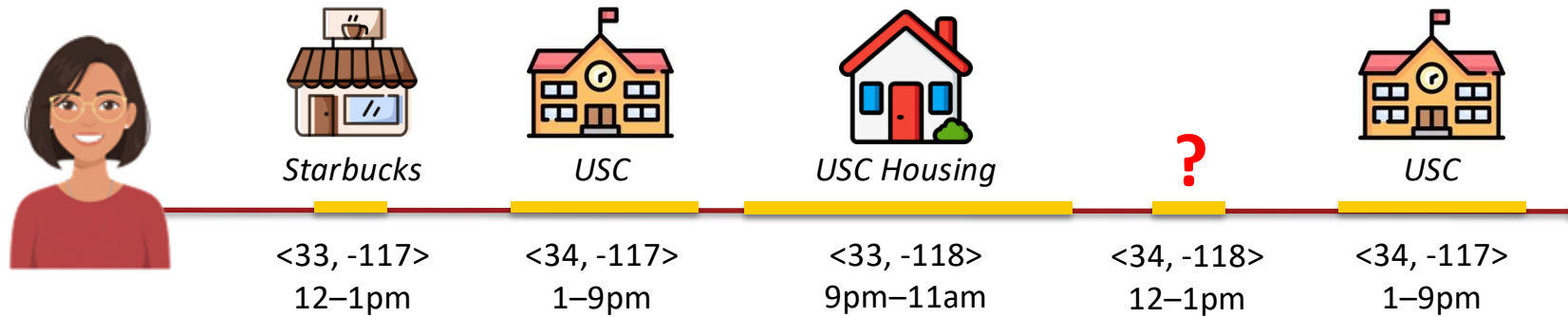
Problem Definition: POI Attribution

Input



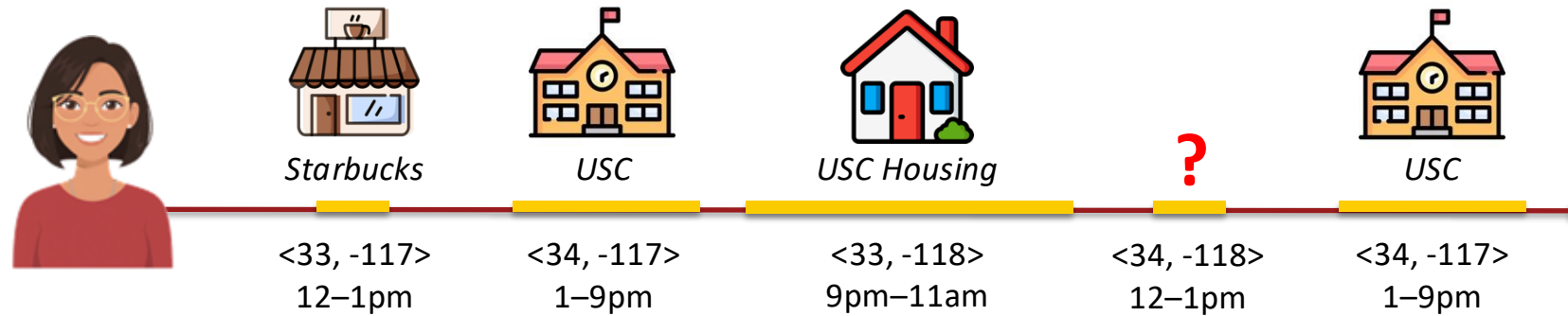
Problem Definition: POI Attribution

Input



Problem Definition: POI Attribution

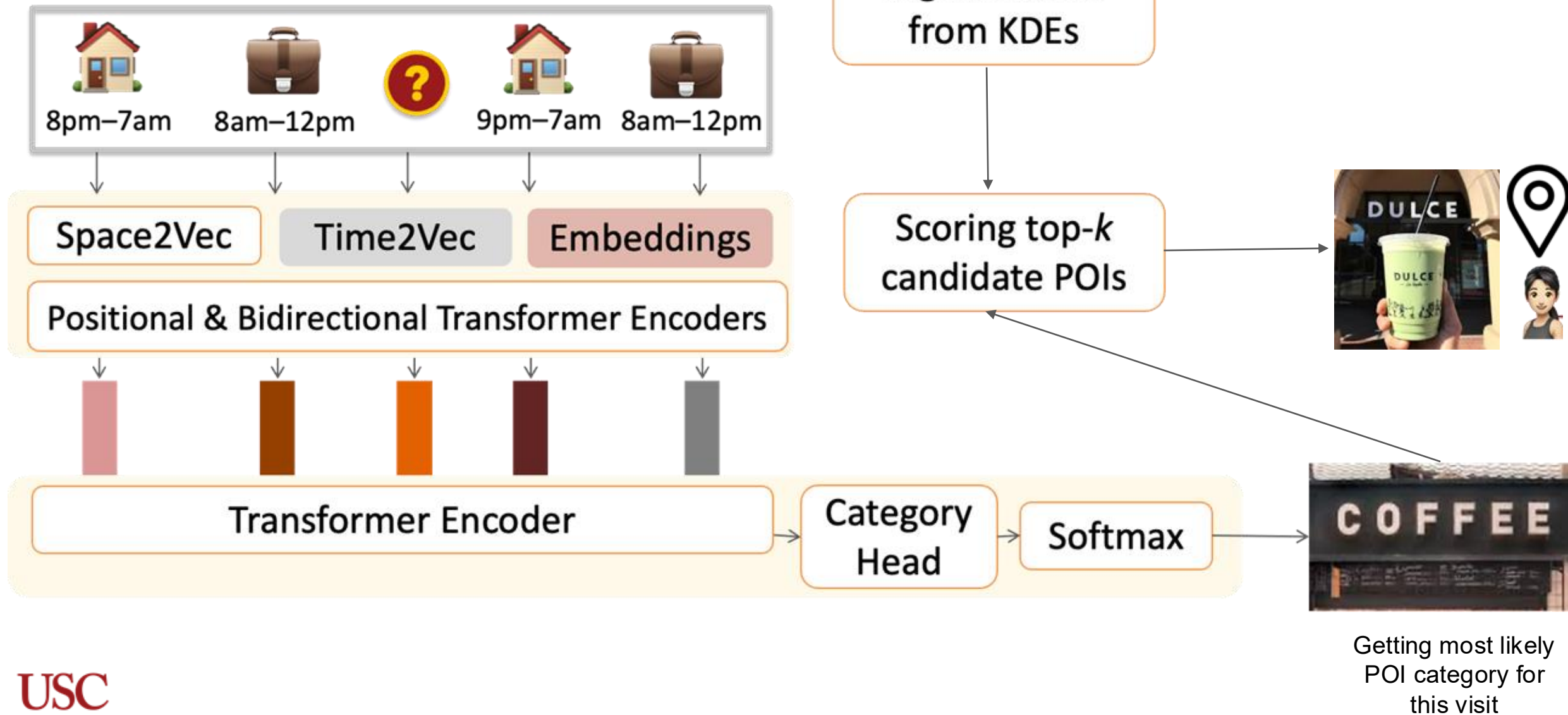
Input



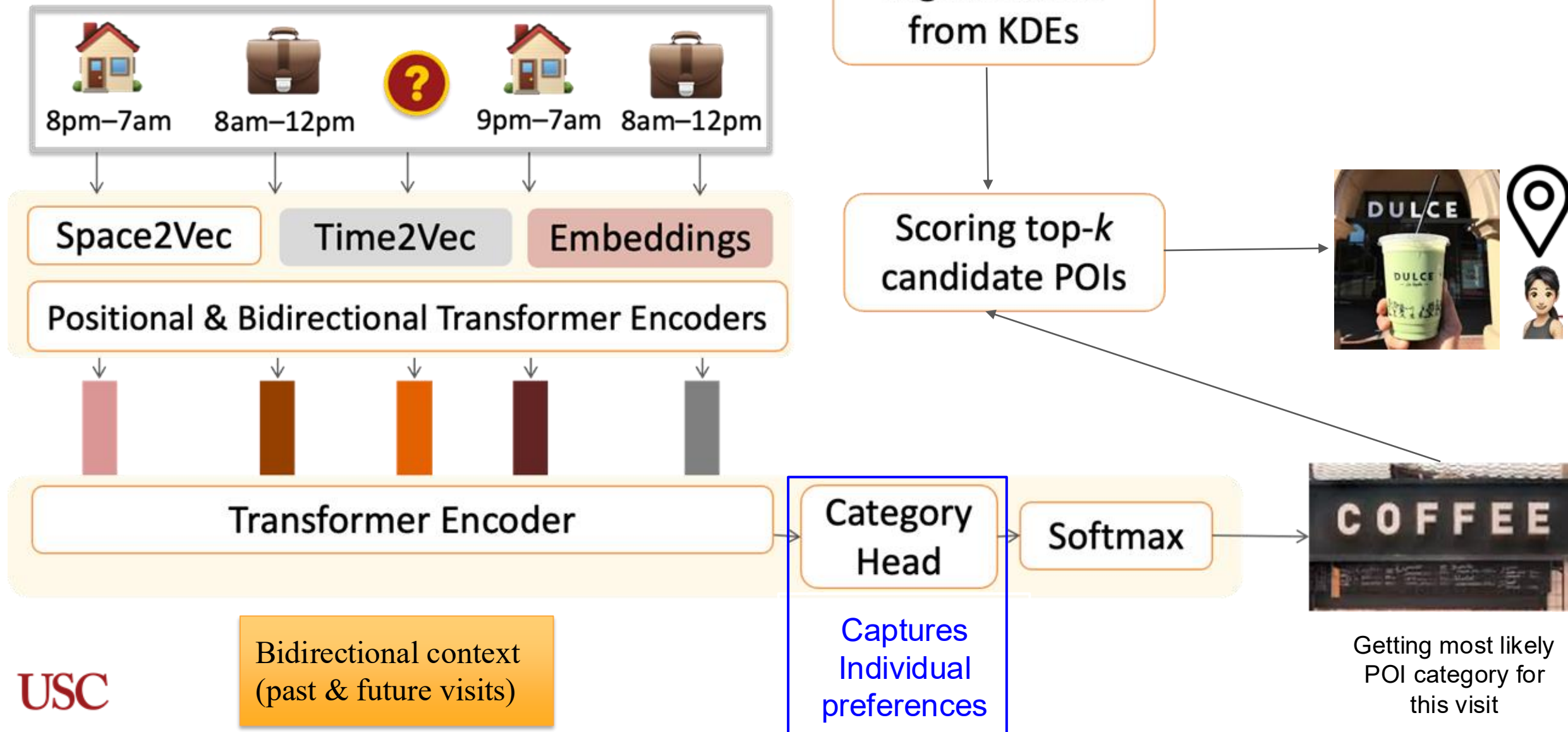
Output



TrajGPT



TrajGPT



TrajGPT



Space2Vec

Time2Vec

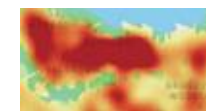
Embeddings

Positional & Bidirectional Transformer Encoders

Transformer Encoder

Captures crowd-level preferences

Log likelihood from KDEs



Coffee shops



Restaurants

Scoring top-k candidate POIs



Category Head

Softmax



Getting most likely POI category for this visit

Captures Individual preferences

Bidirectional context (past & future visits)

TrajGPT



Space2Vec

Time2Vec

Embeddings

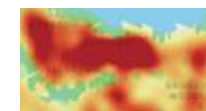
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Coffee shops



Restaurants

Crowd
Time &
Location
given
Category

$$\Pr(p|t,l,H) = \prod_{c \in C(p)} \Pr(t, l | c) \Pr(c | H)$$

$$p = \text{ArgMax}_{\{p \in P\}} \Pr(p|t, l, H)$$

Individual
Category
given
History

Category
Head

max

Captures
Individual
preferences



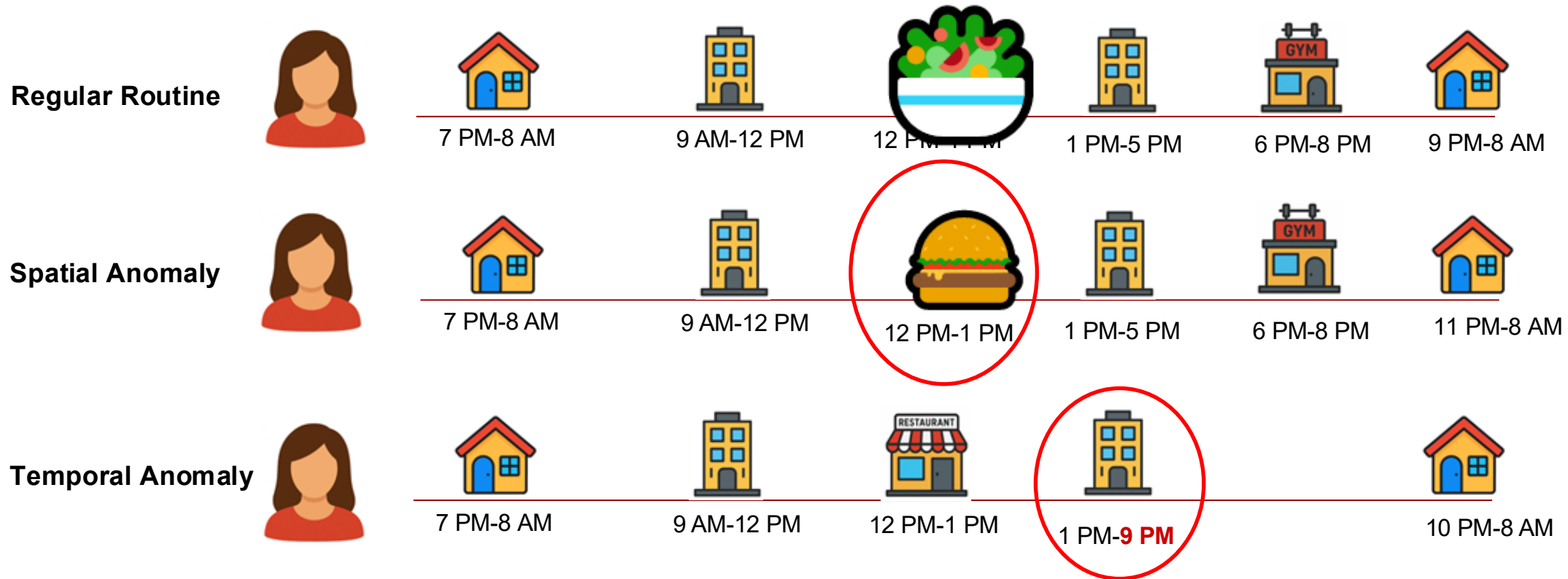
Getting most likely
POI category for
this visit

OUTLINE



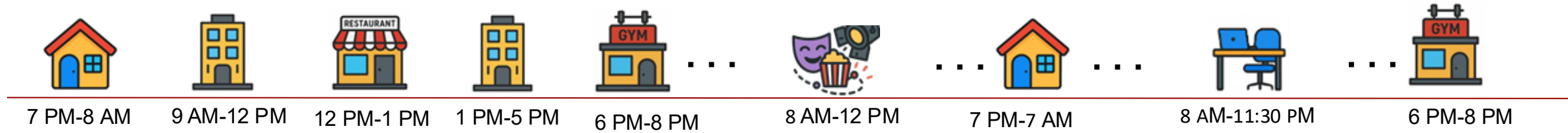
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- **Towards a Mobility-Foundation Model (Open Problems)**

Problem: Visit-Level Anomaly Detection

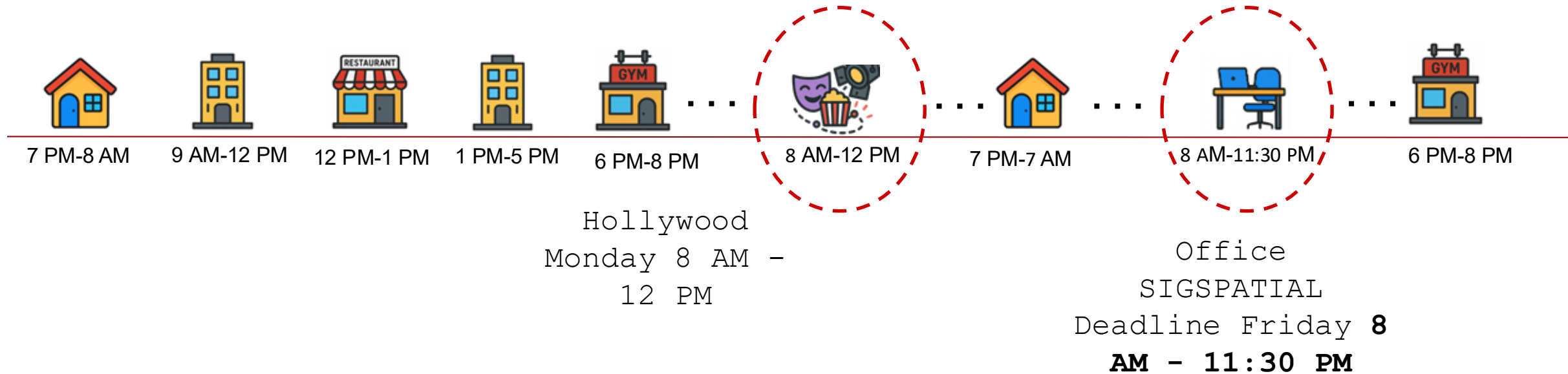


Human mobility anomalies are **multi-context**. They can be **spatial** and **temporal**.

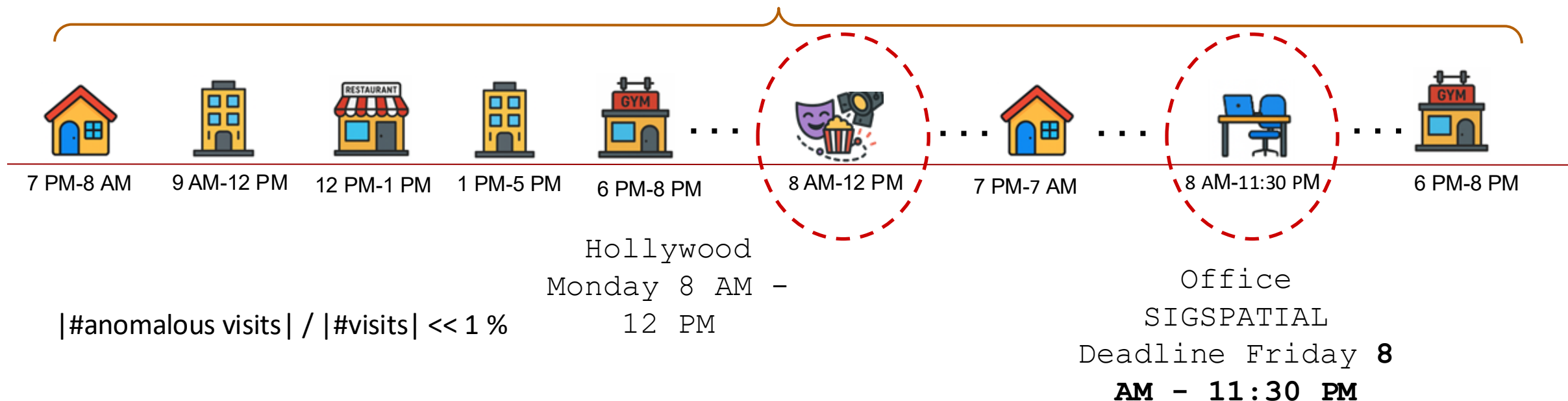
Challenge of Visit-level Anomaly Detection



Challenge of Visit-level Anomaly Detection



Challenge of Visit-level Anomaly Detection



- Sequence-level anomaly detection may miss fine-grained visit-level anomalies.

Related Work



Category	Method	Visit-level	Sequence-level	Temporal Modeling	Interpretability
Computational	iBAT [UbiComp '11]	✗	✓	✗	✗
	Visit Rate [GeoAnomalies '24]	✓	*	✗	✗
Reconstruction/Adversarial	ATD-RNN [ADMA '18]	✗	✓	✗	✗
	IGMM-GAN [TSAS '20]	✗	✓	✗	✗
	GM-VSAE [ICDE '20]	✗	✓	✗	✗
	DeepTEA [VLDB '22]	✗	✓	✓	✗
	NCF-AHSTD [GeoAnomalies '24]	✗	✓	✗	✗
Contrastive Learning	ATROM [IJCAI '23]	✗	✓	✗	✗
	TOD4Traj [SIGSPATIAL '24]	✗	✓	✗	✗
Autoregressive Generation	LMTAD [SIGSPATIAL '24]	✓	*	✗	✗
	DeepBayesics [GeoAnomalies '24]	✓	*	✓	✗
	ICAD [SIGSPATIAL '25]	✓	*	✓	✓

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Auto

Trajectory anomaly detection mostly studied for **taxi trajectories**.

Related Work



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Related Work



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	GM-VSAE [ICDE '20]	X	✓	X	X
	DeepTEA [VLDB '22]	X	✓	✓	X
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Related Work

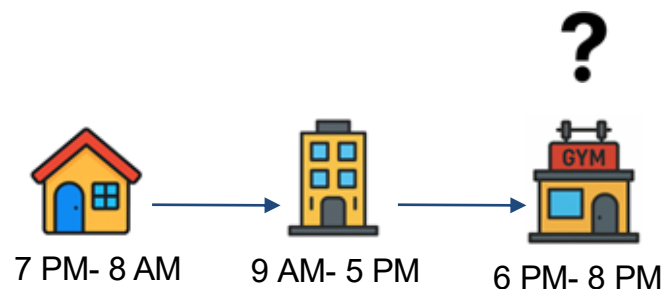


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	ICAD [SIGSPATIAL '25]	✓	*	✓	✓

Our Approach: TrajGPT-based



TrajGPT's Output

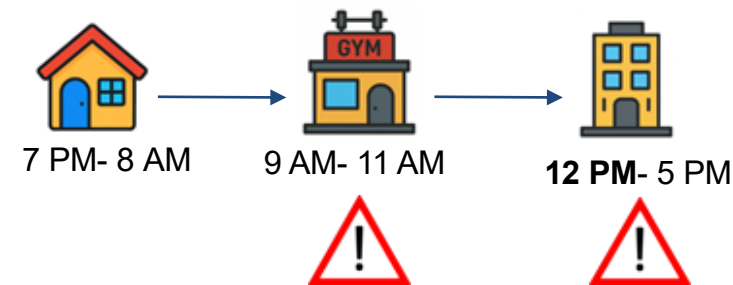


Normal



Compare visits **relative** to
learned distribution of normal
behavior obtained through next
visit prediction.

Anomalous Behavior



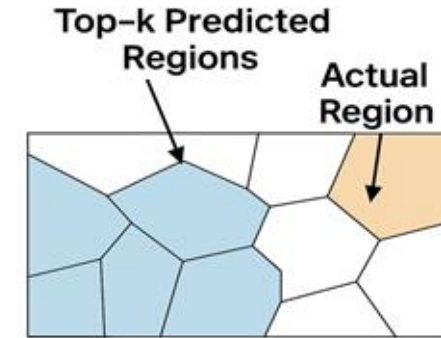
Anomaly



Relative Multi-context Anomaly Scoring

- Region Score (Discrete Space)

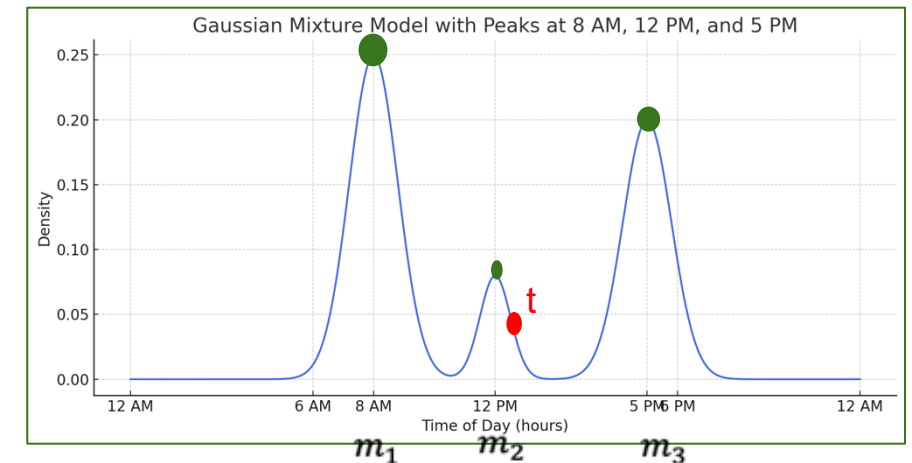
$$S_{v_i}^R = \begin{cases} 0, & \text{if } r_i \in \mathcal{T}_k, \\ \frac{1}{k} \sum_{r_j \in \mathcal{T}_k} \log \hat{P}(r_j | V_{i-1}) - \log \hat{P}(r_i | V_{i-1}), & \text{otherwise.} \end{cases}$$



- Arrival and Departure Score (Continuous Space)

$$S_{v_i}^t = -\text{Log} \left(\frac{\sum_k \pi_k f_k(t)}{\sum_k \pi_k f_k(m_k)} \right)$$

The **higher** the distance of an observed point to all modes of Gaussian components $\rightarrow$ **Higher** anomaly score



Experimental Insights



NUMOSIM Dataset (Novateur's public data with labels)

Model	Visit-level		Agent-level	
	AP	AUROC	AP	AUROC
STOD	0.00024	0.5940	0.00182	0.5180
RioBusData	0.00019	0.5050	0.00164	0.5010
GM-VSAE	—	—	0.00192	0.5070
DeepBayesics	0.0042	<u>0.6598</u>	<u>0.0121</u>	<u>0.7258</u>
LMTAD	<u>0.0196</u>	0.5472	0.0020	0.5020
TOD4Traj	—	—	0.0022	0.5256
ICAD	0.1610	0.8332	0.2561	0.9010

Consistently outperforming baselines by a significant margin in NUMOSIM synthetic data with **16M** visits and **200K** agents.

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Consistently outperforming baselines by a significant margin in NUMOSIM synthetic data with **16M** visits and **200K** agents.

Scoring: Unlike all baselines, which rely on absolute scoring, ICAD scores visits relative to highly probable normal behavior in logarithmic space.

Modeling visits autoregressively: conditioning each prediction on all preceding visits, preserves sequential order (per agent)

Continuous modeling of time: Parameterizing arrival and departure times using GMMs (implicitly capturing both travel time and dwelling time)

Feature encoding and representation: Poly2Vec preserves spatial distance between region cells , and Time2Vec captures both periodicity and temporal shifts.

OUTLINE



- **Motivation**
- **Background**
- **TrajGPT: Transformer for Next-Visit Prediction**
- **Downstream Applications**
- **Towards a Mobility-Foundation Model (Open Problems)**

Toward Foundation Models for Mobility Enriched Geospatially Embedded Objects

Maria Despoina Siampou[†], Shang-Ling Hsu[†], Shushman Choudhury[‡], Neha Arora[‡], Cyrus Shahabi[†]

[†] University of Southern California, Los Angeles, California, USA

[‡] Google Research, Mountain View, California, USA

{siampou,hsushang,shahabi}@usc.edu,{shushmac,nehaarora}@google.com

Abstract

Recent advances in large foundation models (FMs) have enabled learning general-purpose representations in natural language, vision, and audio. Yet geospatial artificial intelligence (GeoAI) still lacks widely adopted foundation models that generalize across tasks that require joint reasoning over geospatial objects and human mobility. Such tasks are crucial as mobility, along with satellite imagery, street view, and text, is a core modality for understanding the physical world. We argue that a key bottleneck is the absence of unified, general-purpose, and transferable representations for geospatially embedded objects (GEOs). Such objects include points, polylines, and polygons in geographic space, enriched with semantic context and critical for geospatial reasoning. Much current GeoAI research compares GEOs to tokens in language models, where patterns of human movement and spatiotemporal interactions yield contextual meaning similar to patterns of words in text. However, modeling GEOs introduces challenges fundamentally different from language, including spatial continuity, variable scale and resolution, temporal dynamics, and data sparsity. Moreover, privacy constraints and global variation in mobility further complicates modeling and generalization. This paper formalizes these challenges, identifies key

enter the correct conference title from your rights confirmation email (Conference acronym 'XX). ACM, New York, NY, USA, 6 pages. <https://doi.org/XXXXXXX.XXXXXXX>



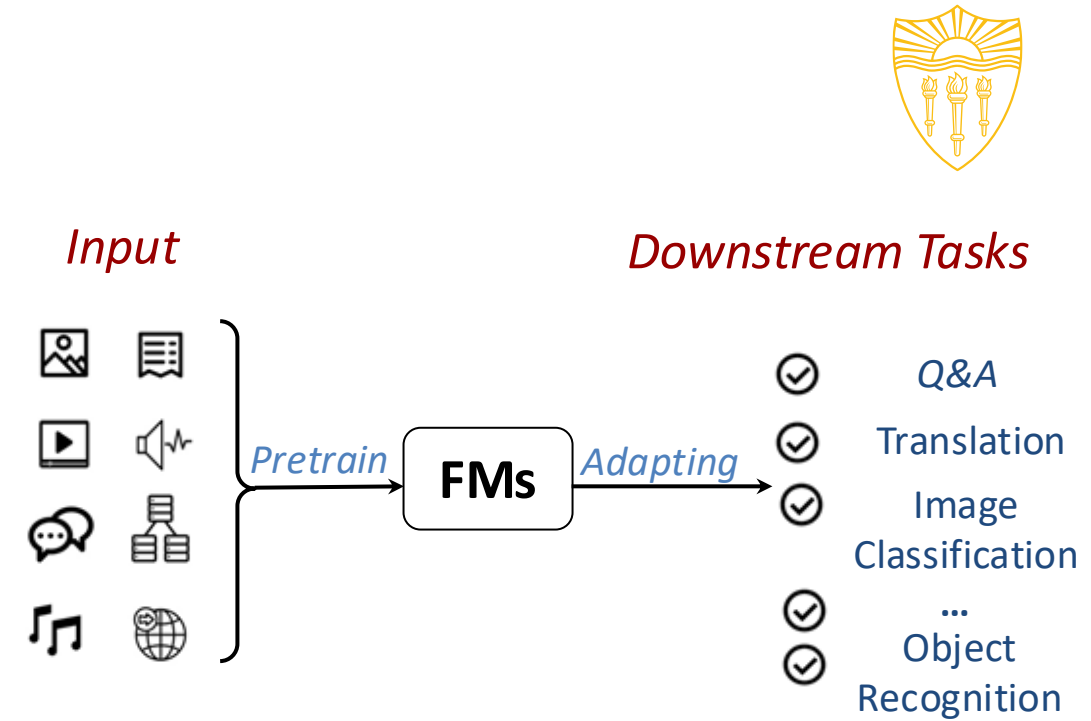
Figure 1: Pipeline overview for mobility-enriched GEOs: approach, applications, and modeling considerations.

The success of Foundation Models

Achieved **breakthrough performance** across many domains (e.g., NLP, vision).

Recipe for success:

- *Transferable representations* of their **core units** (e.g., words, pixels).
- *Self-supervised pretraining* on massive unlabeled datasets.
- *Universal backbone* adaptable to many downstream tasks.



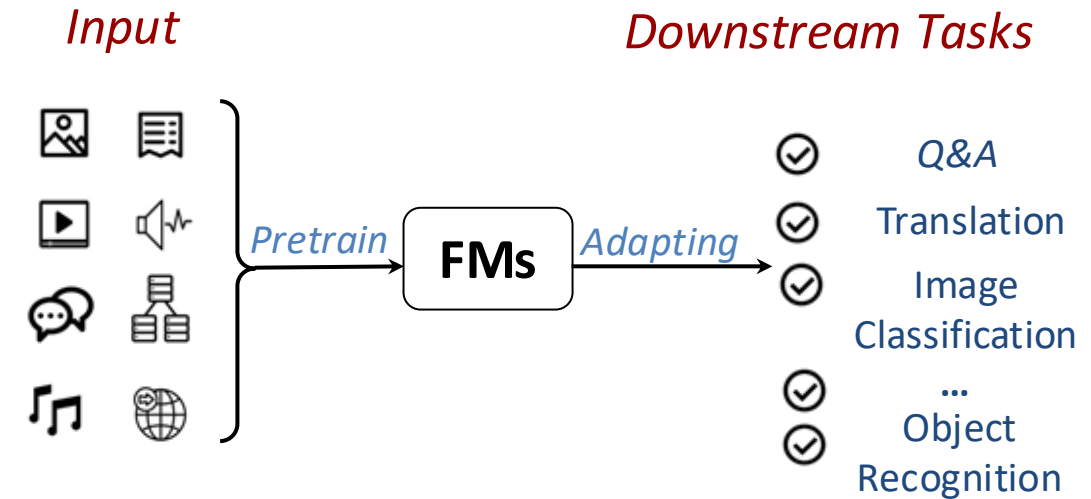
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- *Self-supervised pretraining* on massive unlabeled datasets.
- *Universal backbone* adaptable to many downstream tasks.

🌐 **GeoAI has yet to see comparable success.**





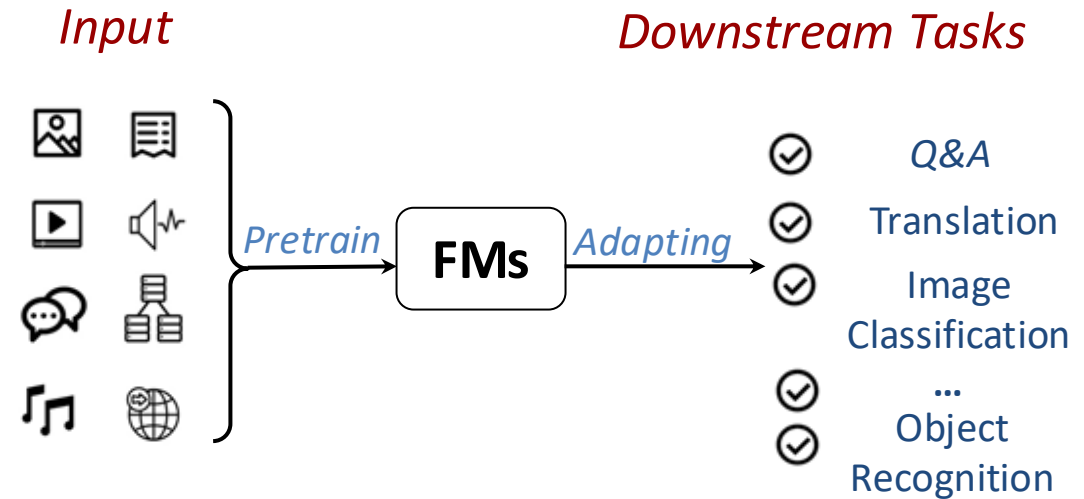
The success of Foundation Models

Achieved **breakthrough performance** across many domains (e.g., NLP, vision).

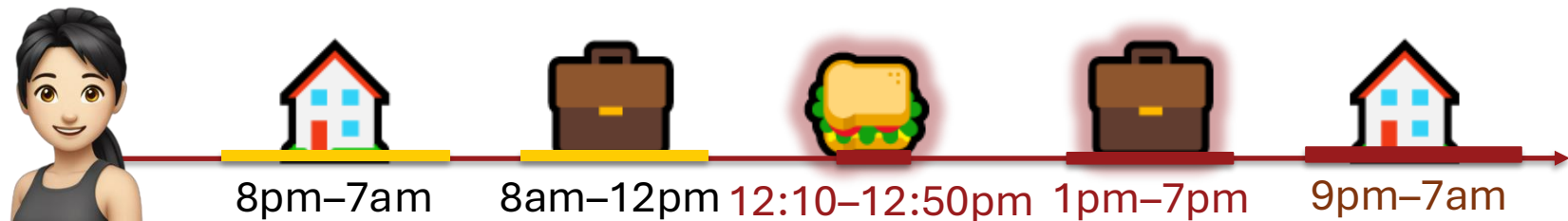
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TrajGPT



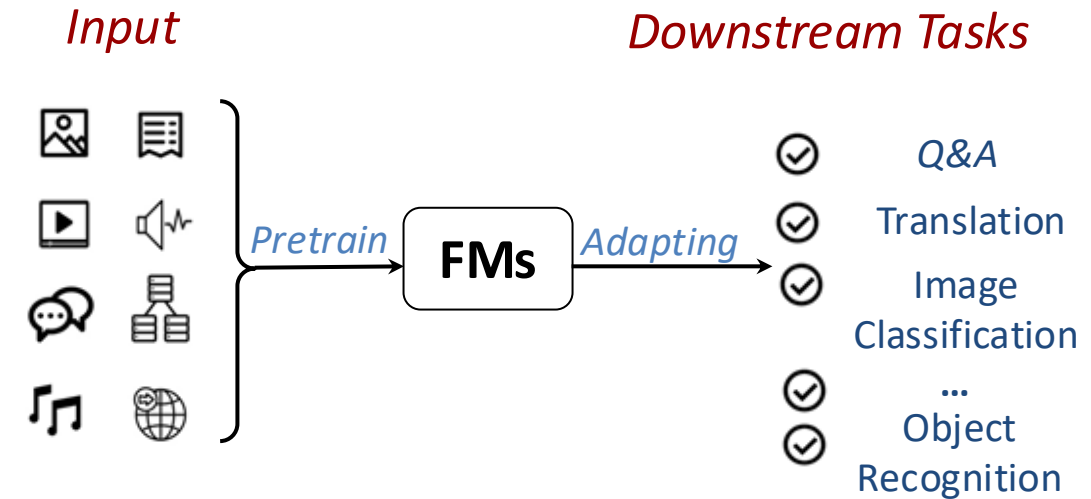


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1. Next Visit Prediction
2. Synthetic Trajectory Generation
3. Filling missing visits (gap imputation)
4. POI attribution
5. Mobility outlier detection

The success of Foundation Models

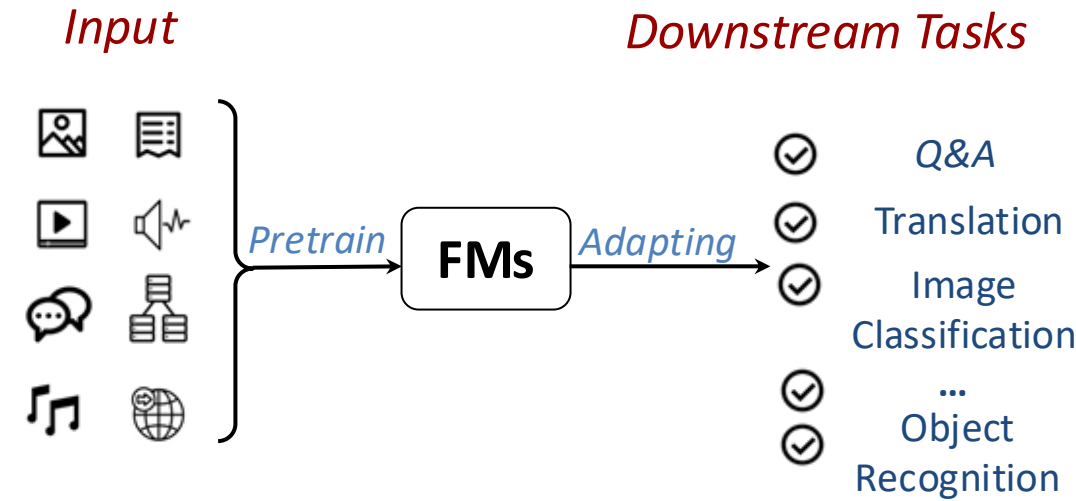


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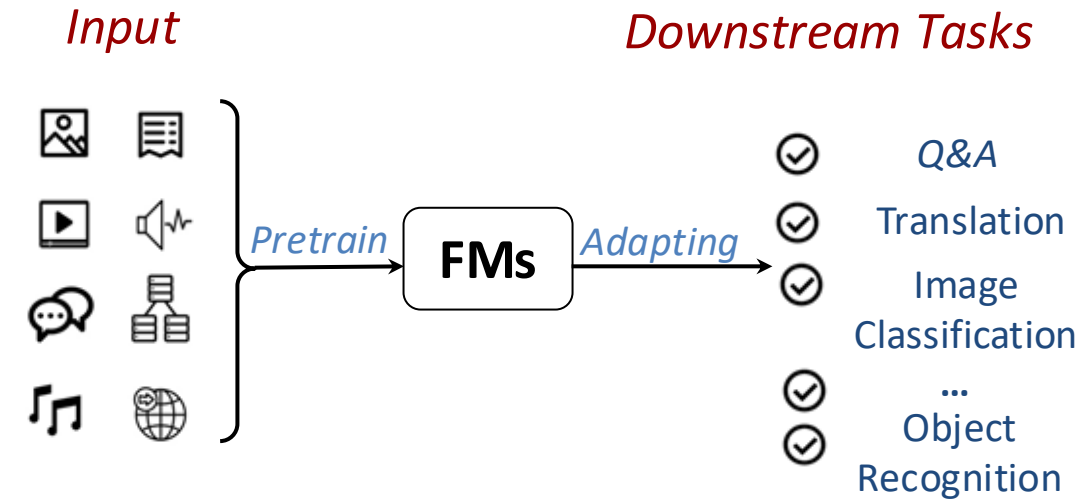
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Geo Tokens $\neq$ Words in Language; Key differences remain.

- GEOs are continuous, sparse, and have complex context—unlike discrete words

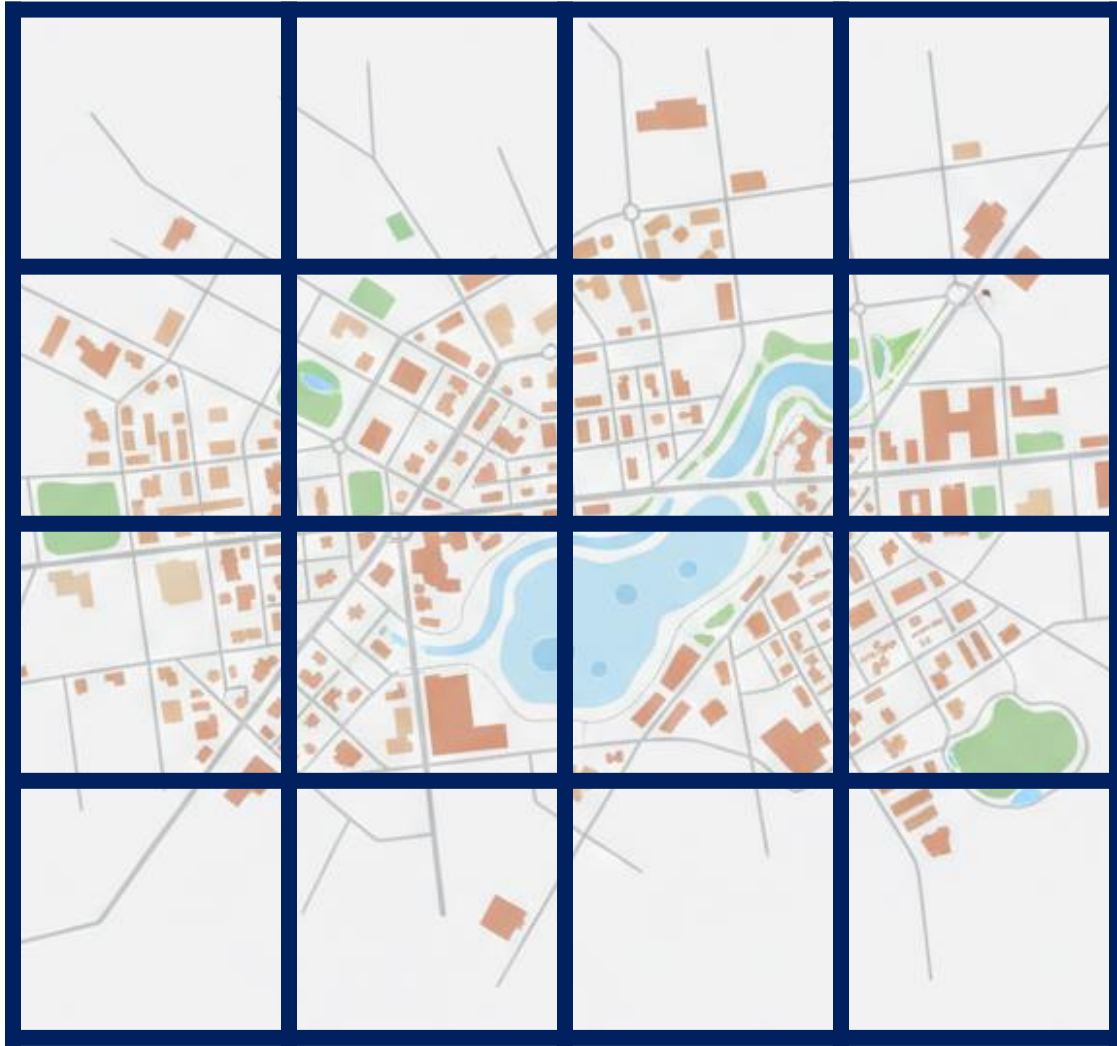
Open Problem: Location Vocabulary



Can we “*speak*” trajectories?



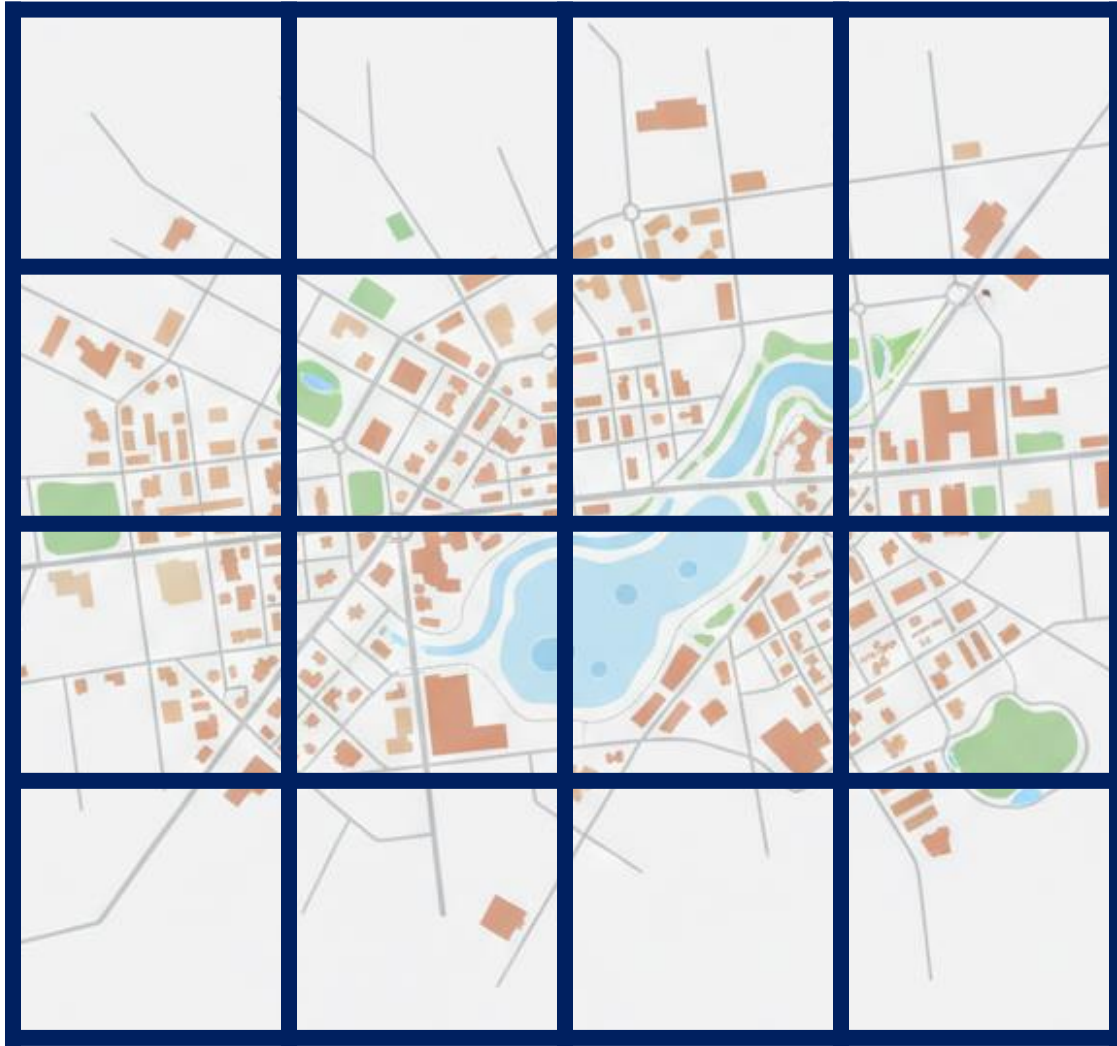
Open Problem: Location Vocabulary



Can we “*speak*” trajectories?

- Grid-based

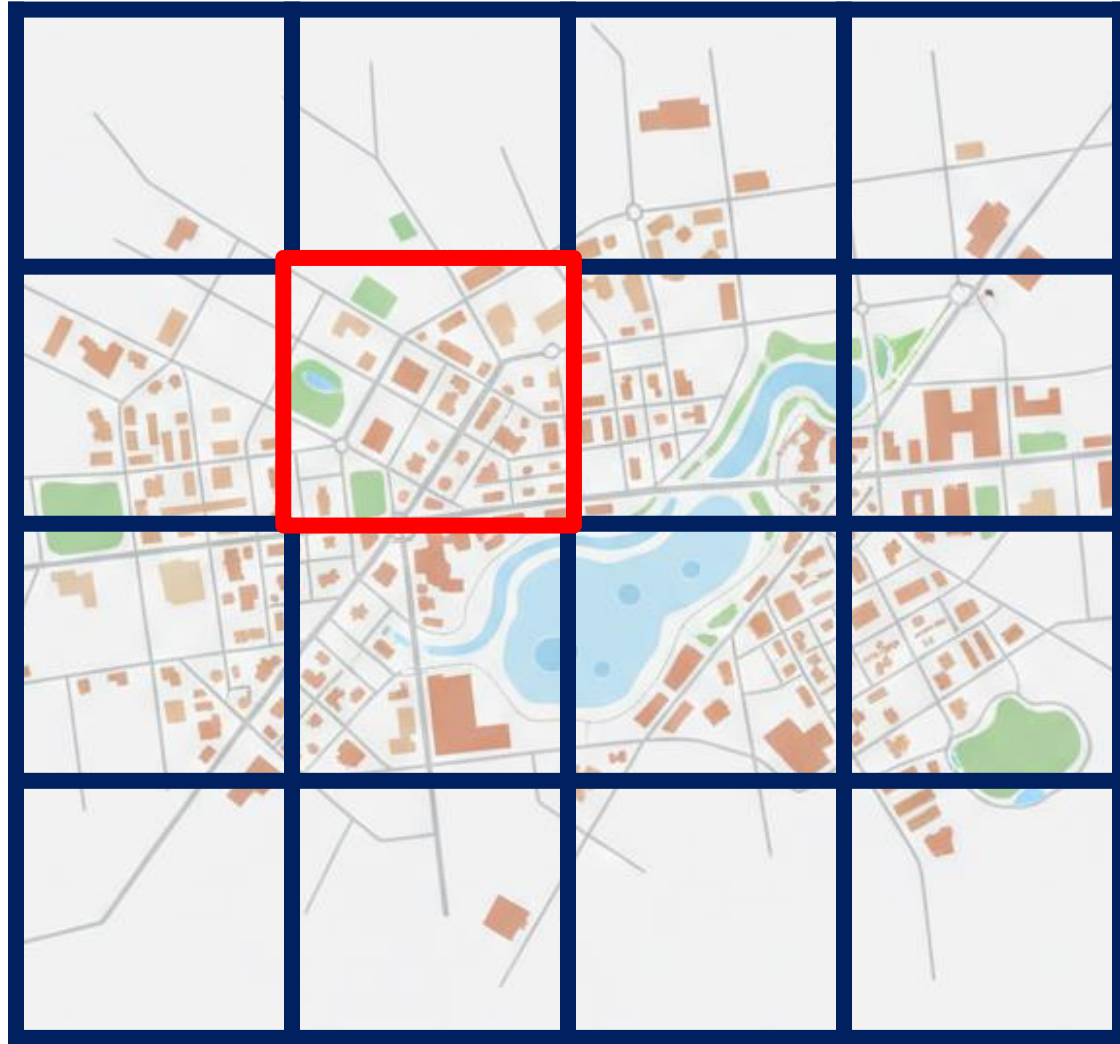
Open Problem: Location Vocabulary



Can we “*speak*” trajectories?

- Grid-based
 - Resolution trade-off

Open Problem: Location Vocabulary

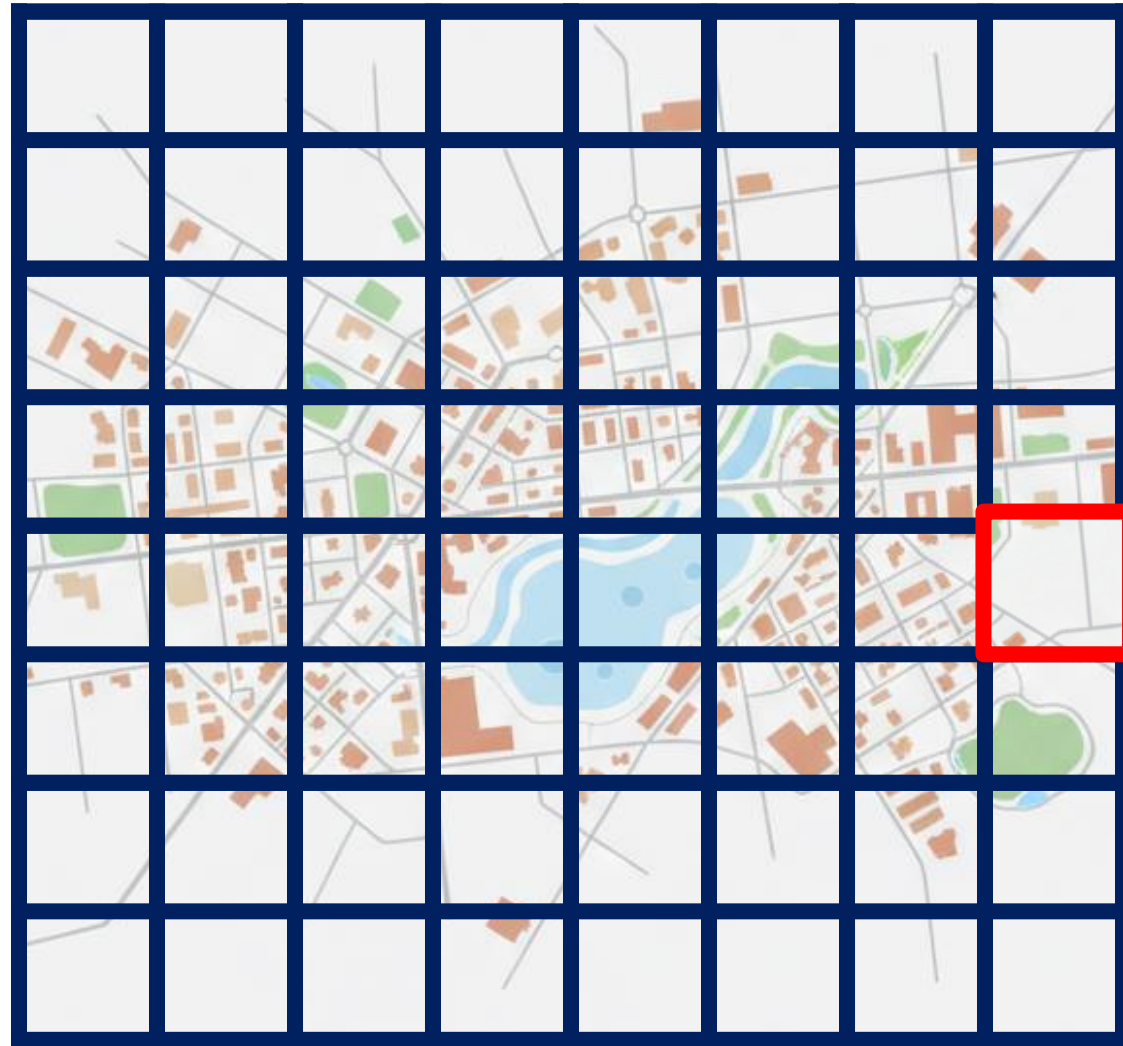


Can we “*speak*” trajectories?

- Grid-based
 - Resolution trade-off

Where in the cell does it mean?

Open Problem: Location Vocabulary



Can we “*speak*” trajectories?

- Grid-based
 - Resolution trade-off

Not enough data to learn a good representation

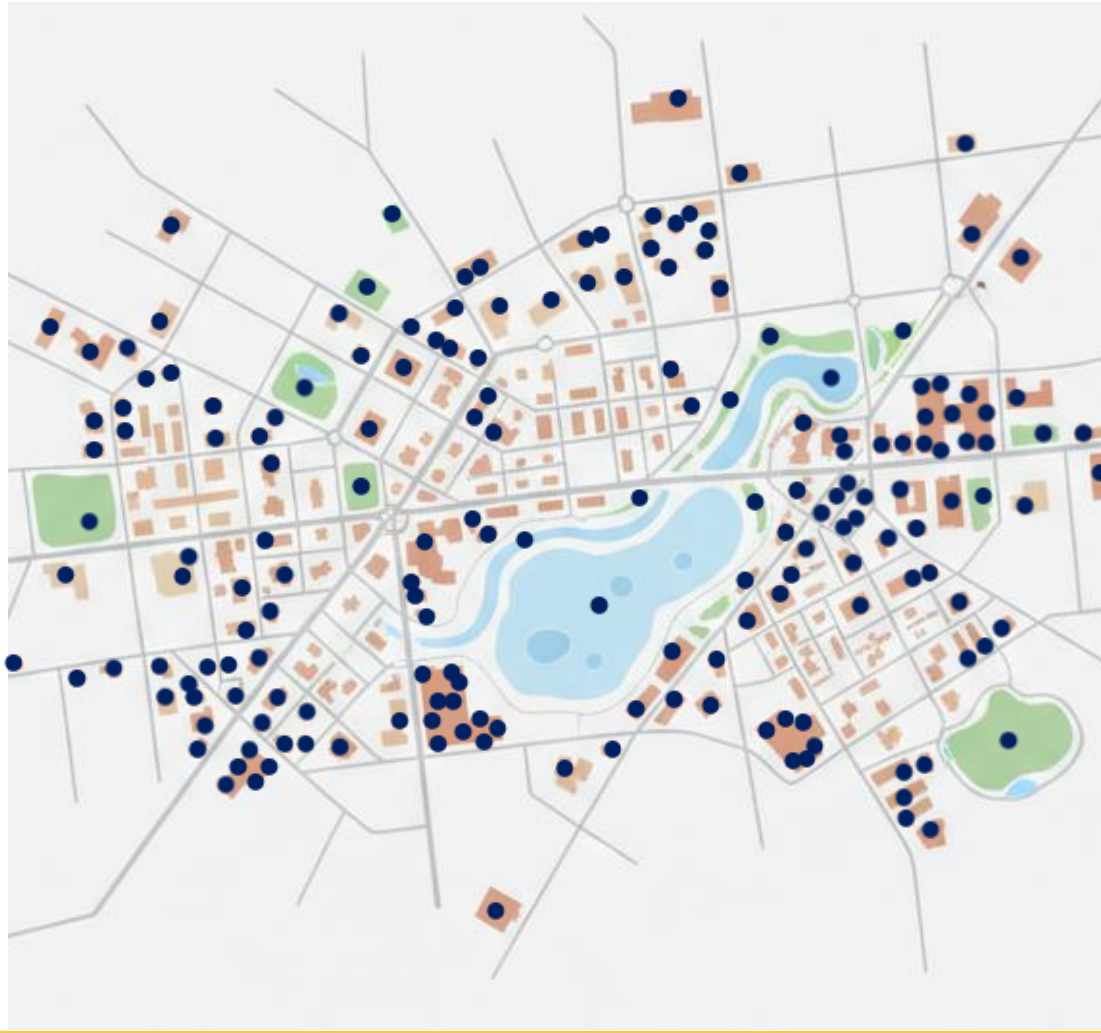
Open Problem: Location Vocabulary



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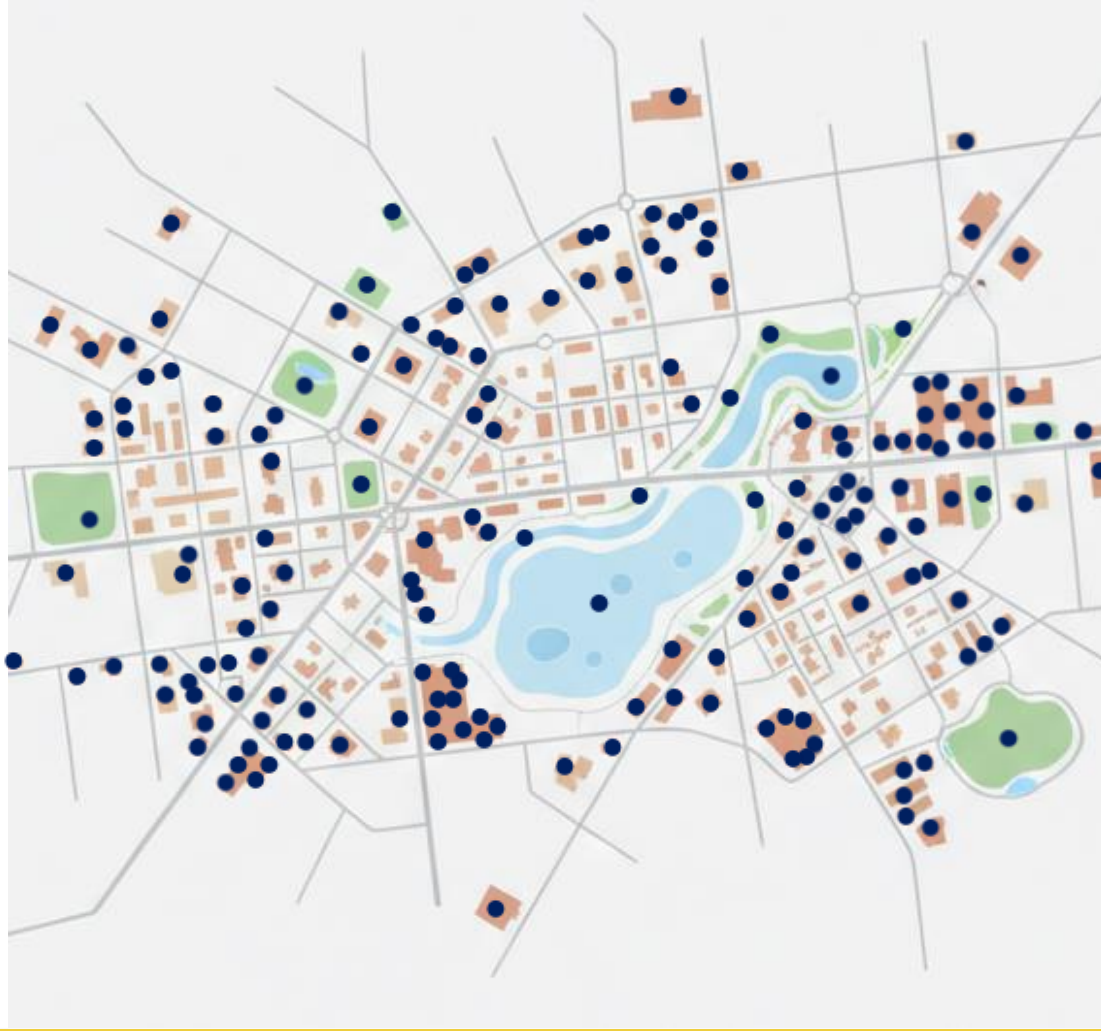
Open Problem: Location Vocabulary



Can we “*speak*” trajectories?

- Grid-based
 - Resolution trade-off
- Object-based

Open Problem: Location Vocabulary



Can we “*speak*” trajectories?

- Grid-based
 - Resolution trade-off
- Object-based
 - Limited scalability

Open Problem: Location Vocabulary



Can we “*speak*” trajectories?

- Grid-based
 - Resolution trade-off
- Object-based
 - Limited scalability
 - Extensive preprocessing

Open Problem: Location Vocabulary



Can we “*speak*” trajectories?

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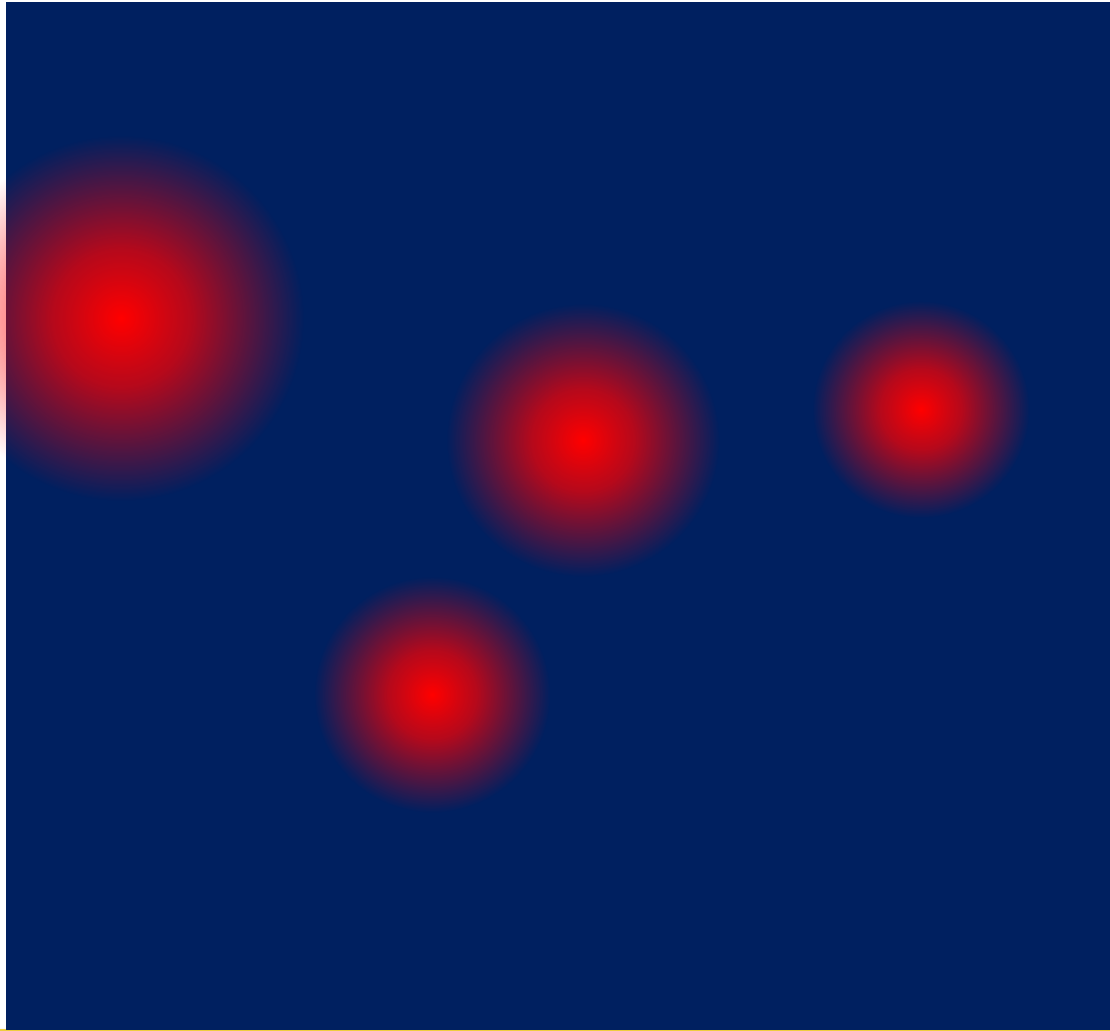
Open Problem: Location Vocabulary



Can we “*speak*” trajectories?

- Grid-based
 - Resolution trade-off
- Object-based
 - Limited scalability
 - Extensive preprocessing
- Continuous-space

Open Problem: Location Vocabulary



Can we “*speak*” trajectories?

- Grid-based
 - Resolution trade-off
- Object-based
 - Limited scalability
 - Extensive preprocessing
- Continuous-space
 - Wastes information from maps

Open Problem: Location Vocabulary



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Open Problem: Location Vocabulary



Can we “*speak*” trajectories?

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 - Wastes information from maps
- Difficulties in unseen places

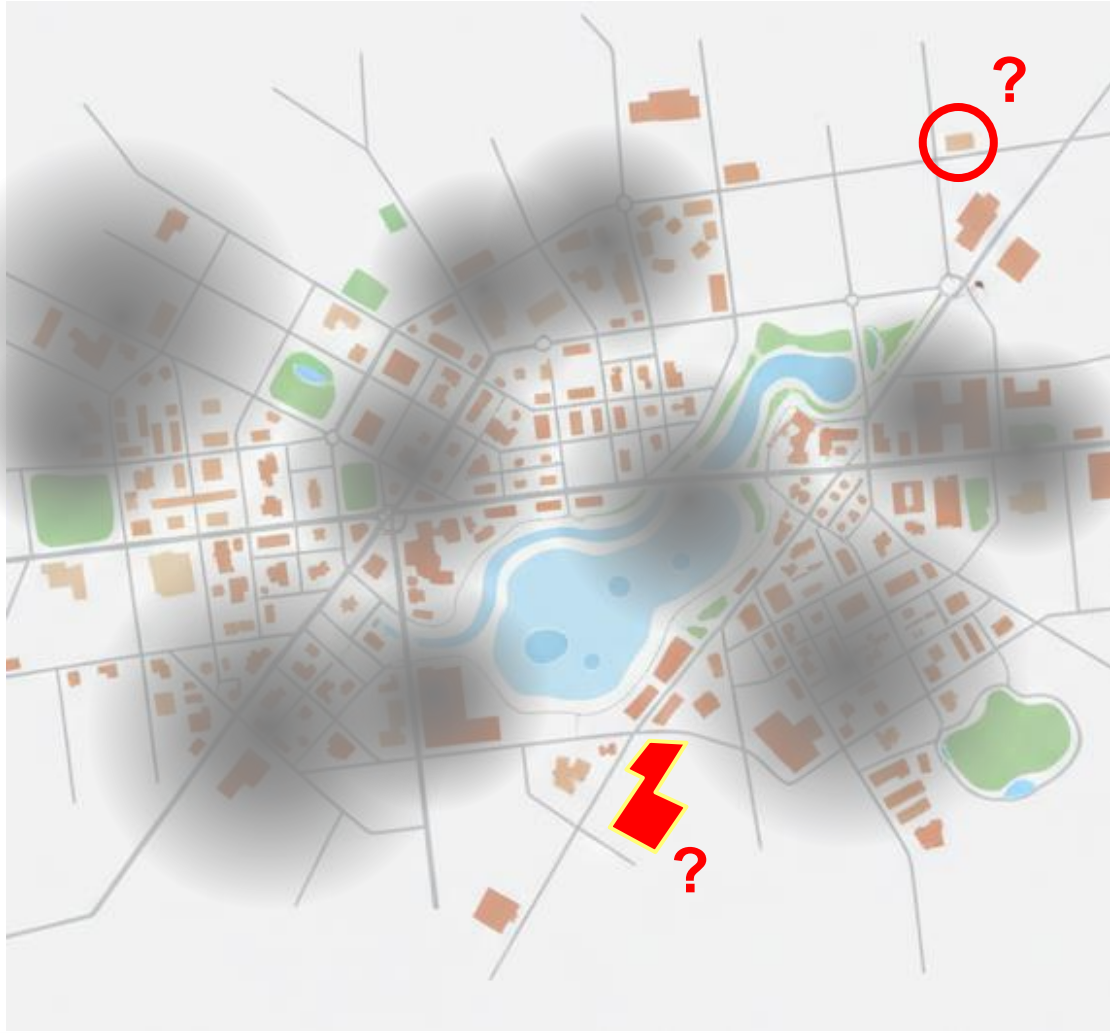
Open Problem: Location Vocabulary



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Open Problem: Location Vocabulary



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- Continuous-space
 - Wastes information from maps
- Difficulties in unseen places
 - e.g., home, new openings

Proposal: Geospatially Embedded Objects (GEOs)



Geospatially Embedded Objects (**GEOs**) = The “**tokens**” of GeoAI

- *Points* , *Polylines* , *Polygons* 
- Each enriched with **geometric** and **semantic** context
 - Enable global spatial reasoning

GEO Encoding

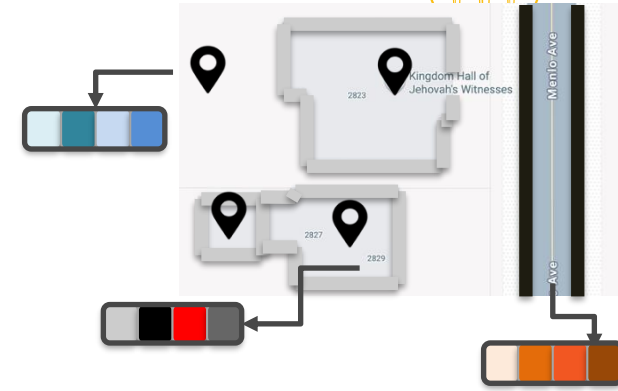
(1) Spatial Dimension^{7,8,9,10} $\approx$ Word Embedding in NLP

- Support **heterogeneous** nature: points, polylines, polygons
- Preserve **rich spatial structure** and **relations**: location, shape, direction, topology
- Robust across **multiple spatial scales**: from objects to neighborhoods to regions
- **Key challenge**: *Unified, geometry- and scale-aware encoders.*



Varying scales

Google Research



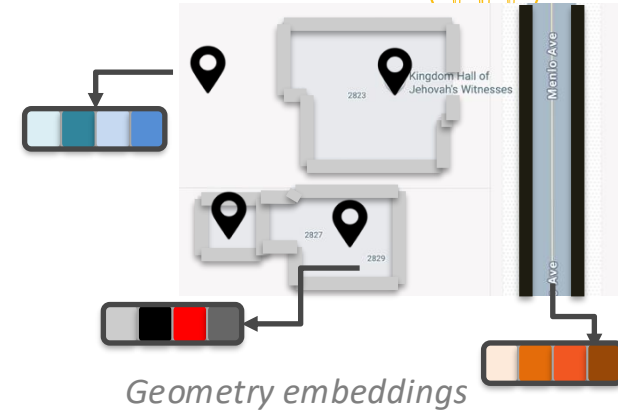
Geometry embeddings

GEO Encoding



Varying scales

Google Research



Geometry embeddings



Category: Coffee shop.
Street Address: 127 35 St.

...

% of **nearby** restaurants: [...]

(1) Spatial Dimension^{7,8,9,10} $\approx$ Word Embedding in NLP

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(2) Contextual Dimension^{11,12,13,14}

- **Object level** functional roles (*e.g., business type*) and metadata (*e.g., road description*)
- **Neighborhood level** information: GEO's functional role within its broader environment
- **Key challenge**: *Model interdependence between geometric features and semantics.*

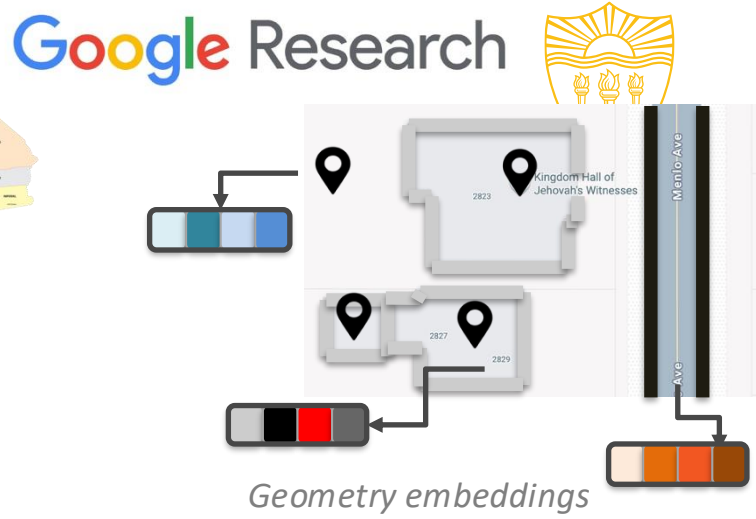
GEO Encoding

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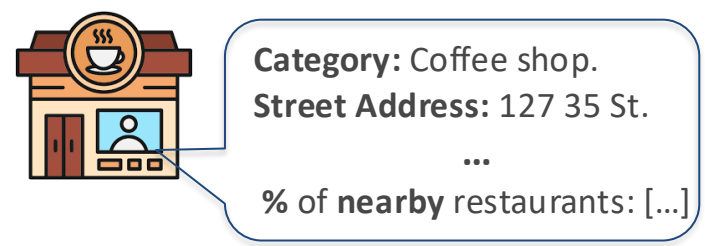
Varying scales



Geometry embeddings

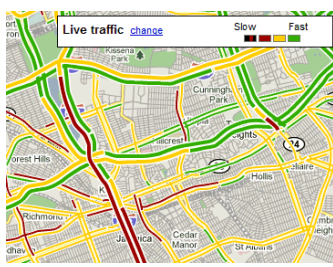
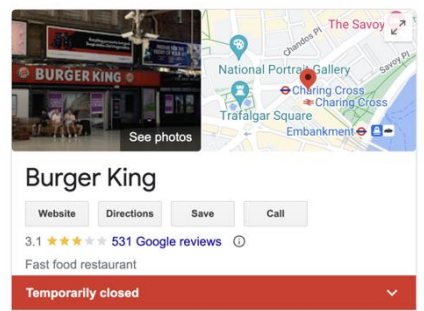
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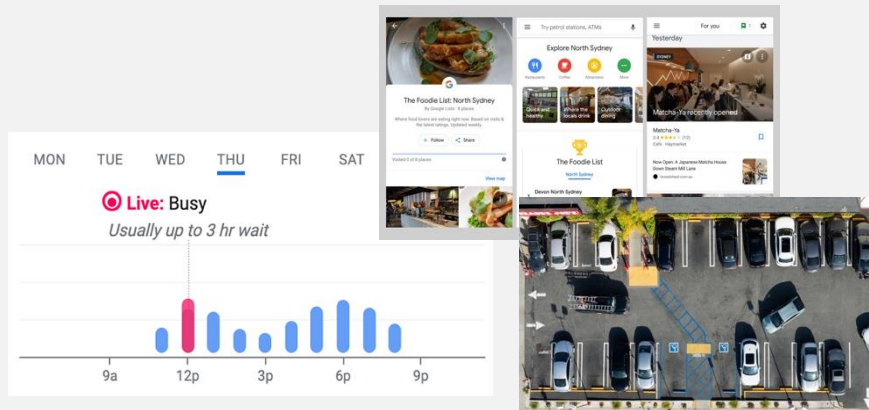
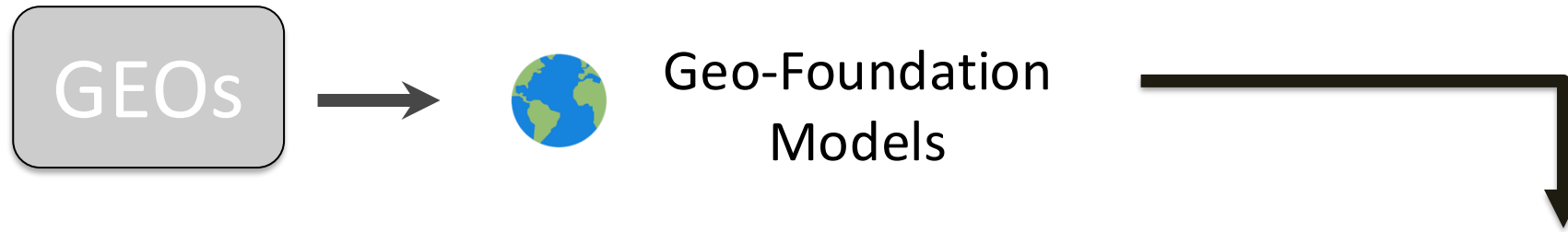


(3) Temporal Dimension^{15,16} $\approx$ Positional Encoding in NLP

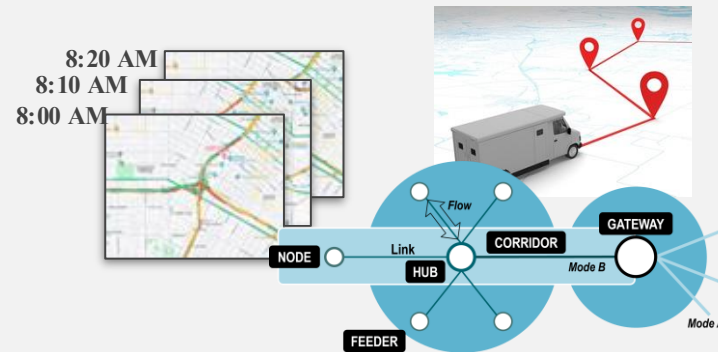
- Derived from the visit order and timing in trajectories
- The “meaning” of GEOs can **change over time** (within a day/ across time)
- **Key challenge**: *Model dynamic nature. Choice of temporal granularity.*



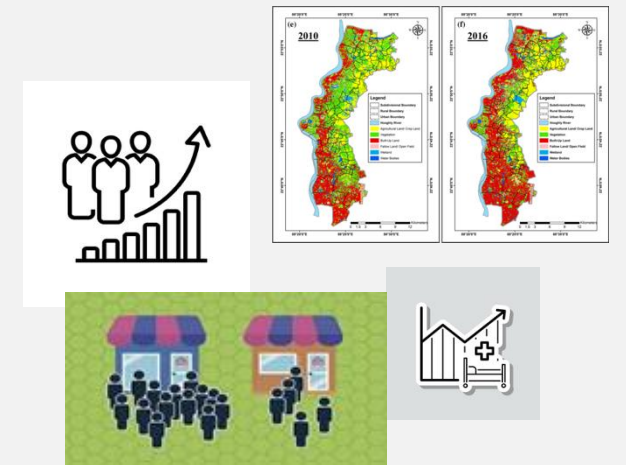
Our Vision (Recap)



GEO-centric tasks



Mobility-centric tasks



Population-centric tasks

Orthogonal Considerations



Privacy^{19,20}:

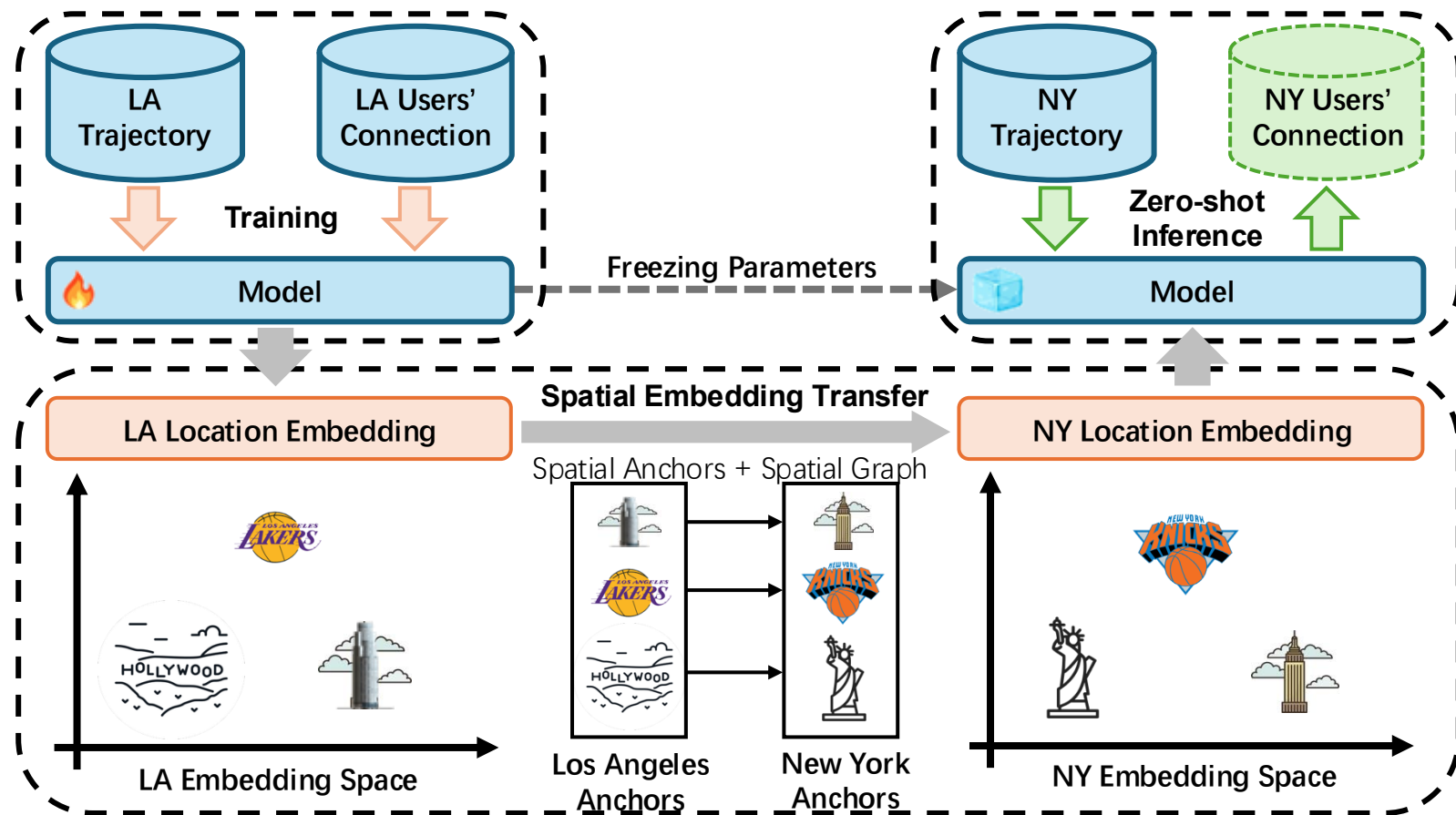
- Human mobility data can reveal **sensitive** personal routines: Re-identification for private homes, hospitals, and places of worship.
- **Need of methods** for filtering or abstracting **sensitive GEOs** to protect individuals' privacy.



Cross Region Transferability²¹:

- Limited data coverage risks overfitting GEO representations to specific regions or periods.
- **Need of methods** for generalizing across regions, urban forms, and mobility behaviors.

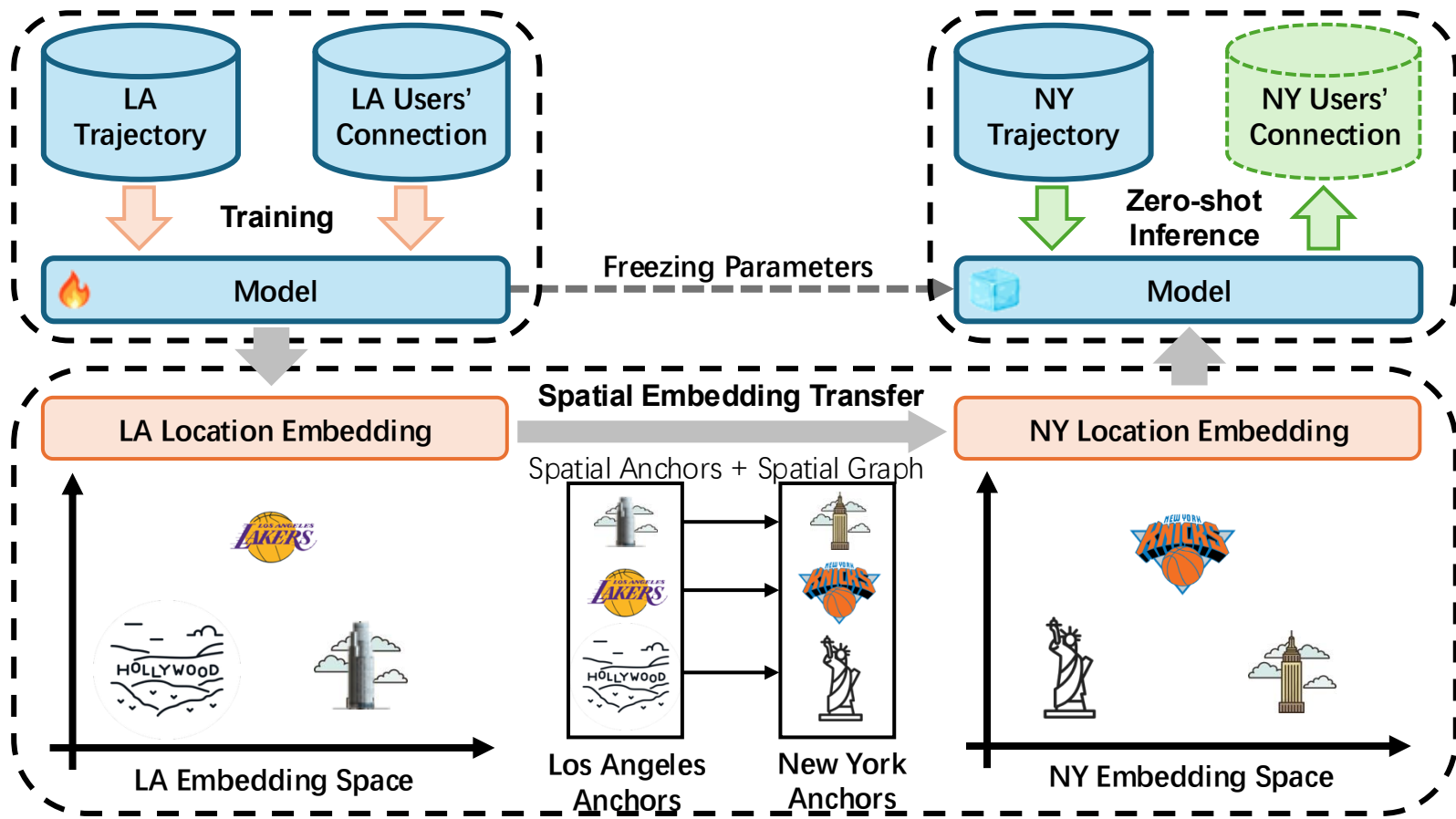
Open Problem: Transferable Cross-Region



Open Problem: Transferable Cross-Region



Transferable social relationship inference

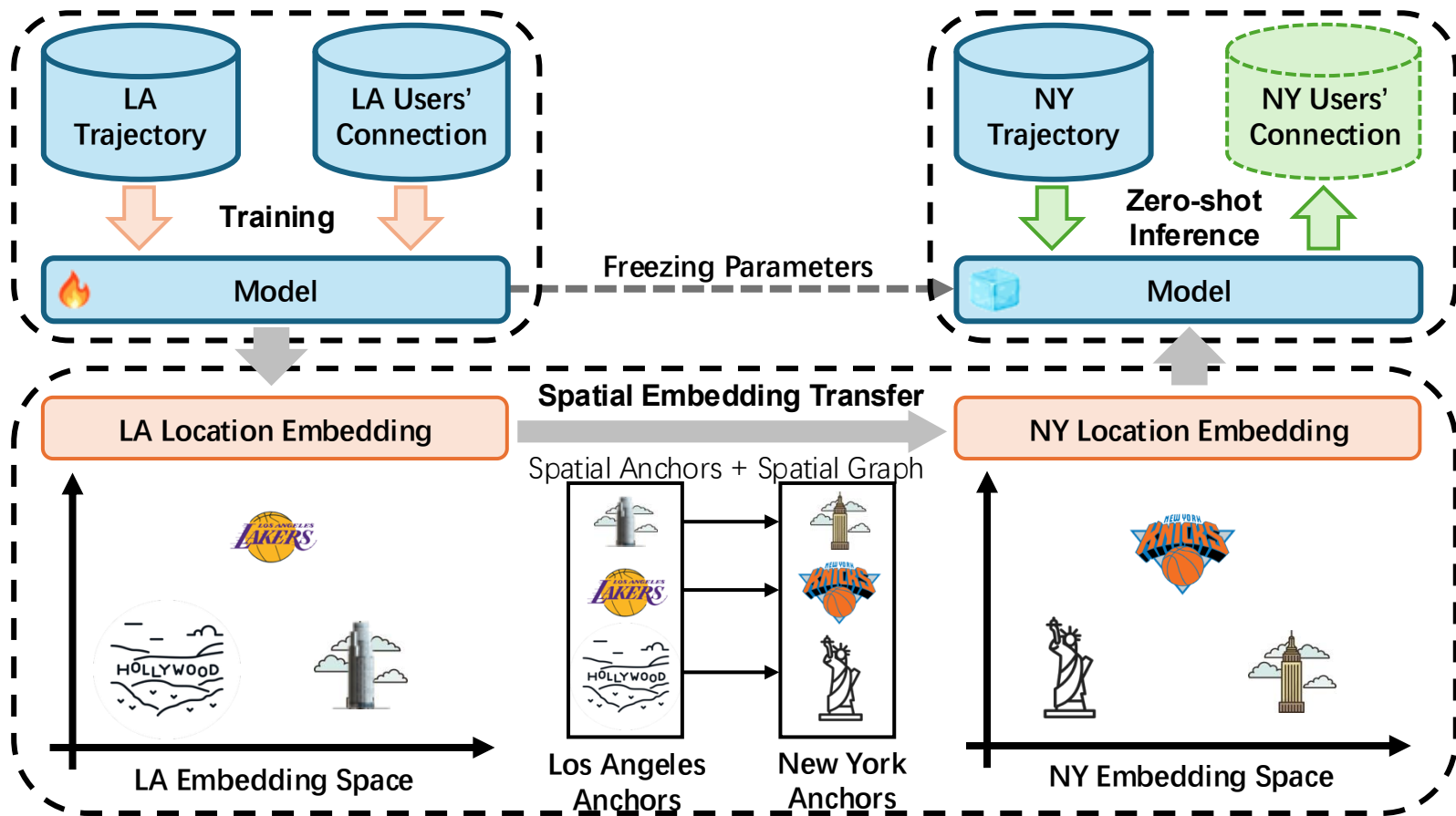


Open Problem: Transferable Cross-Region



Transferable social relationship inference

- Train the foundation model in a data-rich city

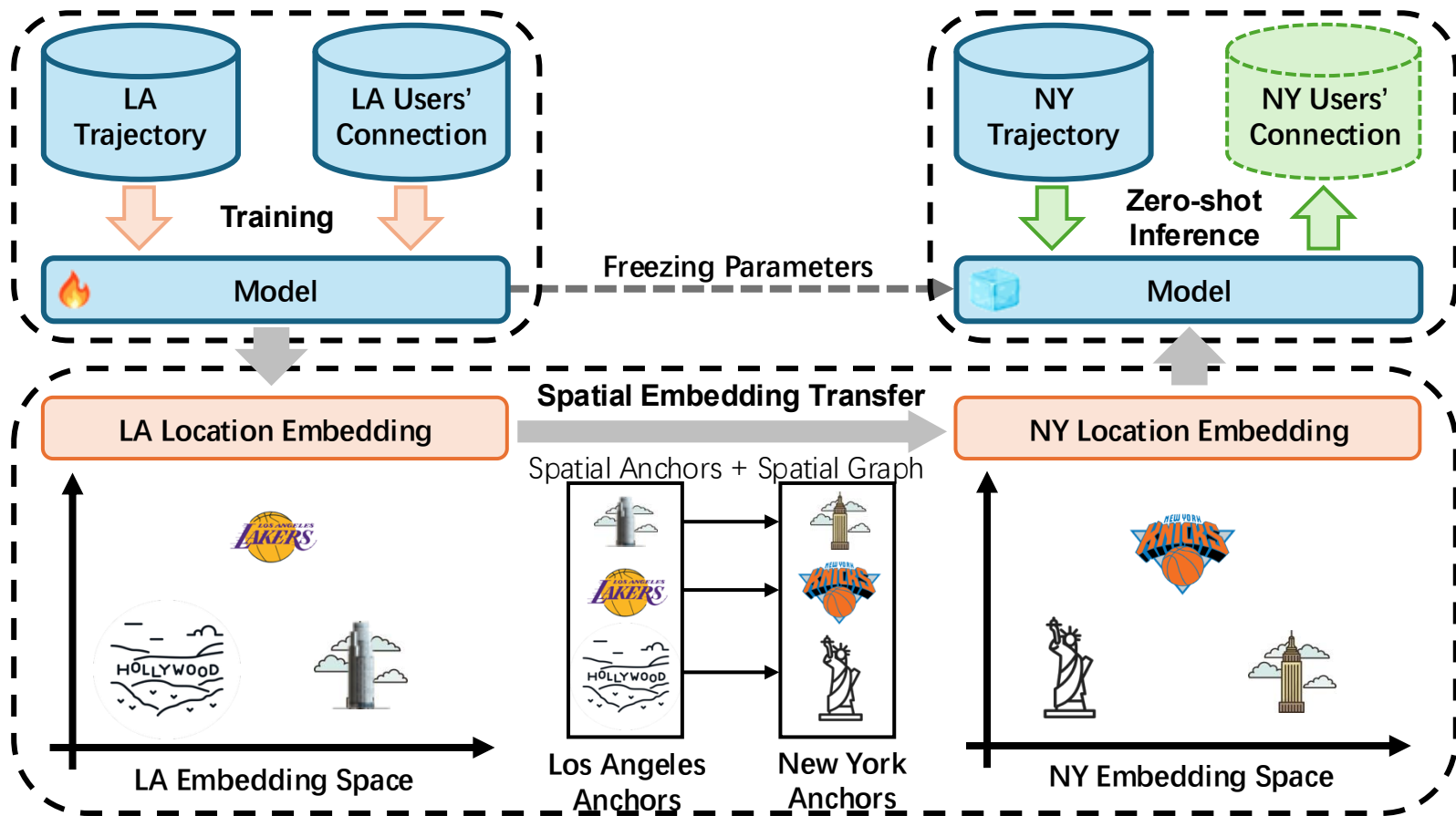


Open Problem: Transferable Cross-Region



Transferable social relationship inference

- Train the foundation model in a data-rich city
- Transfer the spatial embeddings from the source cities to the target city.

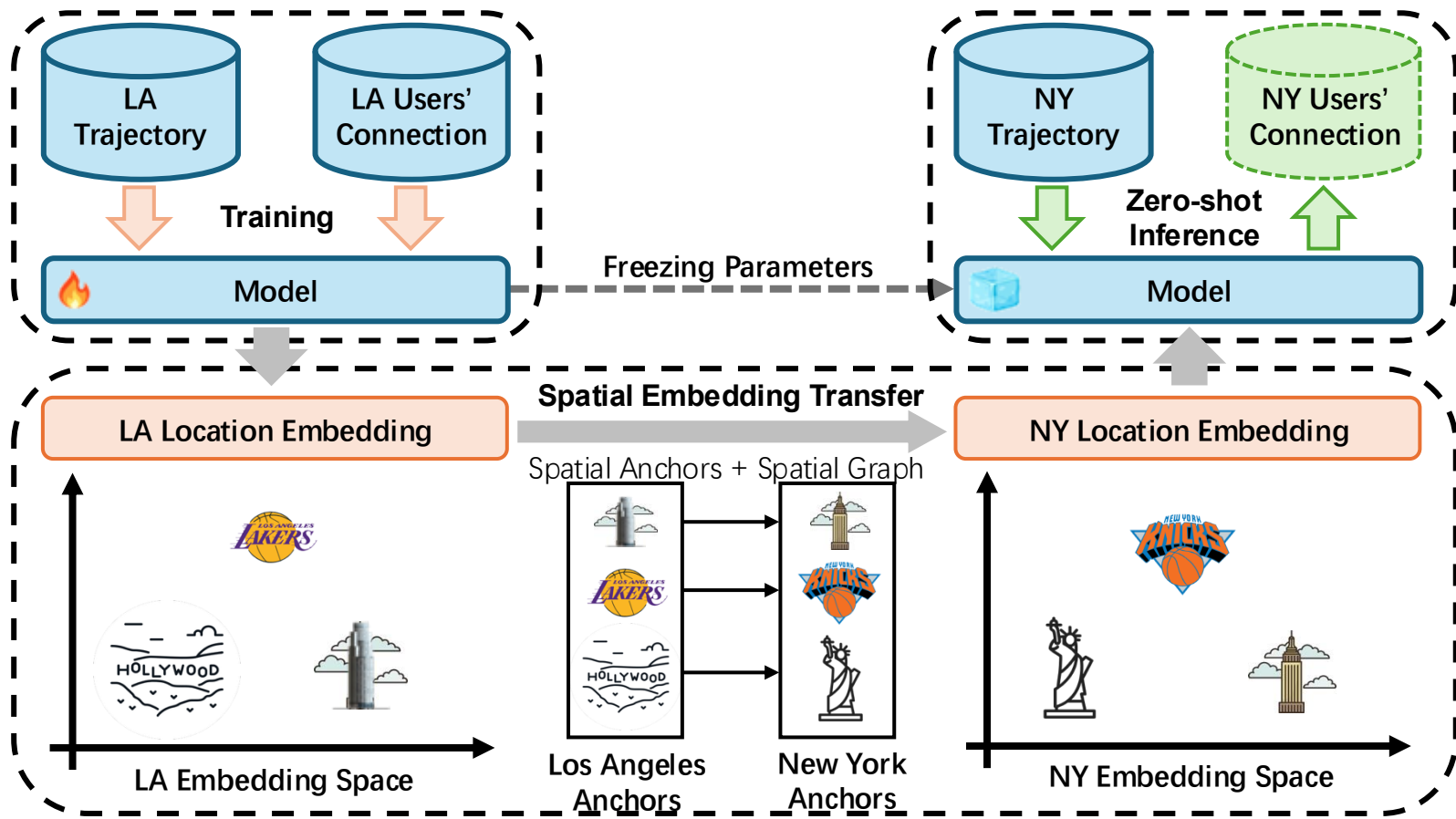


Open Problem: Transferable Cross-Region



Transferable social relationship inference

- Train the foundation model in a data-rich city
- Transfer the spatial embeddings from the source cities to the target city.
- Perform zero-shot social relationship inference on the target city.



Thanks!



John Krumm
(Research Scientist, USC)



Shang-Ling (Kate) Hsu
(PhD)



Nripsuta (Ani) Saxena
(PhD)



Chen Chu (PhD)



**Maria Despoina
Siampou (PhD)**



Bitá Azarijoo (PhD)

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